



**Search for Binary Supernova Ejected
OB Type Runaway Stars**
*in the
neighbourhoods
of*
Young Open Clusters

MASTER'S THESIS

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Abstract

Massive stars of spectral type O and B are often found in binary systems. As the more massive and hence the faster evolving component (primary) of the binary system ends its life in a supernova (SN) event, it produces a neutron star and catapults the other star (secondary) out of the system, thereby producing a runaway star, provided, the binary system is unbound after the explosion. An SN event also leaves behind a supernova remnant (SNR) which consists of the stellar material from the outer layers of the primary expanding into the interstellar medium (ISM). This process, called the binary supernova ejection scenario (BSS), thus produces a runaway-neutron star pair, is one of the main explanations for the origin of runaway stars.

Even though the remnant from the SN event might be undetected or invisible, finding a runaway-neutron star pair provides strong evidence for a BSS event in the past. This work attempts to find such runaway-neutron star pairs around young-open star clusters which host OB-type stars. A sample of young open stars clusters in the galaxy has been chosen and all the stars in the region are traced back to find potential runaway stars. After finding a sample of fast moving stars which can be traced back to the clusters, their spectral types have been ascertained using data from *Gaia* combined with observational data from the Großschwabhausen observatory (GSH) and compared to synthetic stellar populations to select OB-type runaway star candidates. The most promising candidates, some of which can be associated with pulsars in the vicinity, have been discussed in detail in the results section of this thesis.

The region around these candidates can be analysed in detail in future surveys to look for signatures of previously unidentified supernova remnants. An updated set of parameters for the potential runaway stars and their host clusters has been produced using the latest available data, combined with high-resolution spectroscopic observations.

To my parents

Motivation

Massive stars of spectral type O and B have relatively short lifetimes of a few million years. Therefore, OB-type stars are the younger members of open star clusters. It is sometimes found that such stars have a higher-than-normal velocity (> 25 km/s) relative to the surrounding stars (while typical velocity dispersion is ~ 3 km/s) and interstellar medium (ISM). These are called “runaway” stars. They show a lack of multiplicity which is uncommon for early-type stars since more than 50% of early-type stars are found in multiple systems. In addition, they are often found at isolated places, away from star-forming regions, and even outside the disk of the Milky Way. These peculiarities led to several mechanisms to explain their origins, two of which have been relevant to this day: the Dynamical Ejection Scenario (DES) and the Binary Supernova Scenario (BSS). In DES, runaway stars are produced as they are ejected in the early stages of star cluster formation, up to 3 Myr, due to gravitational interactions within the cluster. Once clusters reach a dynamical equilibrium, DES becomes extremely rare. At this point, BSS becomes the primary source of producing runaway stars. This scenario takes place in binary star systems where the primary star, which is more massive ($> 8M_{\odot}$) and hence the faster-evolving component of the binary, reaches the end of its life and undergoes a supernova (SN) explosion. The SN leaves behind a compact object like a neutron star or a black hole depending on the mass of the progenitor. The secondary, less massive star, is catapulted out of the binary system with its orbital velocity, becoming a runaway star.

Supernovae (SNe) events leave behind stellar material in the form of supernova remnants (SNRs). SNRs can be observed for typical lifetimes of about 100 kyr before they dissipate into the ISM and become invisible. This dissipation age can vary depending on the density of the ISM and the energy produced in the SN. Considering the galactic supernova rate of about 2 per century, and the dissipation time, one expects to find 2000 SNRs visible within the Milky Way. However, the number of SNRs visible today in the galaxy stands at around 300. One possible explanation for the missing SNRs comes from “invisible” supernovae. These are expected in low-density

regions where the SNR is unable to sweep up enough interstellar material to enter a radiation or even Sedov phase. On the other hand, SNe which produce lower than the usual 10^{51} erg of energy and/or are in a high-density medium can dissipate earlier than 100 kyr. Considering these, an estimate of the SN rate of our galaxy must take such “invisible SNRs” into account.

This motivates the search for runaway star-neutron stars pairs as evidences for the so-far-undetected SNRs. Young open star clusters are a logical choice for search regions as they are regions of likely SN events due to the presence of early-type stars. An SN creates a region of low density as its shell sweeps out ISM and creates a bubble. If another SN event occurs later inside this bubble, it is likely to be “invisible” because of the low density ISM, and dissipate faster than usual once it hits the edge of the bubble after which the medium has a higher density.

In this thesis, the regions around 399 young open clusters are searched for BSS runaway stars and several candidates have been found. Some of these also show the existence of a nearby pulsar. This, combined with the runaway stars being traced back to the region inside these open clusters provides solid evidence for a BSS scenario in the past 100 kyr. *Gaia* DR3 data has been used to examine the linear 2D motion of the stars with respect to their supposed host clusters in an automated process using a Python program. The program is also used to check the isochrone fits to associate the runaways with clusters and estimate their temperatures, surface gravity values and masses among other quantities. Further, the presence of nearby pulsars and their 2D motions is checked for a trajectory intersection. Associations found in this work would provide target sites for future SNR detections and also possibly regions to look for neutron stars.

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Chapter 1

Introduction

Since our beginning as a species, we tried to find the purpose and meaning of our existence. The most intriguing questions like “Why are we here?”, “How did the universe come into being?”, “How does the universe work?” still remain unanswered convincingly. Several attempts have been made, often taking the shape of religions.

Motivated by religion or otherwise, many great minds across civilisations have made meticulous studies of the night sky and the heavenly bodies to make sense of the universe. People mapped each visible star and tried to find patterns in their positions on the sky. Therefore, the event of the appearance of a “new star” (Latin *nova stella*) or “guest star” was noticed by several independent observers around the world and noted in numerous historical accounts.

Today we know that these are, rather ironically, the spectacular death of stars. The death event of a star is called a *supernova* (pl. *supernovae*), a term coined by Walter Baade and Fritz Zwicky.

Supernovae are extremely luminous events caused by the explosive ends of massive stars’ lives, which release enormous amounts of energy—of the order of 10^{51} erg. Some are so luminous that they can outshine their host galaxies. A supernova leaves behind a *supernova remnant* (SNR) which consists of leftover stellar matter along with elements synthesised in the explosion itself. By studying the aftermath of SN events, including SNRs, it is possible to deduce the initial conditions of the progenitor systems and have a better understanding of stellar evolution processes.

A supernova event in our own galaxy is sometimes visible to the naked eye. SN 1006 was one such supernova event which was visible from Earth and had a brightness comparable to Venus. Some reports claim it to be visible even during the day. Figure 1.1 shows the remnant of SN 1006 as it is seen today in X-Ray by the Chandra X-ray Observatory. Some other “guest stars” that can be reliably associated with SNe (and

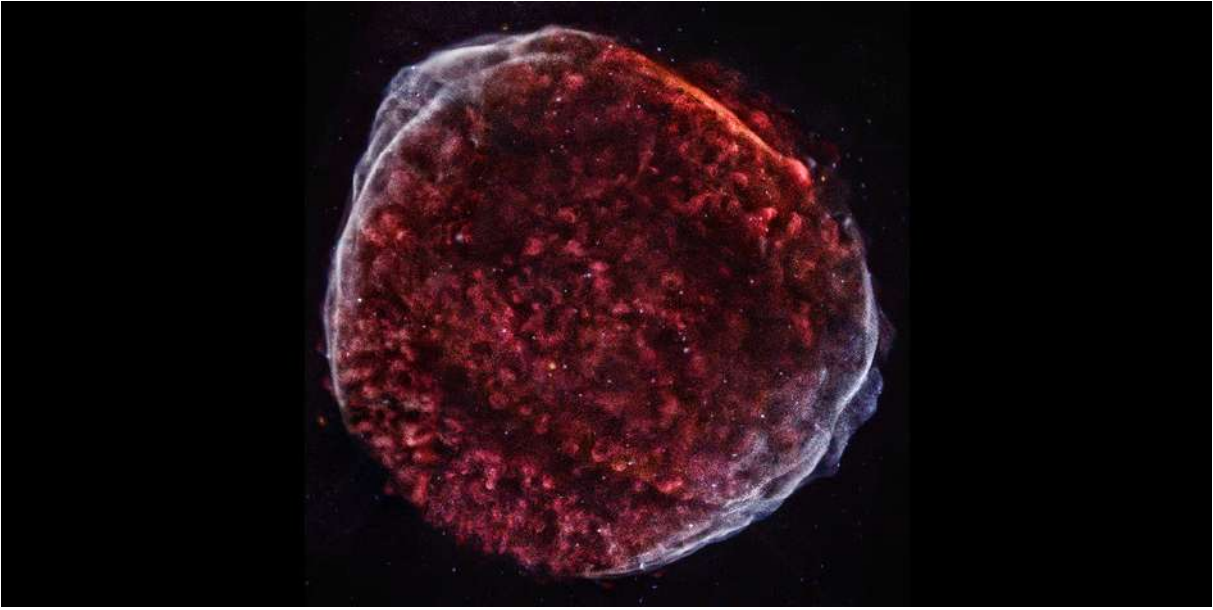


Figure 1.1: Chandra’s false-colour image of SN 1006 shows X-rays from multi-million degree gas (red/green) and high-energy electrons (blue). Credit: NASA/CXC/Middlebury College/F.Winkler

SNRs) are the guest stars of AD 185, AD 1006, AD 1054, AD 1572, and AD 1604.

SNe enrich the interstellar medium (ISM) with the elements that are necessary for life. In fact, today we know that our solar system could be formed thanks to the materials produced by supernova events that most likely shaped the local bubble. Star formation in the solar neighbourhood is triggered by the interaction of the local bubble with the surrounding ISM (Zucker et al. (2022)). It therefore becomes important that we study these events and their aftermath in detail to understand our origins.

This thesis concerns with finding evidences of supernova remnants that have likely been dissipated, or still remain undetected. Supernovae occurring in binary star systems often produce a compact object (like a neutron star) and a *runaway star*. The discovery of such a pair of a neutron and a runaway star, which can be traced back to a common origin, would provide strong evidence for a SN event in the past, even when its SNR remains invisible or undetected. This would also provide promising target sites for further investigation about the nature of the ISM in such regions, and to possibly detect some traces of the SNRs.

It is essential to begin the discussion with a background on the key astrophysical concepts that were used for this work. A good starting point is the formation and evolution of stars.

1.1 Star and star cluster formation

Stars are believed to originate from giant molecular clouds (GMCs). Stars are formed, due to the gravitational interactions between the molecules and dust in GMCs, several at a time in clusters. GMCs collapse under their own gravity and the cloud begins to fragment into parts to create regions of high density which eventually become proto-stars. In the early 1920s, James Jeans calculated that a cloud with a given density and temperature can collapse only if its mass is greater than a threshold called the Jeans mass. Below this mass, the pressure of the gas counteracts collapse. Jeans mass is expressed as:

$$M_J \approx 3 \times 10^4 \sqrt{\frac{T^3}{n}} M_\odot$$

here, n is the number density of atoms/ m^3 and T is the temperature.

Typical values for neutral hydrogen (HI) clouds are $n = 10^6 m^{-3}$ and $T = 100$ K, which give $M_J = 30,000 M_\odot$. As the cloud collapses further, the density rises much faster than the temperatures in the equation, leading to M_J becoming smaller, and therefore smaller clouds begin to collapse individually. As a result, the cloud forms *fragments*. Hundreds to tens of thousands of stars would be formed in several *open clusters* more or less simultaneously from this process (see fig. 1.2).

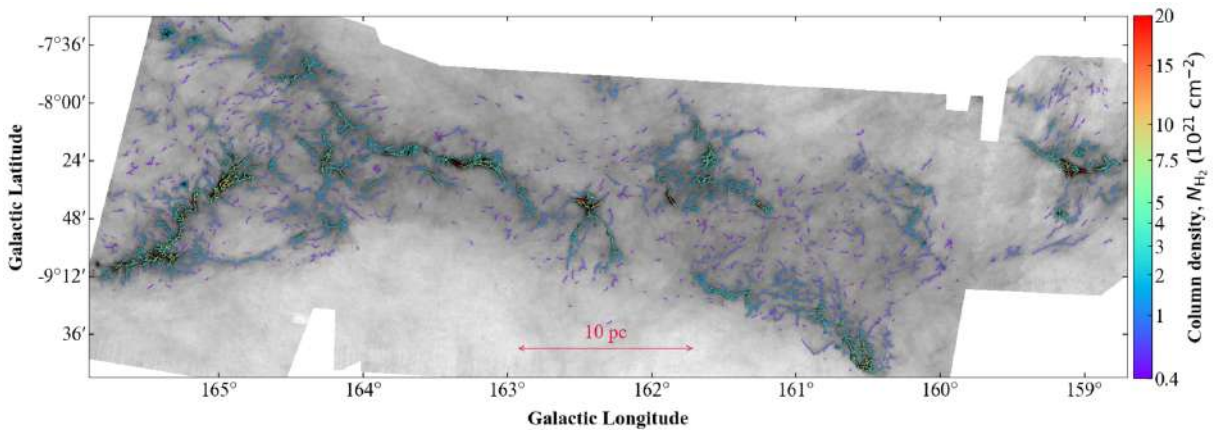


Figure 1.2: The network of filaments in the California GMC showing the initial stages of star and star cluster formation. Credit [Zhang et al. \(2024\)](#)

Contraction and fragmentation continues until densities and temperatures become very high (eventually leading to fusion of H) and the smaller fragments become optically thick. This leads to the formation of *protostars*. Some protostars forming this way rotate so fast that they may split into two, thus forming binary systems. In reality, this involves more complicated processes including considerations for magnetic fields

and magneto-hydrodynamics. An extremely insightful animation has been created by Matthew Bate at the University of Exeter, from which snapshots of the main phases have been taken and shown in figure 1.3.

As can be seen in these snapshots, filamentary structures primarily develop driven by the influence of magnetic fields in the clouds. Each filament, after being Jeans-overcritical, starts to collapse into smaller fragments. The contraction process is nearly isothermal, and density increases until the cores of the stars are formed. Slowly, the protostars attain hydrostatic equilibrium. This process is well supported by observational evidence as in the famous Orion nebula, and in the *Pillars of Creation* in the Eagle nebula etc.

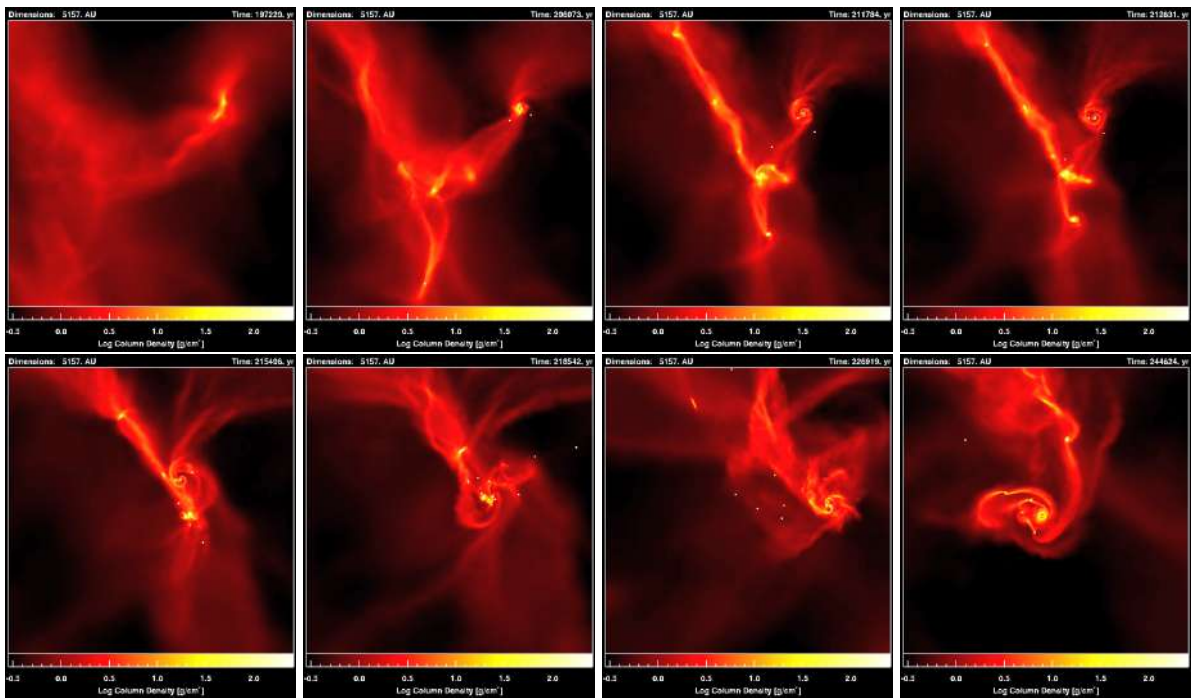


Figure 1.3: Star formation in successive steps from top left to right and then bottom left to right. These show protostars forming from smaller molecular clouds of $50M_{\odot}$ which are 0.375 pc in diameter. Notice the overdensities being formed, these become stars. Credit: [Matthew Bate, University of Exeter](#).

1.2 Star Clusters

Stars are formed in giant molecular clouds (GMCs), also known as stellar nurseries in a process described in section 1.1. The first catalogue of star clusters was prepared by Charles Messier in 1784 which included globular clusters and open clusters along with gas nebulae and galaxies. A more comprehensive catalogue called the New General Catalogue of Nebulae and Clusters of Stars (NGC) was published in 1888 by John Louis Emil Dreyer [Dreyer \(1888\)](#). Many globular and open clusters have a naming scheme

starting with NGC from this catalogue.

Although most stars in the Milky Way are observed in isolation, it is believed that most (possibly all) stars form in clustered environments and spend at least a short amount of time gravitationally bound with their siblings [Portegies Zwart et al. \(2010\)](#). Groups of stars can be mainly classified into three types: stellar associations, open clusters and globular clusters.

1.2.1 Stellar Associations

Associations are groups of very young stars which are usually difficult to identify based on their appearance in the sky. These are very loosely bound but have a common origin which can be determined by spectroscopy and isochrones. They become gravitationally unbound over the years and drift apart from each other. These usually have a few tens of members. These associations can be classified based on their ages into OB associations, consisting of OB-type stars, or T-Tauri associations (named after the prototypical T-Tauri star, a young star in the Taurus star-forming region) consisting of pre-main sequence young stellar objects.

1.2.2 Open Clusters

Open clusters usually contain from a few tens to a few hundreds of stars which are bound by their mutual gravitational attraction. The strength of this bond depends on the number of members in the cluster: denser and larger populations are more strongly bound. These originate from the same GMCs and have roughly the same age. Open clusters serve as ideal laboratories for understanding stellar formation and evolution. These clusters generally survive for a few hundred million years up to a few billion years. They are found in active star-forming regions of galaxies and therefore are more commonly found in the spiral arms of spiral galaxies and are not found in elliptical galaxies.

Open clusters can be identified in a field of stars based on their tight distribution in position, common astrometric (proper motions in right ascension μ_α^* and declination μ_δ , and parallaxes π) and photometric properties (magnitudes in different filters). [Cantat-Gaudin & Anders \(2020\)](#) find 1229 clusters and their members using these properties in the Milky Way using data from [Gaia Collaboration et al. \(2018\)](#).

The kinetic energy of the cluster members, combined with the differential rotation of the galaxy and external gravitational disturbances tend to gradually disperse the open clusters. Dynamical ejections also take place in the early stages of star cluster

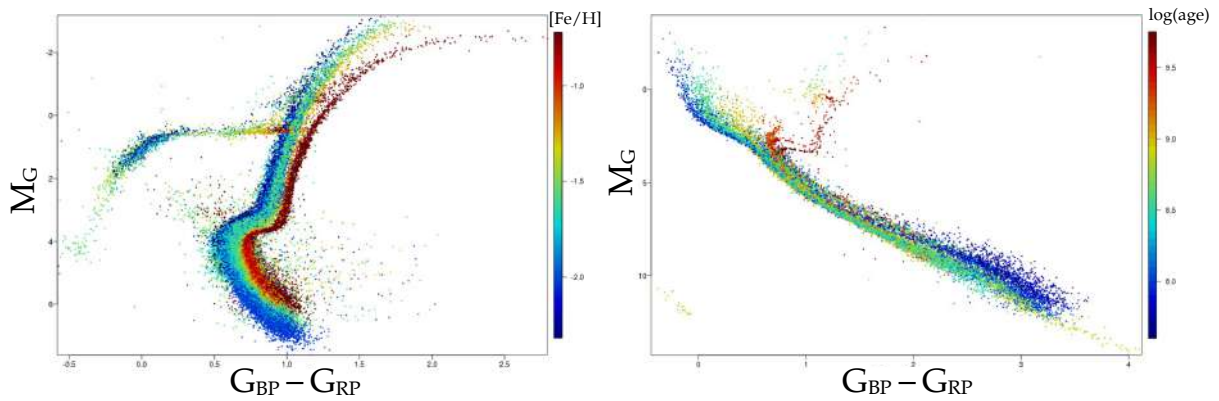


Figure 1.4: Figures taken from *Gaia Data Release 2. Observational Hertzsprung-Russell diagrams* (2018)
Left: Composite HRD for 14 globular clusters, coloured according to metallicity.
Right: Composite HRD for 32 open clusters, coloured according to $\log(\text{age})$.

formation, up to 3 Myr [Oh & Kroupa \(2016\)](#) which disperse the clusters and remove stars from the them at high velocities.

1.2.3 Globular Clusters

Globular star clusters (GCs) usually contain roughly about 10^4 to 10^6 stars. The Milky Way hosts about 150-200 GCs. The distribution of the stars in GCs is spherically symmetric, and the central densities are about ten times larger than in open clusters. Stars in globular clusters are among the oldest in the Milky Way, and therefore they are of great importance for studies of stellar evolution. In the Milky Way, these are found in the galactic halo. They are the largest and most massive type of star cluster, tending to be older, denser, and composed of lower abundances of heavy elements than open clusters. These are similar to dwarf spheroidal galaxies, and their distinction is not always clear [Van Den Bergh \(2008\)](#).

Their colour-magnitude diagrams (see fig. 1.4 left) show a prominent giant branch, along with horizontal and asymptotic branches. Their main sequences consist of only faint red stars.

1.3 Stellar Evolution

1.3.1 Pre-Main Sequence Phase

After the protostar or pre-main sequence star is formed, the increase in its density causes its opacity to increase. A large fraction of the energy released due to gravitational collapse cannot escape the protostar as light and transforms into heat. The

luminosity of the star, which increased to a peak after starting the process of collapse, now begins to reduce, while the temperature of the star increases. The reduction in luminosity and the increase in temperature lead to a movement of the star on the Hertzsprung–Russell (HR) diagram on a *Hayashi track*. These Hayashi tracks are shown for stars of different masses in fig. 1.5. Depending on the mass, the evolution of the star takes different amounts of time and different Hayashi tracks. Stars move on these tracks until they start H fusion in their cores as they settle down on the *main sequence* of HR diagrams. A $15M_{\odot}$ star condenses to the main sequence in 60,000 years, whereas for a $0.1M_{\odot}$ star, the time is hundreds of millions of years. At the end of this phase, H burning starts in cores with the p-p chain as the core temperature reaches about 4 million K.

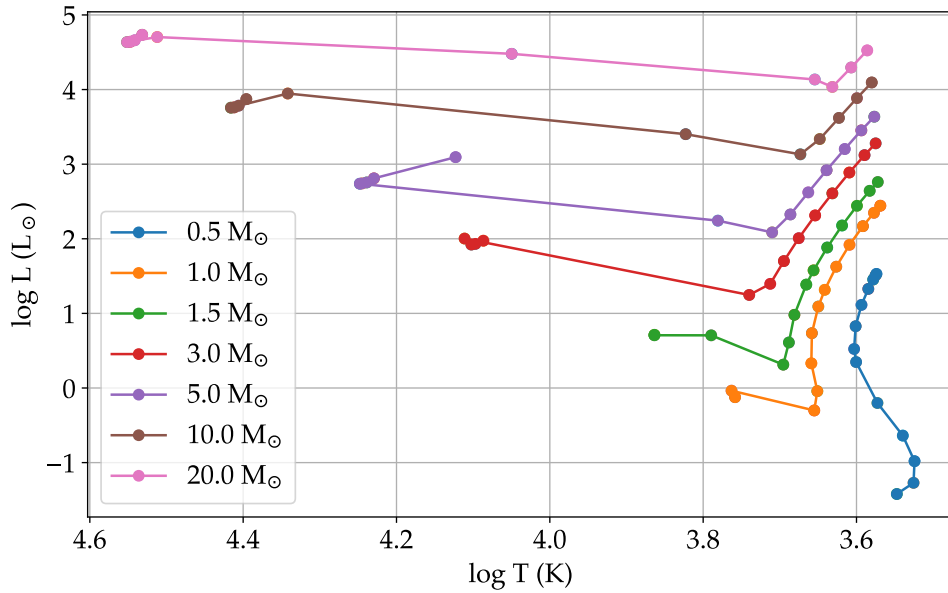


Figure 1.5: Hayashi tracks generated using PARSEC evolutionary tracks [Bressan et al. \(2012a\)](#) for different mass stars for ages 10^2 yr to 10^8 yr. Each consecutive point on a track, starting from the top left, denotes an increase by a factor of $10^{0.5}$ in the age of the star. Note the phase of sharp luminosity decline and temperature rise as the star moves towards the main sequence.

1.3.2 The Main Sequence Phase

The star is said to be “born” when it reaches the main sequence. This is why the main sequence at this point is sometimes also called the zero-age main sequence (ZAMS). H burning in the core is the only source of energy in this phase. The stars remain on the main sequence for most of their lives in hydrostatic equilibrium as they burn H into He.

While the star is on the main sequence, it burns H either through p-p chain or through CNO-cycle depending on the temperature that its core can reach. Stars more

massive than $1.3 M_{\odot}$ have the CNO cycle as the primary energy generating mechanism.

There is little change in a star's luminosity on the main sequence. The best example of a main-sequence star is the Sun. It is known that the luminosity of the Sun has not changed much during its lifetime of 5 billion years. The main sequence lifetimes (τ) and luminosities (L) of a star can be roughly given by empirical relations $\tau \approx 10^{10} \left(\frac{M_{\odot}}{M} \right)^{2.5}$ yrs and $L = L_{\odot} \left(\frac{M}{M_{\odot}} \right)^{3.5}$. Table A.3 shows the main sequence lifetimes of these stars with different masses along with some other properties.

While on the main sequence, the core becomes progressively rich in He. Further reactions beyond He are not seen because they require higher temperatures. As a result of He being formed that takes up less volume compared to H, the pressure in the core decreases, and the core must contract under its own weight. The gravitational energy released due to this contraction raises the temperature of H in a thin shell around the core so much that it starts shell burning. The He made in the shell adds to the mass of the core whose contraction is accelerated. The energy produced in the shell and the gravitational energy due to the contracting core push out the envelope of the star due to radiation pressure. As a result, the star expands in size. Its surface becomes cooler, but its luminosity increases enormously because of increased surface area. The star becomes a giant star as it starts to move out of the main sequence phase.

1.3.3 Post-Main Sequence Phase

Once the H is exhausted in the core, stars leave the main sequence and start burning He and then heavier elements. Depending on the mass of the star, there are three main scenarios that can take place:

1. Stars less massive than $0.5M_{\odot}$ do not have their core temperatures exceed 100 million K, which is required to start burning the He produced in the p-p chain. They end up as He white dwarfs.
2. Stars with $0.5M_{\odot} < M < 8M_{\odot}$ burn He in the triple α process as they transform into a red giant. These stars shed their outer envelopes as a planetary nebula and become a CO- white dwarf or a ONeMg white dwarf.
3. Stars more massive than $8M_{\odot}$ successively burn heavier fuels in the later stages after the main sequence until the core becomes gravitationally unstable due to two scenarios: Electron capture or Fe core-collapse. Stars with mass greater than $8M_{\odot}$ up to about $10M_{\odot}$ or even $12M_{\odot}$ undergo an electron capture SN (ECSN),

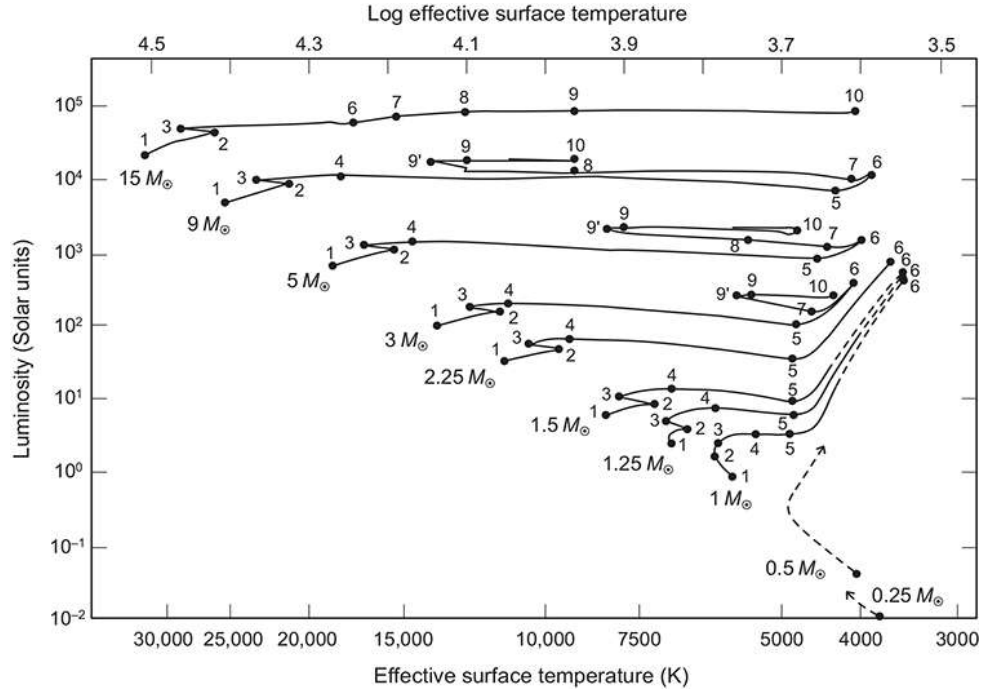


Figure 1.6: Evolution off the main sequence for stars of different initial main sequence mass. Dashed lines are estimates. Times to evolve through each numbered interval are given in table 1.1 Adapted from Iben (1967)

discussed further in section 1.4.1. More massive stars undergo core-collapse supernova explosions (CCSN), also discussed in section 1.4.1. The evolution further than the main sequence is shown in fig. 1.6 along with the duration it takes to move between the different phases in table 1.1.

Depending further on the metallicity, the outcome of an SN may be a remnant along with a neutron star, or the direct formation of a black hole. Various outcome cases have been shown in fig. 1.7.

Table 1.1: Duration of the numbered intervals displayed in fig. 1.6

Mass ($M_{\odot}$)	1.0	1.25	1.5	2.25	3	5	9	15
1→2	7×10^9	2.8×10^9	1.6×10^9	4.8×10^8	2.2×10^8	6.5×10^7	2.1×10^7	1.0×10^7
2→3	2×10^9	1.8×10^9	8.1×10^7	1.6×10^7	1.0×10^7	2.2×10^6	6.1×10^5	2.3×10^5
3→4	1.2×10^9	1.0×10^9	3.5×10^8	3.7×10^7	1.0×10^7	1.4×10^6	9.1×10^4	-
4→5	1.6×10^8	1.5×10^8	1.1×10^8	1.3×10^7	4.5×10^6	7.5×10^5	1.5×10^5	7.6×10^4
5→6	$\geq 1 \times 10^9$	$\geq 4 \times 10^8$	$\geq 2 \times 10^8$	3.8×10^7	4.2×10^6	4.9×10^5	6.6×10^4	-
6→7	-	-	-	-	2.5×10^7	6.1×10^6	4.9×10^5	7.2×10^5
7→8	-	-	-	-	-	1.0×10^6	9.5×10^4	6.2×10^5
8→9	-	-	-	-	4.1×10^7	9.0×10^6	3.3×10^6	1.9×10^5
9→10	-	-	-	-	6.0×10^6	9.3×10^5	1.6×10^5	3.5×10^4

Times are in units of years and have been rounded to one decimal place.

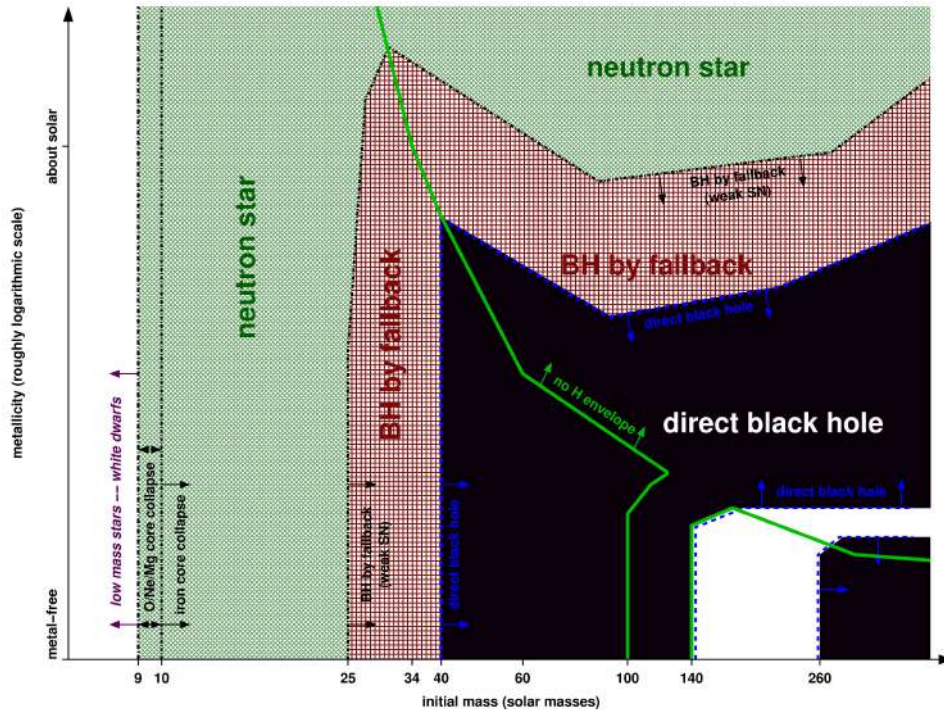


Figure 1.7: Fate of massive stars depends on their mass and metallicity. Figure taken from Heger et al. (2003)

1.4 Supernovae

Supernovae mark the end of the lives of massive stars. They can occur through two very different mechanisms: core-collapse supernovae and thermonuclear (Type Ia) supernovae.

Type Ia (thermonuclear) supernovae occur in binary star systems where a white dwarf accretes matter from a companion star. Once the white dwarf's mass approaches the Chandrasekhar limit ($\sim 1.4M_{\odot}$), the increasing pressure and temperature triggers a runaway thermonuclear reaction, causing the white dwarf to explode. Unlike core-collapse supernovae, Type Ia supernovae completely disrupt the white dwarf, leaving no compact remnant behind. Since this process almost always takes place with the same conditions (reaching the Chandrasekhar limit), the explosions themselves are very similar. Therefore the peak luminosities of the SN events are similar. Type Ia supernovae are of immense importance in cosmology as "standard candles" because of this consistent peak luminosity. This makes them useful for measuring cosmic distances.

Core-collapse supernovae occur in massive stars (typically greater than $8M_{\odot}$) when their cores become unstable and collapse under the force of gravity. This collapse leads to the formation of neutron stars or black holes as compact remnants. The neutron stars often receive a "kick" due to the asymmetric explosions which makes the neutron stars acquire an average velocity of 400 km/s (Hobbs et al. (2005)).

This thesis is concerned with core-collapse supernovae, which produce neutron stars as compact object remnants. In particular, the focus is on those that occur in binary star systems, disrupting the system and producing a neutron star — runaway star pair.

1.4.1 Core Collapse Supernova

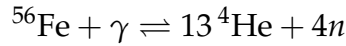
After the burning phases of H, followed by He, C, O, Ne, and Mg, one of two scenarios can take place depending on the mass of the stars:

For massive stars $\gtrsim 8M_{\odot}$ up to $10 - 12M_{\odot}$ an **electron capture SN (ECSN)** takes place Miyaji et al. (1980). In ECSN progenitors, after C burning, develop an ONeMg core. The ONeMg core is supported by electron degeneracy pressure. Shell burning makes the core mass reach close to the Chandrasekhar limit ($\sim 1.4M_{\odot}$). When the density of the core exceeds about 4×10^9 g/cm³, electrons are captured by Mg atoms and the core begins to collapse Takahashi et al. (2013). The collapse leads to the formation of a neutron star core. Energy released in the process drives a shockwave outwards

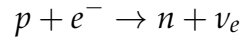
disrupting the star in a relatively less energetic explosion (10% to 100% of the canonical 10^{51} erg). If these SNe occur in binary star systems, the binary most likely remains bound Podsiadlowski et al. (2004). A low kick velocity is imparted on the neutron star.

In stars more massive than $\sim 11 M_{\odot}$, a **core collapse SN (CCSN)** takes place. CCSN progenitor stars burn their fuel progressively starting from H to He all the way up to Si until they reach Fe, forming an iron core with the mass of $\sim 1.4 M_{\odot}$ (Chandrasekhar limit). Further fusion is no longer exothermic and the core becomes gravitationally unstable since it can no longer counteract its gravity by the energy produced with nuclear fusion, and begins to collapse. Because of two critical processes—*photo-disintegration* and *neutronization*—the collapse of the core is further accelerated.

Photo-disintegration occurs when high-energy photons cause Fe nuclei to disintegrate into He and neutrons, leading to a reduction in radiation pressure within the core in the process:



Neutronization is the process where electrons and protons fuse to form neutrons, accompanied by the emission of high-energy neutrinos. These neutrinos escape the core, carrying away energy:



As a result of photodisintegration, decreasing the radiation pressure, and the energy loss due to neutrinos produced in neutronization escaping the star, the core's collapse is further accelerated. In roughly 1 second, the core collapses from a few thousand km, down to ~ 50 km. The neutron degeneracy pressure stops the collapse momentarily as the outer layers of the star fall on the core. This creates a shock wave and the outer layers on the core are “bounce back” and create a reverse shock. As the density of the core increases further, the mean free path of the neutrinos reduces to roughly the size of the core. Neutrinos from the core are kept inside a *neutrinosphere*, which reheats the star. This reheating creates a higher pressure against the infalling matter, which ultimately leads to the shock-wave breaking through the outer layers and producing the SN explosion.

Due to asymmetries in the SN explosions, the neutron stars produced receive a strong kick. Hobbs et al. (2005) showed that the mean velocity magnitude is about 400 km/s but can be as high as 1000 km/s. An SNR is produced as the stellar ejecta which is expelled outwards interacts with the interstellar medium (ISM). SNRs remain visible up to $\sim 10^4$ to $\sim 10^5$ years Vink (2012).

1.4.2 Supernova Classification and Light Curves

Supernovae are classified mainly on the basis of observational features seen in their spectra taken near the peak brightness into 4 main types: Type Ia, Ib, Ic, and II. Type II is further subdivided into II-P, II-L, and II-b. Figure 1.8 shows the classification scheme.

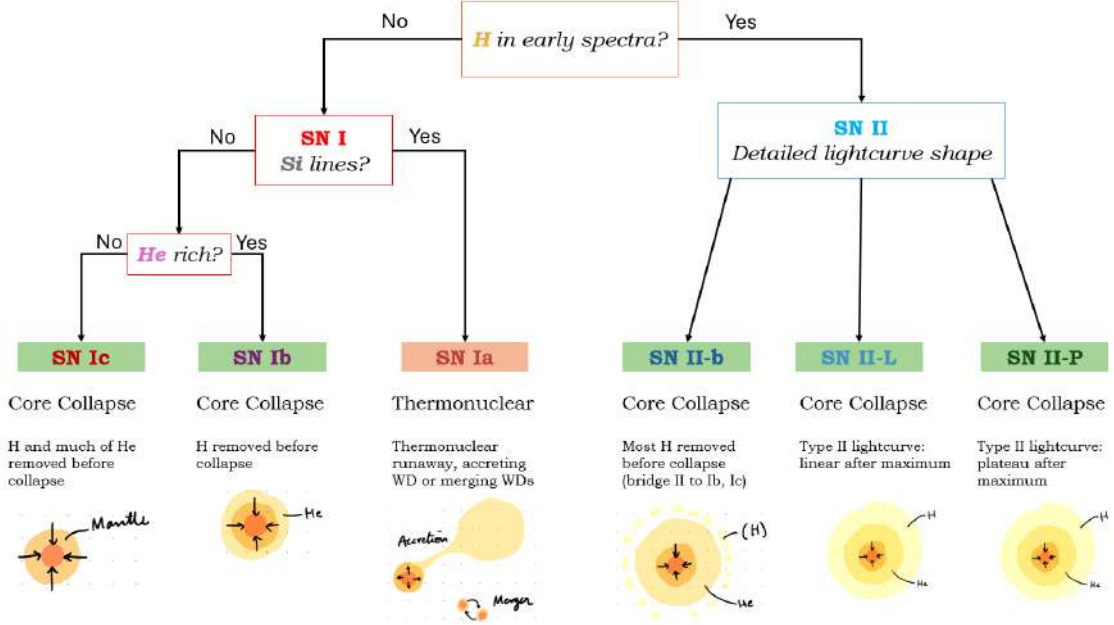


Figure 1.8: SN classification

The spectra from the various types of SN along with the light curves are shown in fig. 1.9. These main types are discussed below:

Type Ia: Type Ia supernovae spectra are identified by the absence of hydrogen lines while having prominent silicon absorption lines near peak luminosity. Some Ca and O lines, also appear in their spectra near the peak brightness. Type Ia supernovae are likely caused by the thermonuclear explosions of white dwarf (WD) stars in accreting systems. The WD accretes mass from its companion until it reaches the Chandrasekhar limit ($\sim 1.4M_{\odot}$). When this limit is reached, the WD explodes in a thermonuclear runaway reaction as a type Ia supernova. This makes almost every type Ia supernova have the same characteristic light curve. This class of supernovae is sometimes termed a thermonuclear supernova, to distinguish it from all other classes that derive their power from gravitational collapse and not from thermonuclear reactions. Type Ia supernovae are found in all types of galaxies and their standard candle properties make them a valuable distance-measuring tool.

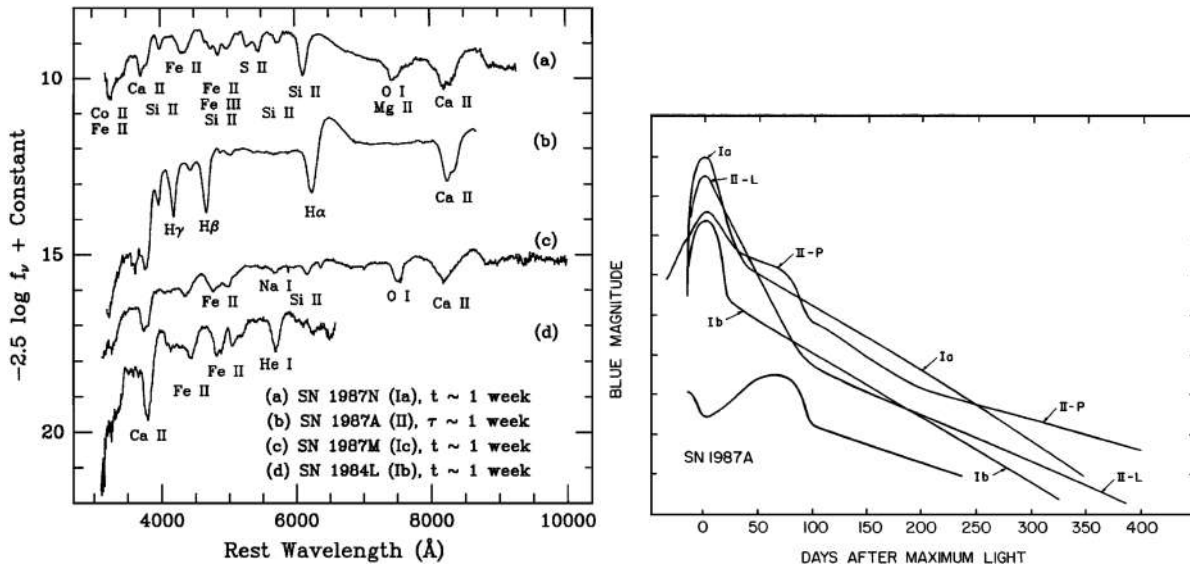


Figure 1.9: *Left:* Examples of spectra from various types of supernovae: a Type Ia, Type II, Type Ic and a Type Ib supernova. *Right:* Examples of light curves of various types of supernovae. Credit: Filippenko (1997)

Type Ib: These are formed when a massive star collapses under its own gravity. Their progenitors lose much of its outer H envelope due to strong stellar winds or interactions with a binary companion, hence no H are seen but He lines are present in their spectra. Their progenitors are called Wolf-Rayet stars.

Type Ic: These are similar to type Ib, except their progenitors lose much of their outer H as well as He envelope due to strong stellar winds. Therefore, these show neither H nor He lines in their spectra. Their progenitors are also Wolf-Rayet stars.

Type II: These supernovae are characterised by prominent hydrogen lines and are thought to be associated with the core collapse of a massive star. They are found only in regions of active star formation. Based on the light curves, type II supernovae may be further subdivided into II-P, II-L, II-b based on the light curve evolution. Type II-P show a plateau in the light curve after the peak brightness while type II-L show a linear decline in the light curve after the peak brightness. In a Type II-b event the spectrum contains prominent hydrogen lines initially, but it transitions into one similar to that of a Type Ia,b supernova. The suspected II-b mechanism is core collapse in a red giant that has lost most but not all of its hydrogen envelope through stellar winds or interaction with a binary companion.

1.5 Supernova Remnants

A supernova remnant (SNR) is a supersonically expanding nebula consisting of stellar ejecta that is produced as a result of a supernova that sweeps up its surrounding ISM. There are 294 known SNRs in our galaxy according to Green’s catalogue [Green \(2019\)](#). Their galactic distribution is shown in fig. 1.10. As expected, most of the SNRs lie close to the galactic plane.

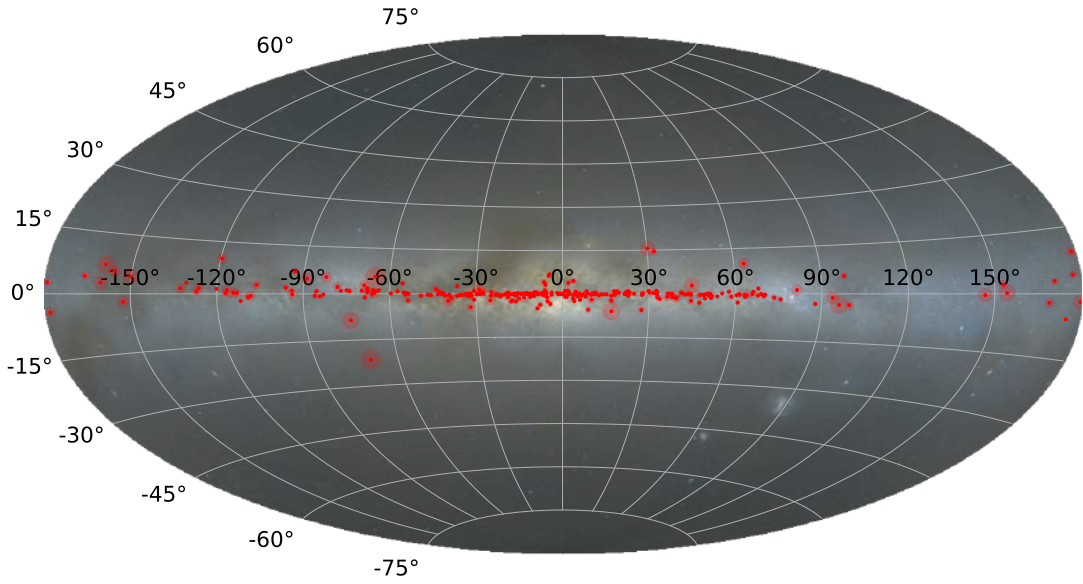


Figure 1.10: Galactic SNR locations based on Green’s catalog [Green \(2019\)](#). Scatter plot of the 294 SNRs is overlaid on top of *All-Sky Panorama Image of the Milky Way* [Mellinger \(2009\)](#) making use of [hips2fits](#), a service provided by CDS.

1.5.1 Supernova Remnant Evolution

The evolution of SNRs can be divided into four main phases [Woltjer \(1972\)](#): the ejecta dominated phase, the Sedov-Taylor phase, the pressure-driven or “snow-plough” phase, and the merging phase. It must be noted that this categorisation is not strict and it is often hard to ascertain the phase of an SNR from observations. It is also possible that different parts of an SNR are in different evolutionary phases, depending on the asymmetries in the interactions with the surrounding ISM. One example of this is CTB 109 in which the analysis of CO observations [Tatematsu et al. \(1987\)](#) has revealed that the western part of the SNR has encountered a giant molecular cloud (GMC) complex, making the the SNR appear semicircular, and the western part of the remnant invisible. A brief description of these phases is given in the following:

The Free-Expansion (Ejecta-Dominated) Phase

This phase is also known as the free-expansion phase since the expansion velocity of the shell is close to $V_s = R_s/t$, where R_s is the radius of the shock and t is the time since the SN (SNR age). This can also be written in terms of the mass ejected M_{ej} and the energy released E as $V_s = \sqrt{\frac{2E}{M_{\text{ej}}}}$. With typical values of $M_{\text{ej}} = 1M_{\odot}$ and the energy released $E = 10^{51}$ erg, $V_s = 10^4$ km/s. There is no heating and the ejecta is cold, with the remnant being faint in all wavelengths. Only non-thermal synchrotron emission from high energy electrons accelerated in the shock is present. This phase lasts up to about 2000 yr. Most of the historical supernovae belong to this category. In this phase, the mass of the supernova ejecta, M_{ej} , dominates over the swept-up mass M_{sw} . The early phase is described by models by Truelove & McKee (1999) and Chevalier (1982). G1.9+0.3 fig. 1.11 which is an example for an SNR in the free expansion phase Reynolds et al. (2008). The acceleration of electrons up to GeV energies is sufficient to produce radio synchrotron emission. A few SNRs in this phase also emit in X-rays (> 10 TeV).

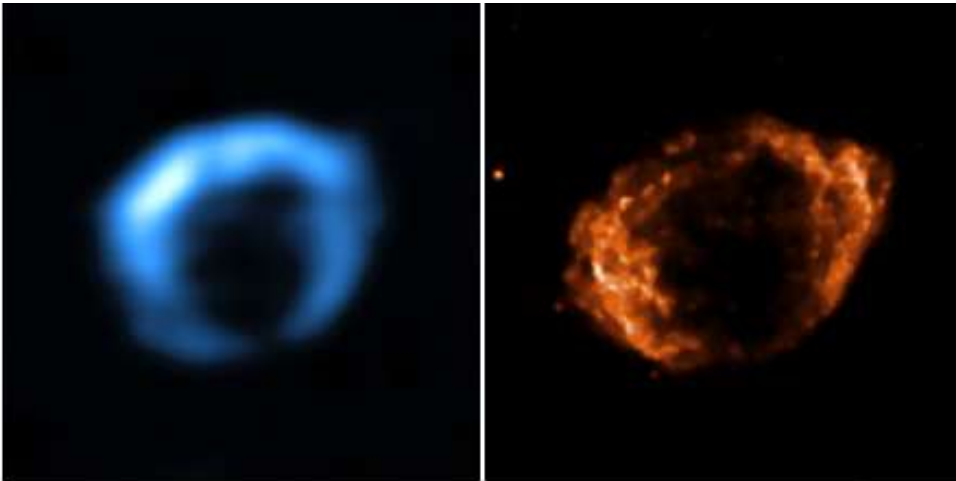


Figure 1.11: *Left:* SNR G1.9+0.3, the youngest remnant in the Galaxy, seen in radio waves from the VLA in 1985. *Right:* The remnant, in 2007, seen in X-rays with Chandra. In the 22 years between these images, the remnant expanded substantially, allowing the time since the original supernova explosion (about 140 years) to be estimated. *Credit:* X-ray (NASA/CXC/NCSU/S.Reynolds et al.); Radio (NSF/NRAO/VLA/Cambridge/D.Green et al.).

The Sedov-Taylor (Adiabatic expansion) Phase

This phase starts when the mass swept up by the ejecta is no longer negligible for the forward shock to expand freely. The forward shock wave decelerates and leads to the formation of a “reverse shock,” which propagates back into the ejecta towards the centre of the SNR. This phase is also called the adiabatic expansion phase. However, it

must be noted that this can be misleading since there are some energy losses (for eg. by escaping cosmic rays) in this phase. Here, the swept-up mass M_{sw} is greater than the ejecta mass M_{ej} , but radiative losses are not energetically important.

In this phase, $\sim 70\%$ of the initial energy gets converted into thermal energy of the swept-up matter [Chevalier \(1974\)](#). The Sedov-Taylor self-similar solution ([Sedov \(1959\)](#); [Taylor \(1950\)](#)) is a good approximation for the shock evolution in this phase.

Remnants in the adiabatic or Sedov expansion phase are typically bright at X-ray wavelengths due to the hot ISM gas shocked by the forward shock and the hot ejecta shocked by the reverse shock. Fig. 1.12 shows DEM L71, an SNR in the LMC believed to be few thousand years post-explosion [Rakowski et al. \(2003\)](#).

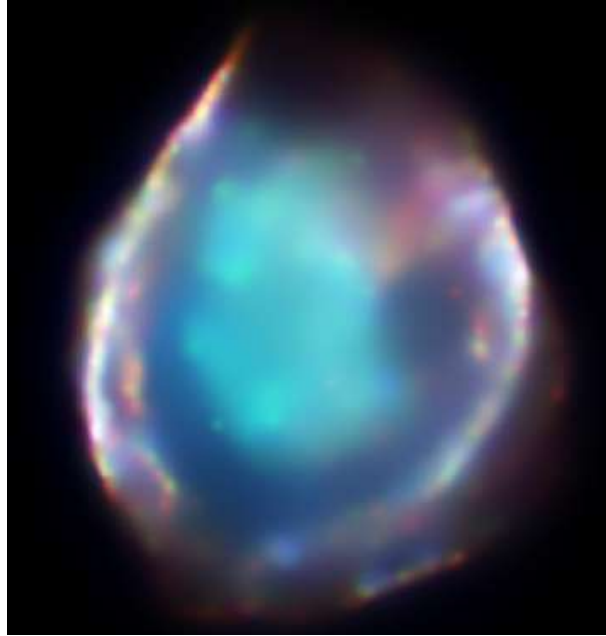


Figure 1.12: SNR DEM L71, located in the LMC. This X-ray image shows a clear separation between the forward shocked material in the ring on the outside and the reverse-shocked material in blue on the inside. Credit: NASA/CXC/Rutgers/J.Hughes et al.

The Pressure-Driven or “Snowplough” Phase

When the shock velocity decelerates to approximately 200 km s^{-1} , the temperature of the gas drops below $\sim 10^6 \text{ K}$, $M_{\text{sw}} \gg M_{\text{ej}}$, and radiative cooling becomes energetically significant. The cooling process accelerates as the gas cools, leading to an increased cooling rate. The shock front’s density peaks where the radiative cooling timescale is shortest, resulting in a cool, dense shell just behind the shock front. Meanwhile, the inner gas remains hot and under high pressure due to its lower density. In this phase, the remnant emits brightly in optical wavelengths from recombining atoms, such as

$H\alpha$, [S II], [N II], and [O III]. Cygnus Loop (fig. 1.13) is believed to be 10,000-20,000 years old, illustrating this phase. The cool shell continues to expand, gathering interstellar medium (ISM) gas in a manner similar to a snowplough, hence the term "snowplough phase." Initially, the gas expands adiabatically, with the pV term remaining constant, and the shock follows a time dependency of $R_s \propto t^{2/7}$. As the temperature drops further, pressure becomes negligible, and the cool shell expands with constant radial momentum, exhibiting a time dependency of approximately $R_s \propto t^{1/4}$.



Figure 1.13: A portion of the Cygnus Loop SNR, seen with HST. Narrowband images of [O III], [S II], and $H\alpha$, plus V and I continuum bands combine to make this image. Credit: NASA/ESA/Hubble Heritage Team.

The evolution of the shock radius is best described using momentum conservation. Models such as from [Toledo-Roy et al. \(2009\)](#) describe the evolution in this phase with simulations.

The Merging or Dissipation Phase

In the final stage of evolution, the SNR's expansion velocity becomes even smaller and the shock velocity becomes comparable to the sound velocity. The temperature of the shock also cools down close to that of the ISM as the SNR dissipates or blends into it. The radius of the SNR expands to the order of 10^2 pc. The SNR becomes practically invisible in all bands of the electromagnetic spectrum. This phase is typically reached in about 10^6 yrs. This number can vary depending on the density of the ISM and the energy released in the SN.

The evolution of the radius and velocity of shock wave has been summarised in fig. 1.14.

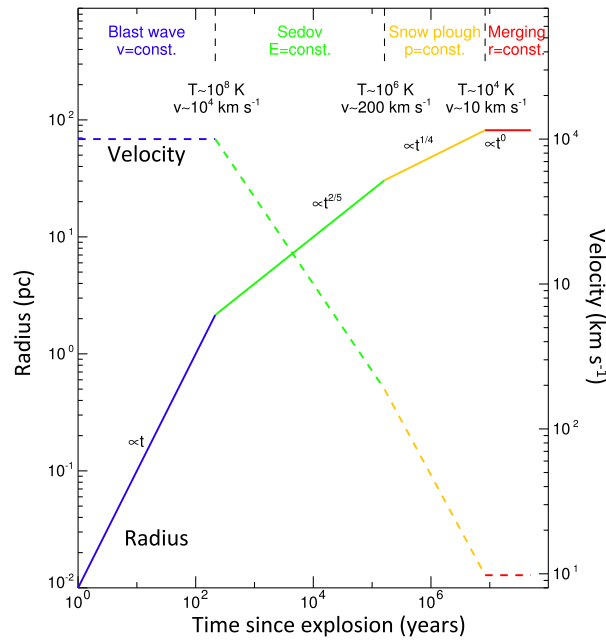


Figure 1.14: Phases of supernova remnants resulting from CCSN. The radius values were published by Padmanabhan (2001), and that of the velocity has been added by Wilms (2012).

1.5.2 Invisible/Undetected SNRs

The detailed study of supernova remnants (SNRs) has traditionally relied heavily on observations in the radio wavelength. However, radio telescope surveys for SNRs have faced three major selection biases: (1) they often miss SNRs with low surface brightness, particularly in regions with a high background, (2) they struggle to detect SNRs with small angular sizes, and (3) they lack uniform sky coverage. Recent advancements in radio astronomy, including telescopes with improved sensitivity, higher angular resolution, and more comprehensive sky coverage, alongside space-based X-ray observations, have significantly enhanced our ability to detect SNRs. These advancements have led to the discovery of previously undetected SNRs with lower surface brightness and smaller angular sizes, greatly increasing the catalogue of known SNRs.

According to the estimates of galactic supernova rates from Diehl et al. (2006) and Tammann et al. (1994), our galaxy has a core-collapse supernova rate of about 2 per century. At this rate, considering the typical supernova remnant dissipation times of 100 kyr, one can expect to find $\frac{2 \text{ SN}}{100 \text{ yr}} \times 100 \text{ kyr} = 2000 \text{ SNR}$ to be visible at any given point in time.

However the lower number of detections at about 300 compared to this expected 2000 highlights observational biases and calls for possible revisions to the theories. One of the reasons, for instance, can be that the observational time limit of 100 kyr could be overestimated and the dissipation times might be lower by up to an order of

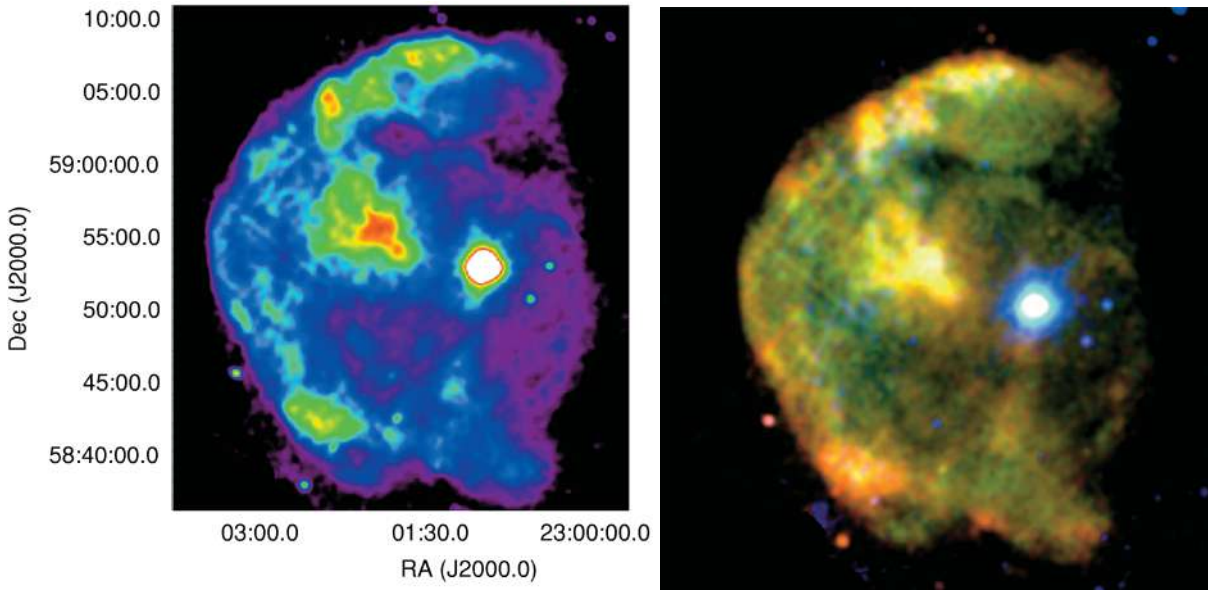


Figure 1.15: *Left:* False colour image using *XMM Netwon* data (X-ray 0.3 – 4.0 keV). The very bright point source is the pulsar 1E 2259+586. *Right:* Mosaic RGB image created from European Photon Imaging Camera (EPIC) data: $R = 0.3 - 0.9$ keV, $G = 0.9 - 1.5$ keV, $B = 1.5 - 4.0$ keV. Credit: [Sasaki et al. \(2004\)](#)

magnitude especially in higher-density regions of the ISM. Additionally, some SNRs might not be visible if they do not sweep up enough ISM or are less energetic than the canonical 10^{51} erg. These low-density regions are created by either a recent supernova in the vicinity where the associated SNR has already dissipated or by depletion of material through star formation. For instance, a faster dissipation of the SNR can take place when it interacts with a molecular cloud. This can be seen in SNR CTB 109. CTB 109 is located near a giant molecular cloud (GMC) complex. It has an unusual semicircular morphology in both X-ray and radio and an extended X-ray bright interior region known as the “Lobe”. This semicircular morphology is explained in [Sasaki et al. \(2004\)](#) as being caused due to its interaction with the GMC complex in the western part. In the eastern part, the shock wave is expanding into a lower density medium. This has been shown in fig. 1.15.

SN explosions of massive progenitors result in supernova explosions that occur in extended low-density wind bubbles. An SNR will be unobservable while expanding into such a bubble and then would fade rapidly after it collides with swept-up material at the bubble’s edge [Braun et al. \(1989\)](#). This possibility would explain the lack of SNR associations for several pulsars or soft gamma-ray repeaters (SGRs), and would also explain why many SGRs are associated with molecular clouds and massive star clusters [Gaensler \(2004\)](#).

For such cases, in which the SNR is invisible, this work aims to find indirect evi-

dence in the form of runaway star and neutron star pairs. These pairs originate from the disruption of a binary system when the more evolved component goes SN. As a result, a neutron star is produced and the other star leaves the system with its orbital velocity. This is the “Binary-Supernova Ejection Scenario (BSS)” (see section 1.7.2). This would contribute to a better estimate of the SN rate of our galaxy by partly overcoming the observational bias for the detection of associated SNRs.

1.6 Neutron Stars

Neutron stars (NS) are formed as compact object remnants of SN after the cores of massive stars collapse. These are highly compact objects with a mass of order that of the Sun but only about 10 km in radius. Neutron stars are supported against gravity by the pressure of the degenerate neutron gas. The idea that neutron stars could be the remnants of supernovae was first proposed by [Baade & Zwicky \(1934\)](#). Later, Robert Oppenheimer and George Volkoff developed theoretical models to describe the structure and properties of neutron stars, applying Einstein’s Theory of General Relativity to determine the maximum mass a neutron star could have. They derived equations describing the hydrostatic equilibrium of a static, spherically symmetric object supported by neutron gas pressure [Oppenheimer & Volkoff \(1939\)](#).

In the 1960s, further advancements were made in understanding neutron stars. Hoyle, Narlikar, and Wheeler suggested that neutron stars might rotate rapidly and possess strong magnetic fields [Hoyle et al. \(1964\)](#). In 1967, Jocelyn Bell Burnell and Antony Hewish discovered the first pulsar, PSR B1919+21, which emitted regular pulses of radio waves at 81 MHz [Hewish et al. \(1968\)](#). [Gold \(1968\)](#) later proposed that pulsars were rapidly rotating, strongly magnetised neutron stars, and predicted that their rotation periods would gradually increase due to energy loss through electromagnetic radiation. The discovery of the pulsar PSR B0531+21 in the Crab Nebula in 1968, which showed a measurable increase in its spin period, supported Gold’s theory and solidified the connection between neutron stars and core-collapse supernovae, as initially proposed by Baade and Zwicky.

Neutron stars, typically observed as pulsars, emit periodic radio or X-ray radiation that corresponds to their rotation period. Along with the rotation period, the magnetic field is a key characteristic used to classify neutron stars. Based on their periods, a $P - \dot{P}$ diagram is made in which these pulsars are found in certain preferred regions. The classification of pulsars is generally made based on this diagram as shown in fig. 1.16.

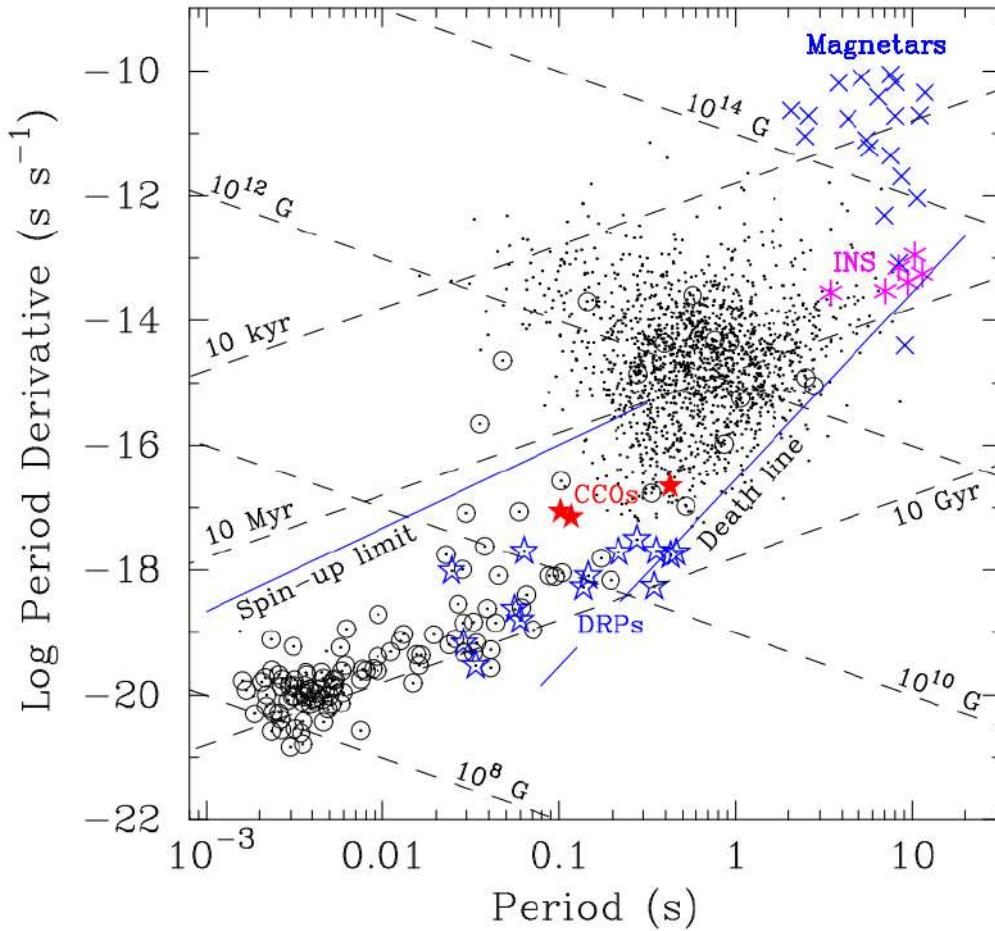


Figure 1.16: $P - \dot{P}$ diagram of the pulsar population with Magnetars (blue crosses), INSs (magenta asterisks), CCOs (red stars), DRPs (blue stars), recycled pulsars (black circles with dot) and not allocated NS (black dots). The maximum Spin-up limit and the death limit is highlighted with a blue line. (Taken from [Gottlieb et al., 2013](#)).

The high spins of neutron stars are easily explained by the conservation of angular momentum of the star during a collapse. Further, the high magnetic fields of neutron stars can also be simply explained by the conservation of magnetic flux. Precise observations of the spin periods have concluded that the spin periods of neutron stars increases gradually over time. The conversion of rotational power ($\mathcal{P}_R$) of the neutron star to the power radiated by electromagnetic radiation ($\mathcal{P}_{EM}$) causes the increase in the rotational periods of neutron stars. The power loss in the form of electromagnetic radiation can be described by the Larmor formula:

$$\mathcal{P}_{EM} = \frac{2m^2\omega^4 \sin^2 \alpha}{3c^3} \quad (1.1)$$

where $m = B_p R_{NS}^3$ is the magnetic moment (B_p is the polar magnetic field magnitude at the NS surface while R_{NS} is the radius of the neutron star), $\omega = 2\pi/P$ is the angular

velocity of the spin, and α is the angle between the magnetic axis and the rotational axis. On the other hand, the loss in the rotational energy is given by,

$$\mathcal{P}_R = I\omega\dot{\omega} \quad (1.2)$$

where I is the moment of inertia of the neutron star and $\omega, \dot{\omega}$ represent the angular velocity and its derivative. Since the energy lost in radiation is supplied by the slowing down of the spin,

$$\mathcal{P}_{\text{EM}} + \mathcal{P}_R = 0 \quad (1.3)$$

which gives:

$$\dot{\omega} = -\frac{2B_p^2 R_{\text{NS}}^6 \omega^3 \sin^2 \alpha}{3c^3 I}. \quad (1.4)$$

This relation then can be generalised as

$$\dot{\omega} = -A\omega^n \quad (1.5)$$

where A is a constant (if we assume that the magnetic field magnitude is time-independent and the α angle does not change significantly) and n is called *breaking index*. This can be found by taking the first derivative of eq. (1.5):

$$n = \frac{\ddot{\omega}\omega}{\dot{\omega}^2}. \quad (1.6)$$

However, the second derivative of the spin period is known for a small number of NSs, because of time noise and glitches [Jiang et al. \(2013\)](#). One common parameter adopted in the literature, and taken into consideration in this thesis, is the *characteristic age*. This can be found by expressing eq. (1.5) as a function of spin period and integrating it on both sides:

$$\int_{P_0}^P P^{n-1} dP = \int_0^t A(2\pi)^{n-1} dt, \quad (1.7)$$

ending up with

$$t = \frac{P}{(n-1)\dot{P}} \left[1 - \left(\frac{P_0}{P} \right)^{n-1} \right]. \quad (1.8)$$

With the assumption that the initial spin period of neutron stars was much smaller than the spin period today, $P_0 \ll P$, and the breaking index $n = 3$ (as typically observed), then the characteristic age is:

$$\tau = \frac{P}{2\dot{P}} \quad (1.9)$$

This is an upper limit of the *true age*, which is observationally difficult to constrain.

The period P and period derivative $\dot{P}$ further provide information about the surface magnetic field, which comes from rearranging eq. (1.4):

$$B_P = \sqrt{\frac{3c^3 I P \dot{P}}{8\pi^2 R^6}} \approx 3.2 \times 10^{19} (P \dot{P})^{1/2} \text{ G} \quad (1.10)$$

1.7 Runaway stars

Runaway stars are high-velocity stars that travel through interstellar space. They have been known for several decades, but it has always been a problem to explain their high velocities. The first runaway stars (RW) were found by Humason and Zwicky (1947) [Humason & Zwicky \(1947\)](#). It is found that 10% to 30% of the O stars and 5% to 10% of the B type stars are RW stars [Stone \(1991\)](#); [Gies \(1987\)](#). A velocity threshold for runaway stars of 25 km/s has been set by [Tetzlaff et al. \(2011\)](#); [Tetzlaff \(2013\)](#) corresponding to the intersection point of the distribution curves of the low-velocity and high-velocity stars (see fig. 1.17).

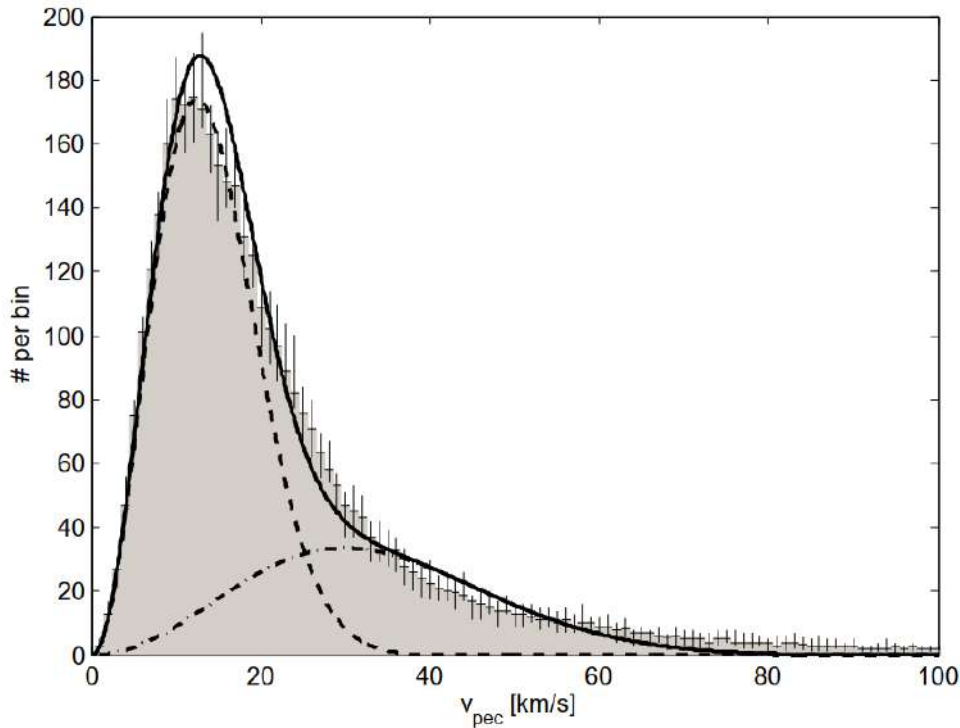


Figure 1.17: The distribution of the peculiar 3D space velocity of stars measured by the *Hipparcos* satellite. The distribution is a linear combination of two separated group distributions: the distribution of stars which are not ejected as RW or WW star (*left peak*) and the distribution of ejected RW stars (*right peak*). (Taken from [Tetzlaff et al., 2011](#); [Tetzlaff, 2013](#)).

Runaway stars are often found outside the Milky Way disc and show a complete

lack of multiplicity, which is otherwise very common in early type stars: more than 50% of OB type stars occur in multiple systems. These peculiarities about runaway stars led to two possible scenarios for explanation:

1. Dynamical Ejection Scenario (DES)
2. Binary-Supernova Ejection Scenario (BSS)

1.7.1 Dynamical Ejection Scenario (DES)

Poveda et al. (1967) proposed another mechanism for the production of runaway stars: the DES scenario. In this scenario, gravitational interactions between stars in clusters disrupt binary or multiple systems and produce runaway stars. Simulations have shown that these runaways can have velocities up to 200 km/s Leonard (1991). For the DES to be efficient the initial binary fraction in clusters needs to be large. DES runaways are formed most efficiently in a high-density environment, e.g., in young open clusters. The birth sites of massive stars are unbound stellar groups and expand in size, so DES runaways must have been ejected very soon after their formation. The kinematic age and the age of the parent association are thus nearly equal. Dynamical ejections occur in the early stages of cluster formation. Star clusters almost fully undergo dynamical ejections up to 3 Myr after formation Oh & Kroupa (2016). After 3 Myr, DES events are extremely rare and BSS is the primary source of producing runaway stars.

1.7.2 Binary-Supernova Ejection Scenario (BSS)

This scenario was first proposed by Blaauw (1961). BSS occurs in a binary star system in which the more massive, and hence faster evolving star ends its life in an SN event. This process often disrupts the binary which catapults the companion star out of the system with a velocity equal to the orbital velocity at the moment. The dynamics of the exact process depend on the binary system's parameters and the mass loss at explosion. If more than half of the mass of the binary system is lost in the SN, the binary is disrupted. The first confirmed evidence of a binary supernova ejection scenario was presented by Dinçel et al. (2015), where they discovered that the star HD 37424 was a pre-supernova binary companion to the progenitor of PSR J0538+2817 inside SNR 147 as shown in fig. 1.18.

This work deals with the runaway stars that are produced by BSS. The following section provides more insight into binary systems and their disruption to produce a BSS.

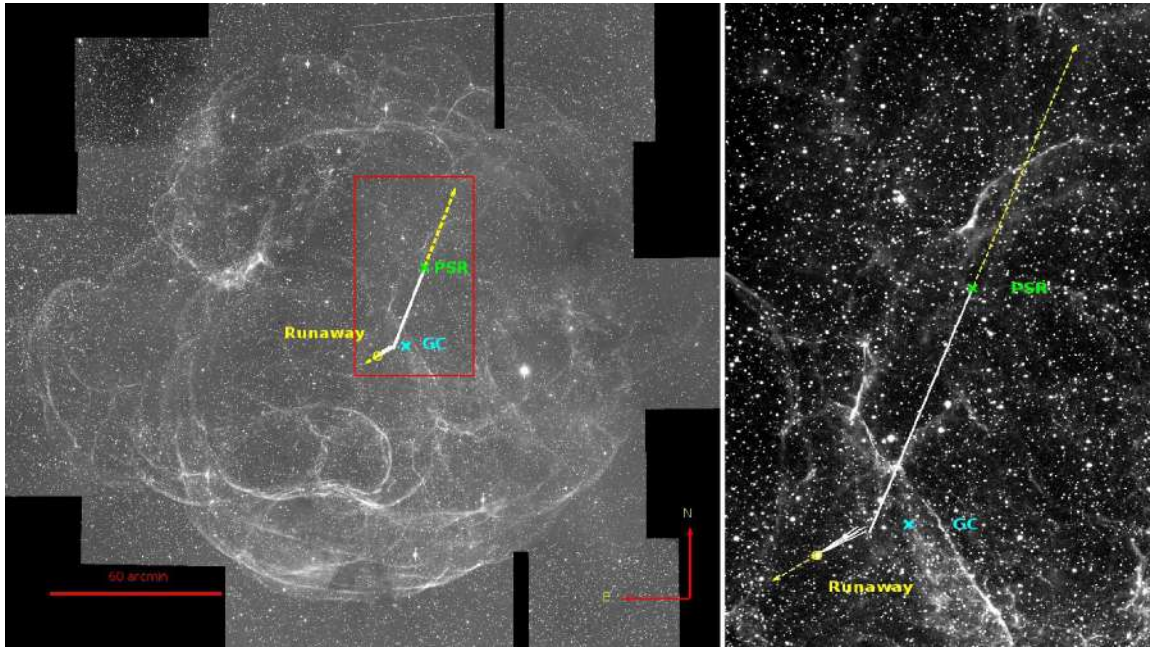


Figure 1.18: The $H\alpha$ image of SNR S147 taken at the University Observatory Jena. The green cross represents the pulsar, the cyan cross shows the geometric centre of the SNR and the yellow circle is HD 37424. The yellow vectors show the pm of the objects. The red box is the zoom-in area shown on the right. The white arrows are the tracing back cones of the pm for 29,300 yr.

1.8 Binary Disruption

In a system of massive binary stars, the SN explosion of one of the stars can lead to the binary being disrupted. The sudden mass loss of the system results in the neutron star receiving a kick velocity known as the ‘Blaauw kick’. Additionally, if the SN takes place asymmetrically, which is mostly the case, the neutron star receives an additional strong kick velocity [Blaauw \(1961\)](#) which makes it reach velocities up to 1000 km/s.

Assuming a circular orbit with a symmetric explosion, a binary system gets unbound if the mass lost during the SN is at least 50% of the total mass ($\Delta M \geq M_{total}$). If the kick w imparted on the neutron star is not large enough, the system can remain intact but change its orbital eccentricity and semi-major axis of the orbit. The general expression for the change in the semi-major axes for an asymmetric explosion is:

$$\frac{a_f}{a_i} = \frac{1 - \Delta M/M}{1 - 2\Delta M/M - \left(\frac{w}{v_{rel}}\right)^2 - 2\cos\theta\left(\frac{w}{v_{rel}}\right)}. \quad (1.11)$$

where a_f and a_i are the final and initial semi-major axes, ΔM is the mass lost in the SN and M is the mass of the system before the SN. $\vec{w}$ denotes the kick velocity and v_{rel} is the relative velocity of the two components of the binary. This is depicted in [fig. 1.19](#) for a symmetric explosion, $\vec{w} = 0$ and a simpler expression can be derived.

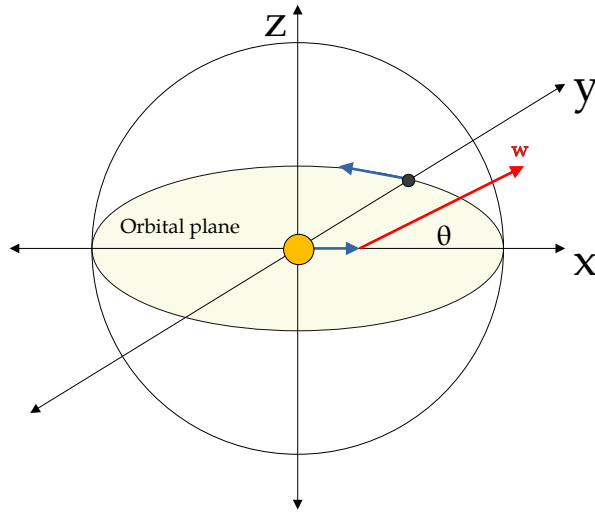


Figure 1.19: Orbital plane of the binary after the second SN explosion and the kick angles θ . The black dot represents the NS, and the blue one the exploding star. The kick vector $\vec{w}$ is shown in red. Adapted from [Tauris et al. \(2017\)](#).

$$\frac{a_f}{a_i} = \frac{1 - \Delta M/M}{1 - 2\Delta M/M} \quad (1.12)$$

In this case, the probability of disruption is then 1 if $\Delta M/M > 0.5$; otherwise, the system remains bound.

1.9 Stellar Spectroscopy

Stellar spectra refer to the patterns of absorption lines in the light emitted by stars, which are used to classify stars based on their spectral characteristics such as temperature and luminosity. Stars can be approximated as blackbodies due to the high temperatures and opacities of the outer layers. These layers contain mostly H and He along with the heavier elements that are produced in the stellar cores. The spectra of a star consists of a blackbody continuum with absorption and emission features depending on the elements present in its atmosphere. For instance, the spectrum of the Sun in the visible range is shown in Fig 1.20.

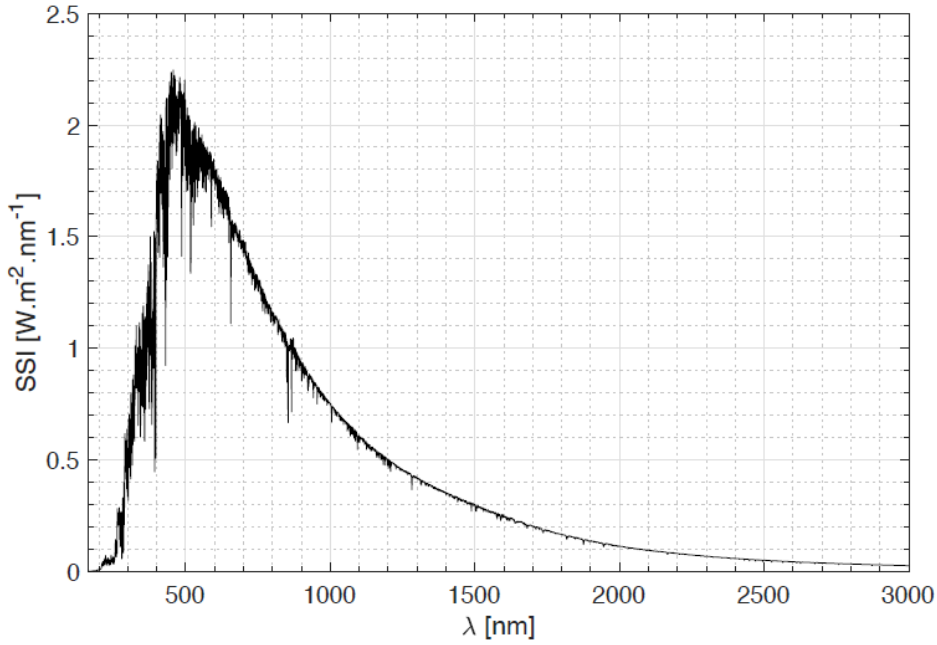


Figure 1.20: Spectrum of the Sun

A blackbody spectrum's peak lies at a wavelength that is given by Wein's law:

$$\lambda_{\max} = \frac{b}{T} \quad (1.13)$$

where $\lambda_{\max}$ is the peak wavelength, T is the temperature of the blackbody and b is a proportionality constant. The Sun's temperature can be estimated using 1.13. We see the peak at roughly 500 nm. This implies $T = \frac{0.0029 \text{ Km}}{500 \times 10^{-9} \text{ m}} = 5800 \text{ K}$ is the temperature of the Sun.

The dips in fig. 1.20 represent the absorption features. These are unique for different stars and are often called the “fingerprints” of stars. These are caused by the presence of different elements in the atmosphere of a star undergoing certain transitions. The temperature of the star also determines these absorption features. Depending upon the temperatures, certain transitions of the elements become more or less likely. These features can be used to distinguish and precisely determine stellar temperatures. Fig. 1.21 shows the spectrum of stars with different temperatures and spectral types.

1.9.1 Spectral features of OB stars

Since this work focuses on massive OB-type stars, this subsection focuses on some particular features of these stars which determine the spectral type, temperature and

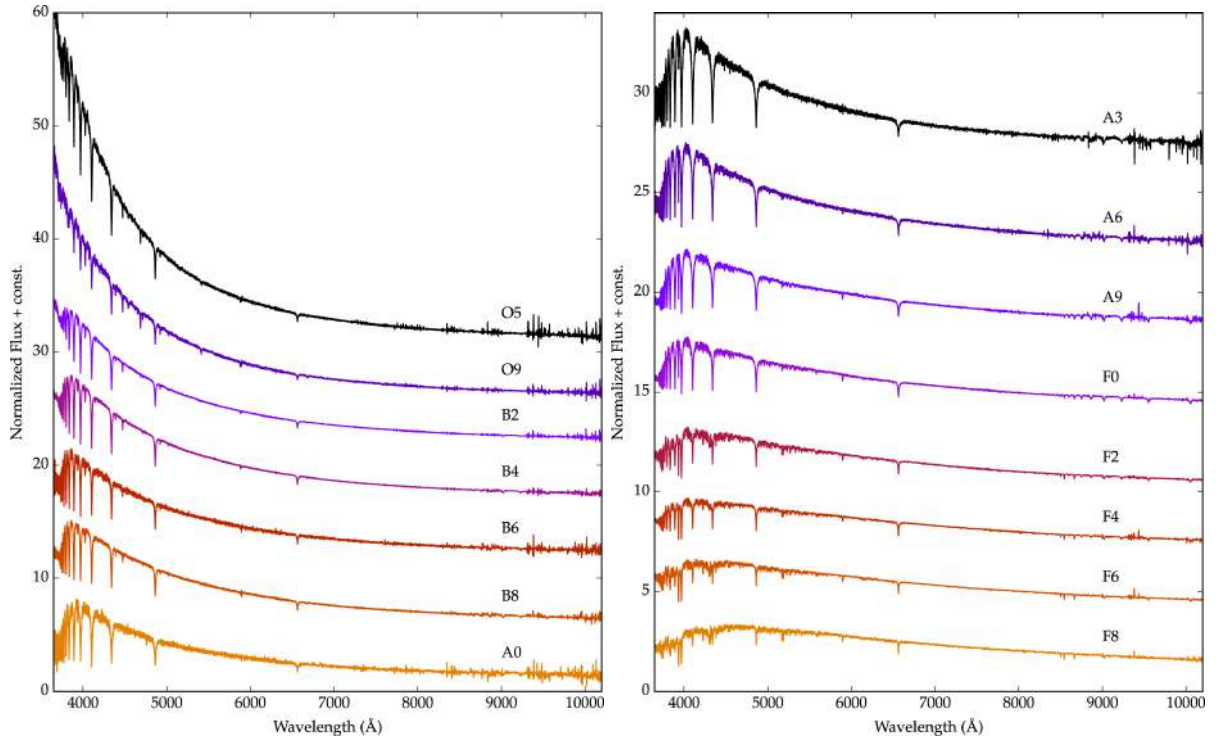


Figure 1.21: Sample of the different spectral types. The spectra are all normalized at 8000 Å, and a constant is added to each template to improve readability. Figures taken from Kesseli et al. (2017).

other properties of the stars.

O type stars

O-type stars have temperatures greater than 25,000 K. Their spectra display a strong ultraviolet emittance, with absent to weak He I and He II (neutral and singly ionised helium) and strong Balmer (ionised hydrogen) lines, which become weaker in earlier types at higher temperatures. Lines of N IV and N III (triply or doubly ionised nitrogen) and Si III (doubly ionised silicon) are apparent but weak. Their spectral classification is based on the line strength of N IV (Å4058) (O2 to O3.5) and the line strength ratios of He II (4200 Å)/He I (Å4026) (1.0 = O6), He II (Å4541)/He I (Å4471) (1.0 = O7), and He II (Å4541) = Si III (Å4552) (1.0 = O9.7). The temperature classification is based on helium ionisation. Stellar winds become increasingly significant in the earlier subtypes.

Examples: zeta Orionis (O9Ib), zeta Puppis (O5), iota Orionis (O9III)

B type stars

B-type stars have temperatures greater than 10,000 K. Their spectra show strong He I (neutral helium) and Balmer hydrogen lines, Si I and Si II, O I (neutral oxygen) and Mg

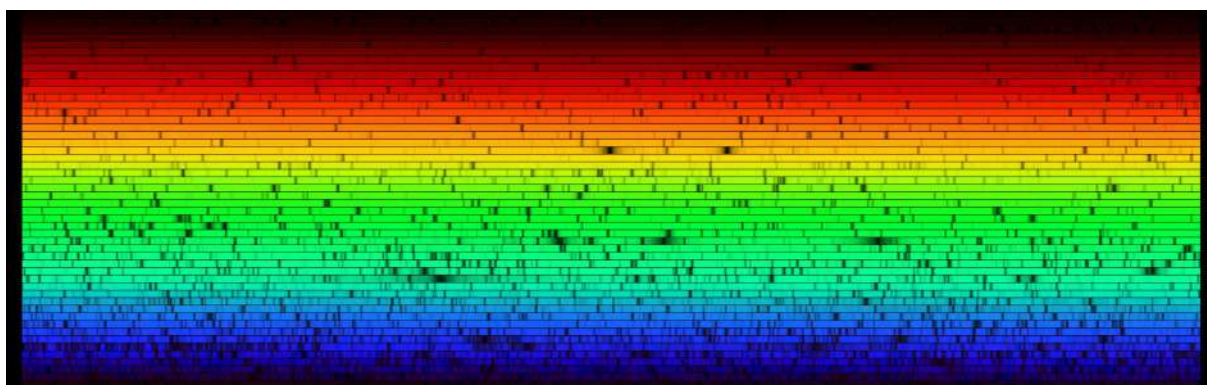


Figure 1.22: Cross-dispersed Solar spectrum covering almost the entire optical range, obtained using the HIRES echelle spectrograph on the 10 m Keck telescope, Hawaii. Credit: [Vik Dhillon](#)

I (neutral magnesium). Their spectral classification is based on absence of He II in blue violet, weakening of He I and strengthening of Balmer hydrogen lines, the ratio S IV/S III ($1.0 = B0.7$) and strengthening of the Mg II ($\text{\AA}4481$) line. The high-excitation line Mg II $\lambda 4481$ gradually strengthens through the B-type stars, making the He I $\lambda 4471$ /Mg II $\lambda 4481$ ratio a useful indicator for assessing temperature in helium-normal stars of type B3 and later. The Type B stars present many spectral and luminosity classification anomalies affected by the proportion of helium in the photosphere.

Examples: beta Orionis (B8Ia), tau Scorpii (B0V), alpha Virginis (B1III), alpha Leonis (B7V)

1.9.2 Échelle Spectra

The French word “Échelle” means stairs or ladder. The Échelle is a reflection grating, in which light is reflected at each of its grooves. As the width of each groove is comparable to the wavelength of light, optical diffraction occurs. Each groove is like a slit, which directs the diffracted light into the way of the geometric reflection. An échelle spectrograph typically consists of the following components: a collimator to make the incoming light parallel, an échelle grating as the primary dispersive element, a cross-disperser to separate overlapping orders, a camera to focus the light onto a detector, and a detector (usually a CCD) to record the spectrum. Thus, an Échelle-Spectrograph provides high-resolution spectra of several orders which overlap each other. With an additional dispersive element – like a prism or a grating – the spectra is stretched at a right angle to the Échelle grating, separating its different orders of diffraction.

These orders are sorted in two dimensions as they cover the complete spectrum. A complete spectrum between ca. 392nm and 800nm without gaps – depending on the size of the camera sensor can be seen in fig. 1.22.

Chapter 2

Data Sample

This chapter describes the various data sources that have been used in this work. This includes a sample of young open star clusters in the galaxy from [Dias et al. \(2021\)](#) and the stars in the region around them from *Gaia* ([Gaia Collaboration et al. \(2023\)](#)). The data about the stars includes their astrometric and photometric information. For the analysis and possible associations, neutron stars from the ATNF catalogue [Manchester et al. \(2005\)](#) are found in the neighbouring regions. Colour-magnitude diagrams are made using synthetic stellar populations from [Oh & Kroupa \(2016\)](#) IMF functions, evolved on PARSEC evolutionary tracks [Nguyen et al. \(2022\)](#). Promising candidates are observed from the university observatory at Großschwabhausen. The spectroscopic data thus obtained is compared to model stellar atmospheres from TLUSTY [Hubeny & Lanz \(2011\)](#) and Kurucz models [Kurucz \(1993\)](#).

2.1 Clusters

A sample of 1743 open clusters listed in [Dias et al. \(2021\)](#) were used in this work. These clusters and their members were accessed using the VizeiR service from the table [J/MNRAS/504/356](#). Among the complete sample, the clusters of interest here are the young ones with an age of < 50 Myr (corresponding to a $\log \text{Age} < 7.7$) because they are likely to still contain OB type stars which can be companions of supernova progenitors. Further, clusters which have a parallax $\pi > 0.3$ mas (or $d \leq 3333$ pc) have been chosen for reasonable astrometric accuracy with the possibility of follow-up observations. This leaves behind 399 out of the 1743 clusters which were analysed and are tabulated [online](#)¹ and in the appendix table [A.4](#).

A view of the selection sample in the Milky Way has been presented in [fig 2.1](#).

¹<https://github.com/surodeep26/Runaway-Stars>

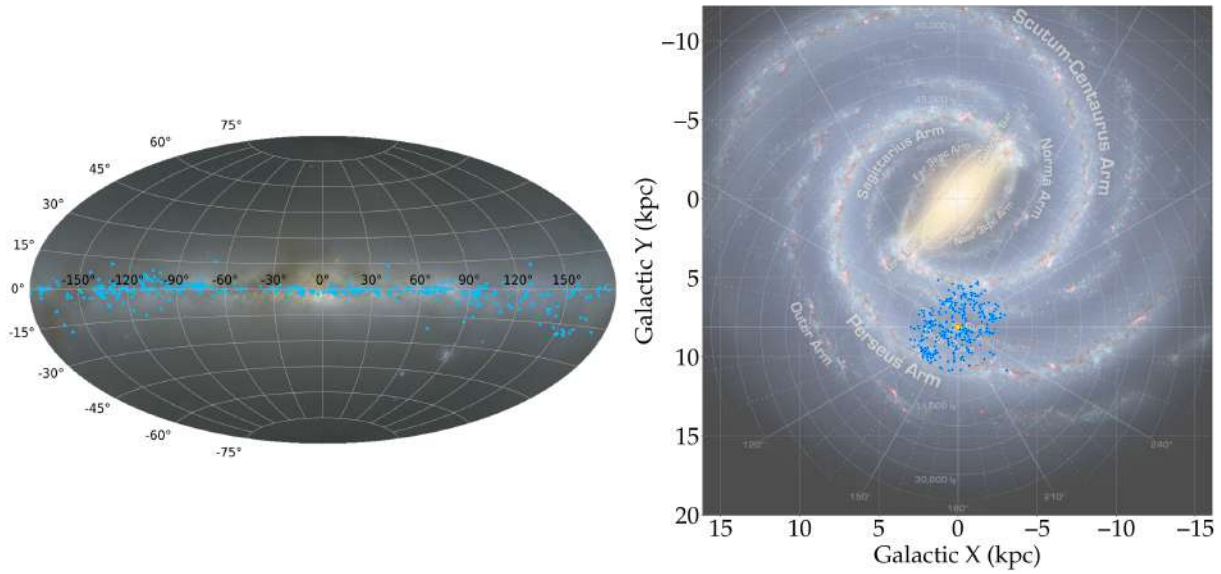


Figure 2.1: Sample of clusters

2.2 Stars from *Gaia* DR3

Gaia is a European space mission that collects detailed data on over a billion stars in the Milky Way, including their positions, motions, and brightness. It also gathers information on a significant number of extragalactic objects and Solar System bodies. *Gaia* data includes precise measurements of positions, parallaxes, proper motions, and radial velocities. Additionally, it provides low-resolution spectra of the sources. These cannot, however, be used for very accurate analysis and hence further observations are required. These spectra can give rudimentary estimates of the temperature, surface gravity, and radial velocity values.

The astrometric information about the stars in and around the clusters has been retrieved from *Gaia* Data Release 3 released on 13 June 2022 [Gaia Collaboration et al. \(2023\)](#) using the `astroquery` module to access the corresponding [VizieR table](#) (henceforth table 1).

The distances are obtained similarly from the [VizieR table](#) (henceforth table 2) for the catalogue [Bailer-Jones et al. \(2021\)](#). This catalogue provides refined distance estimates by employing a probabilistic approach that mitigates the limitations of direct parallax inversion, particularly for stars that are distant or faint. Two types of distances are provided: geometric, which uses parallax data with a direction-dependent prior, and photogeometric, which additionally incorporates stellar colour and apparent magnitude to enhance accuracy, especially for stars with poor parallax measurements. This comprehensive dataset includes over 1.47 billion geometric and 1.35 billion pho-

togeometric distances, offering robust posterior probability quantiles that are directly applicable in distance modulus calculations.

2.3 ATNF Pulsars

A table of currently known pulsars in the literature is maintained by the Australian National Telescope Facility [Manchester et al. \(2005\)](#) which was used in this work to find possible pulsar associations for the runaway star candidates. This table is updated with new findings on a regular basis. The latest version as of the writing of this thesis was v2.3.0 containing 3649 pulsars, out of which 3575 had their distances known and were used for this work. The catalogue can be accessed using the web interface at [ATNF](#) or using the python package [psrqpy](#).

The $P - \dot{P}$ diagram shown in figure fig. 2.2 shows these in the plot. For the purposes of this work, the younger pulsars (~ 500 kyr) are relevant. This age is normally derived from the $P - \dot{P}$ relation which assumes that the initial spin period of the pulsars was much smaller than the current period, $P_0 \ll P$ and that the braking index n is 3.

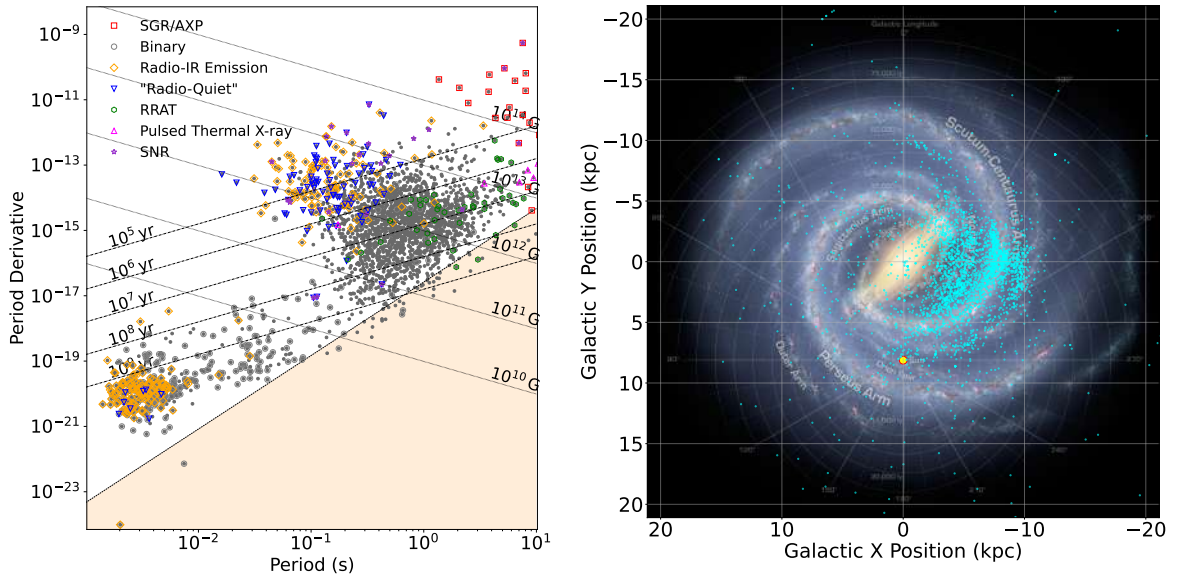


Figure 2.2: $P - \dot{P}$ diagram showing the sample of pulsars used. The right plot shows their positions on the galaxy.

The distance estimates for the pulsars in the ATNF catalogue have default values from the dispersion measure (DM) using the YMW16 model [Yao et al. \(2017a\)](#) for the Galactic distribution of free electrons. First preference for the distance values is given to distances estimated using associations with other objects like a globular cluster or a

supernova remnant. If the parallax is known, then the distances are estimated using the distance parallax relation. In this work, the pulsars with unknown association are used and therefore most of the distance measurements are from the dispersion measure.

2.4 Colour-Magnitude Diagrams

To fit isochrones for each of the clusters involved in this work, synthetic stellar populations have been used as templates. These are then used to vary the estimates of the parameters like extinction (A_V), metallicity ($[\text{Fe}/\text{H}]$) and age ($\log \tau$) of the clusters and select the members in them. These theoretical isochrones or CMDs have been obtained using the [CMD3.7 interface](#) (Girardi (2024)).

PARSEC version 2.0 evolutionary tracks [Nguyen et al. \(2022\)](#) have been used to compute isochrones using *Gaia* passbands [Riello et al. \(2021\)](#) with the data for their zero points from the [Gaia website](#) for the photometric system.

Star-by-star extinction values are used with the extinction curve from [Cardelli et al. \(1989\)](#) and [O'Donnell \(1994\)](#) having $R_V = 3.1$.

IMF functions by [Kroupa \(2001, 2002\)](#), corrected for unresolved binaries have been used to synthesise the cluster populations. The input for the values of ages, metallicities and extinctions vary for each cluster. The configuration used has been tabulated in table 2.1.

Field	Selected setting
Evolutionary tracks	PARSEC version 2.0 Bressan et al. (2012b) COLIBRI S.37 Pastorelli et al. (2020)
Photometric system	<i>Gaia</i> DR2 + Tycho2 + 2MASS Gaia website
Bolometric corrections	YBC + new Vega Chen et al. (2019) ; Bohlin et al. (2020)
Circumstellar dust	None
Interstellar extinction	Adopted extinction curve with Cardelli et al. (1989) ; O'Donnell (1994)
Long Period Variability	Periods from (Trabucchi et al. (2021))
Initial mass function	Canonical two-part-power IMF Kroupa (2001, 2002)

Table 2.1: Selected settings for the fields in the [CMD3.7 interface](#)

The process of obtaining this data would involve accessing the website for each desired isochrone and filling out the form. To automate this process, the python [selenium package](#) for web automation is used.

2.5 Spectroscopic Data from the FLECHAS instrument in GSH

After selecting suitable candidate runaway stars and their potential birth clusters, spectroscopic observations were carried out for the runaway stars and/or the brightest cluster members if they were visible from [Großschwabhausen \(GSH\)](#) and were within the observation limit of about 10 mag for reasonable quality of data.

The university observatory at GSH equipped with a 90 cm telescope was used to obtain high-resolution spectra for the stars. GSH is located at longitude 348.51584° , latitude 50.92888° , and altitude 370 m. Spectra for the stars was obtained with the Fibre Linked Échelle Astronomical Spectrograph (FLECHAS) [Mugrauer et al. \(2014\)](#) instrument. FLECHAS has a resolving power of $R \sim 9200$ and a wavelength range of $3900 - 8100 \text{ \AA}$. Spectra with 2×2 pixel binning were taken for the stars, which reduces the resolving power to $R \sim 7000$. For optimal calibration, three arc (Th-Ar lamp) and flatfield-images (Tungsten lamp), each with a detector integration time (DIT) of 1.5s, were always taken immediately before the science exposures. A set of three dark frames is taken for each DIT used for the arc-, flatfield-, and science exposures and median-combined them for efficient cosmic ray removal. Each spectrum and dark were divided into three frames to remove the cosmic rays.

A 3D model of the FLECHAS [Mugrauer et al. \(2014\)](#) instrument has been presented in figure 2.3 along with a flat field image produced by it.

A total of 52 stars were observed during this work. These are tabulated along with their parameters derived from spectroscopy in the appendix A.1. The runaway star candidates and cluster members lying in the southern night sky could not be observed from GSH.

Using spectroscopic data for stars, it is possible to ascertain several parameters like the temperature, $\log g$, radial velocity and composition of the star. Furthermore, once these parameters are known for members of a cluster, it is possible to fix the parameters for the isochrone of the cluster, enabling for better estimates for the temperatures of the stars which cannot be directly observed.

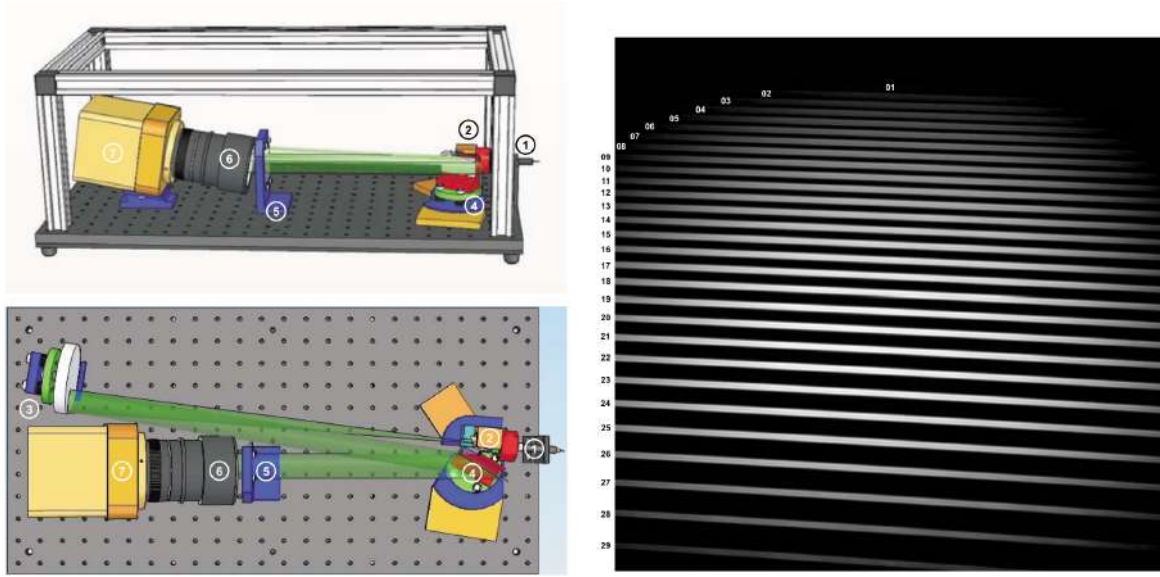


Figure 2.3:

Left: 3D model of FLECHAS with all its internal optical components. All individual optical components of the spectrograph are labelled with numbers: (1) fibre entrance, (2) image slicer, (3) collimator, (4) reflective echelle grating, (5) cross-disperser, (6) objective, (7) CCD-detector.

Right: Flatfield spectrum of FLECHAS taken with a Tungsten lamp. In total 29 echelle orders are imaged on the CCD-detector of the spectrograph. The individual orders are indicated with their order numbers, increasing with wavelength

2.6 Model Stellar Spectra

To determine the spectral types and radial velocities of the stars, theoretical models of stellar atmospheres have been used. These are sourced from two popular stellar atmosphere models: TLUSTY [Hubeny & Lanz \(2011\)](#) and Kurucz [Kurucz \(1993\)](#). A brief description for these models follows:

TLUSTY models

TLUSTY is a package written in FORTRAN77 for modelling stellar atmospheres and accretion disks and a wide range of spectroscopic diagnostics. These are based on a numerical method for computing non-Local Thermodynamic Equilibrium (non-LTE) model stellar atmospheres which is described in detail in [Hubeny & Lanz \(1995\)](#). The models used here are with a micro-turbulence velocity of 2 km/s. Some example models of stellar spectra have been shown in [fig. 2.4](#).

The grid used here consists of model atmospheres with temperatures ranging from 15,000 K to 30,000 K in steps of 1000 K and $\log g$ values ranging from 2.75 to 4.75 in steps of 0.25. The wavelength range from about 3500Å to 7000Å was used for correlation with the observed spectra.

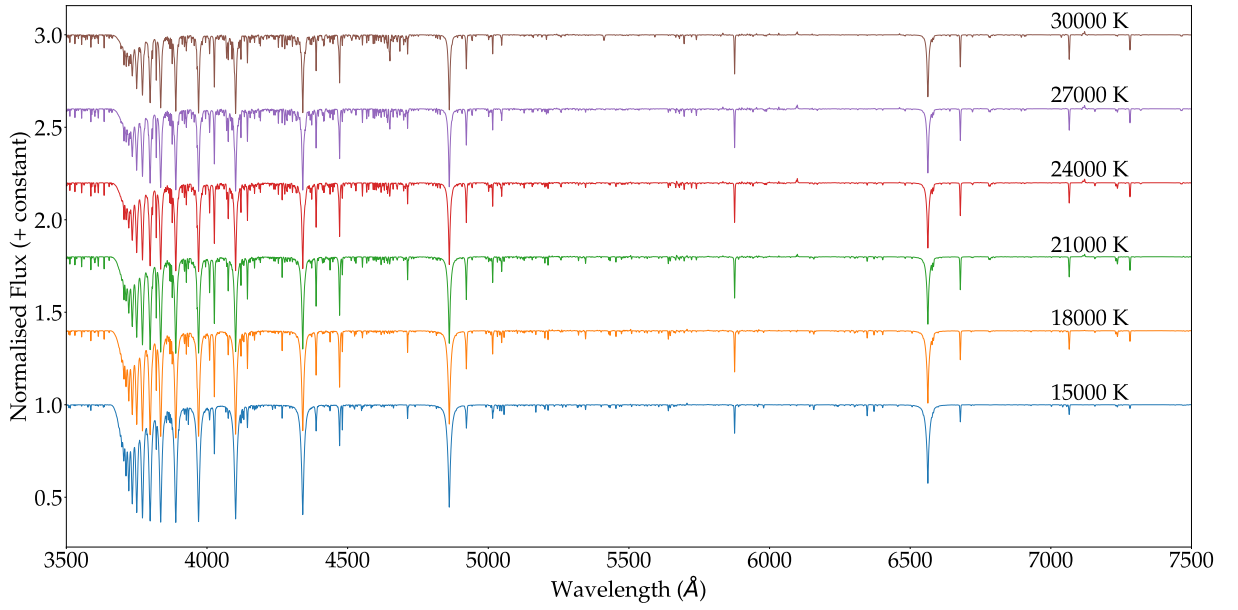


Figure 2.4: TLUSTY stellar atmosphere synthetic spectra. Each spectra has been normalised and has a $\log g$ value of 4.00 and temperatures increasing in steps of 3000 K plotted with a constant added to their fluxes.

Kurucz (1993) models

The Kurucz 1993 ATLAS contains about 7600 stellar atmosphere models covering a wide range of metallicities, effective temperatures and gravities with microturbulent velocity of 2 km/s. These are LTE models based on model stellar atmospheres described in Kurucz (1979). The models were computed by use of a statistical distribution-function representation of the opacity of almost a million atomic lines. Some of the model templates are shown in fig. 2.5.

For the purposes of this work, models ranging from 8000 K to 15000 K have been used in varying stepsizes of 250K to 1000 K. $\log g$ values range from 2.0 to 4.5 in 0.5 steps. The wavelength range from about 4000Å to 5000Å was used for correlation with the observed spectra. This wavelength range contains the most important spectral lines required.

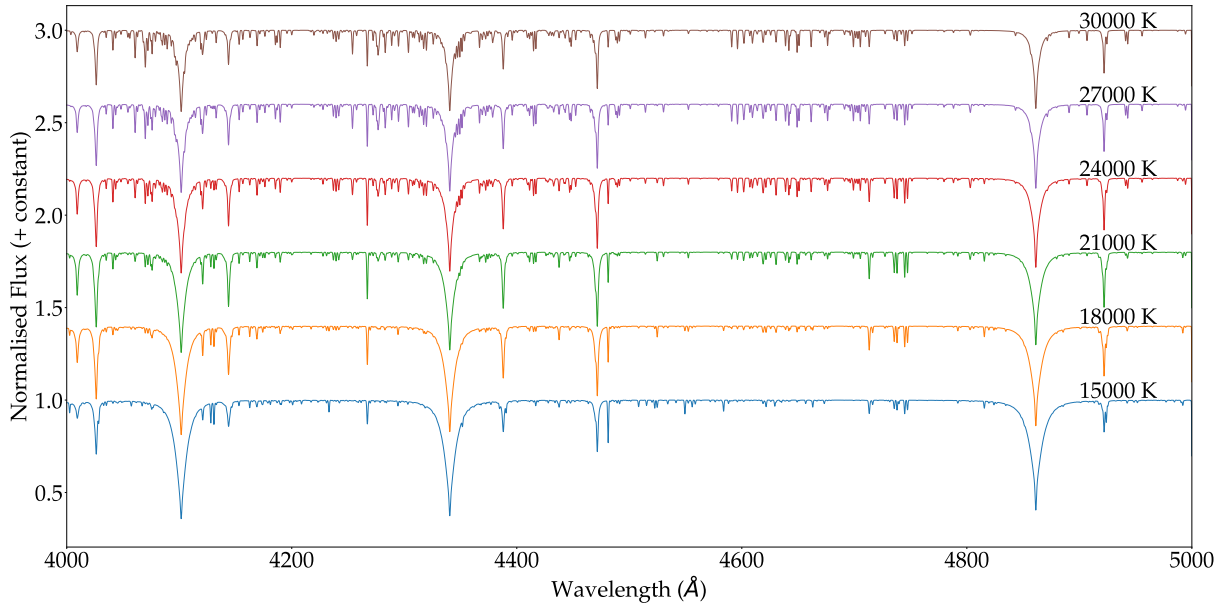


Figure 2.5: Kurucz stellar atmosphere synthetic spectra. Each spectra has been normalised and has a $\log g$ value of 4.00 and temperatures increasing in steps of 3000 K plotted with a constant added to their fluxes.

2.7 SkyView Images

NASA’s SkyView facility <http://skyview.gsfc.nasa.gov> located at NASA Goddard Space Flight Centre is used for generating images of any part of the sky at wavelengths in all regimes from Radio to Gamma-Ray. The python module `astroquery.skyview` has been used to query and obtain images of the regions around the clusters.

For the plots shown in the results chapter, optical Digitized Sky Survey (DSS) images are used and overlaid with the coordinate grids and scatter plots of the stars and their trajectory traces. DSS is a digitized version of several photographic astronomical surveys of the night sky, produced by the Space Telescope Science Institute between 1983 and 2006.

Chapter 3

Methods

With the sample of data as described in chapter 2, a common analysis has been carried out for each of the clusters. This chapter presents an overview of the general procedure that has been applied to the data. This has been carried out using a Python code, making use of several astronomy packages. A GitHub repository with the documentation has been made available at <https://github.com/surodeep26/Runaway-Stars>. Another repository dealing with the spectroscopic part of the data has been made <https://github.com/surodeep26/SpecFit>.

3.1 Querying *Gaia* data

Search region

For each of the 399 selected clusters, a conical search around the cluster centre. Considering a 2D transverse velocity of 100 km/s for a period of 100 kyr, a star can travel up to a distance of:

$$\begin{aligned} d &= v \cdot t \\ &= 100 \text{ km/s} \cdot 100 \text{ kyr} \\ &\approx 10.227 \text{ pc} \end{aligned}$$

Considering this limit, a 10 pc physical distance from the cluster edge was searched around each cluster. To convert from the angular radius r_θ of the cluster to the required search region radius s_θ , a simple formula can be used that converts the angular radius r_θ of the cluster to a physical radius r_{phy} , adds 10 pc to it, and then converts it back to

an angular search s_θ radius. This can be easily achieved using a Python code making use of the `units` class of `astropy`. Note that the radius in this discussion is the `r50` radius of the cluster: the radius within which more than 50% of the members of the cluster lie.

For example, considering the cluster Bochum 13 with a radius of $r_\theta = 6$ arcmin, at a distance of $d = 1611$ pc (ignoring the errors),

$$\begin{aligned}
 r_{\text{phy}} &= r_\theta \cdot d \\
 &= 6 \text{ arcmin} \cdot 1611 \text{ pc} \\
 &= 2.81 \text{ pc} \\
 \text{And, } s_\theta &= \frac{r_{\text{phy}} + 10 \text{ pc}}{1611 \text{ pc}} \text{ rad} \\
 &= 27.639 \text{ arcmin}
 \end{aligned}$$

A search query is made around the cluster centre with 27.639 arcmin as the search radius in the *Gaia* sources table 1 (see section 2.2) with a filter to only keep the sources which have a $G_{\text{mag}} < 17$ and $\pi > 0.3$. Table 2 is queried within the same radius, but the filters of G_{mag} and $\pi > 0.3$ are not available in this table.

Stars in the Cluster Neighbourhood

The query in table 1 returns an `astropy` table containing, in this case, 4649 sources. Table 2 returns 76,962 sources, a higher number since there are no filters applied: foreground and background sources are included. Each of these sources had a *Gaia* source ID, which is used to merge the two tables using the `join` function of `astropy` table. This leaves a total of 1,835 common sources, which are the stars in the region. This selection region is naturally in the shape of a conical frustum.

3.2 Cluster Members and Parameters

For each of the clusters, a list of the cluster members along with their membership probability P_{memb} has been provided. Members with $P_{\text{memb}} > 0.8$ and a parallax quality $(\Delta\pi/\pi) > 10$ are selected. These are cross-referenced with the stars in the search region using their *Gaia* source IDs in *Gaia* DR3 to find the member stars. For Bochum 13, there are 84 members stars in the search region out of the 113 from the Dias catalogue. With the parallax quality and membership probability filters, 29 stars are chosen

as the cluster members for further analysis.

The kinematic parameters of these stars are then averaged over to update the cluster's kinematic parameters of distance (d), proper-motion in right ascension (μ_α^*), proper-motion in declination (μ_δ), and radial velocity v_r . This naturally comes with the advantage of getting the latest and more accurate data about the various parameters of the cluster members and hence the clusters themselves since [Dias et al. \(2021\)](#) uses [Gaia Collaboration et al. \(2018\)](#) to identify open clusters and derive different parameters such as age, metallicity and extinction along with the cluster's kinematics using selected cluster members.

Sometimes, this procedure leaves behind very few or even no stars, in which case any further analysis would have low confidence levels. For such cases, a function `find_cluster` has been developed which works by sigma clipping the kinematic parameters of the stars in the region to find a set of stars which are then called kinematic members. These usually contain all the stars that [Dias et al. \(2021\)](#) mentions as members and some more.

3.3 Runaway Candidate Selection

The relative proper motions of all the stars in the search region with respect to the cluster average is calculated. With this, the relative transverse velocity of the stars has been calculated. For the star to be a runaway, it must have a peculiar velocity greater than the threshold of 25 km/s. The threshold value is the intersection point of the distribution curves of the low-velocity and high-velocity stars ([Tetzlaff et al. \(2011\)](#), [Tetzlaff \(2013\)](#)). If the radial velocity is unavailable, then the 2D counterpart of the 3D velocity of 25 km/s, which is simply $\sqrt{\frac{1}{2}} \cdot 25 \text{ km/s} \approx 17.6 \text{ km/s}$ is taken as the lower limit for selection.

Tracing back

To calculate the trajectories of the stars, their (3+2)-dimensional kinematic parameters, namely, right ascension (α), declination (δ), distance d , along with proper motion in right ascension (μ_α^*), proper motion in declination (μ_δ) were considered. With these 5 parameters, a `SkyCoord` class object was created for each of the stars. Using the `apply_space_motion` method of the `SkyCoord` class, their positions were evolved for a time of -100 kyr. At each step, the angular separation of the star and the cluster centre is stored in a list. If a separation smaller than the radius (r_{50}) of the cluster is found in

the time frame, it is considered as a candidate runaway star since it could be traced to be within the cluster in the past.

The temperatures of these potential runaways are now estimated using an isochrone that fits the cluster members.

If a runaway stars candidate is found, the nearby pulsars are also searched and their possible associations is considered using typical neutron star velocities unless their proper motions are known.

3.4 Isochrones

With the cluster parameters mentioned in [Dias et al. \(2021\)](#), theoretical isochrones are obtained from [Girardi \(2024\)](#) with the configuration described in section 2.4. The members of the cluster are plotted on top of the isochrones. The rest of the stars in the region have also been plotted to see if there are any other populations of stars in the same region. The other populations have different ages and their isochrones can be distinguished in the plot.

Oftentimes, the velocity of the “runaway” found in the process matches with the general direction of motion of the stars in the broader region, in which case it is likely to belong to the other population and not the cluster in question.

The isochrones are used to obtain an estimate of the parameters of the members of the cluster along with the runaways if any. The theoretical isochrones contain information about its constituent stars’ mass M , initial mass at the time of formation M_0 , luminosity L , mass loss rate $\dot{M}$, temperature T_{eff} , surface gravity $\log g$ and photometric measurements of different wavelength bands. These can be used to compare with the spectroscopically derived parameters of the stars to check the quality of the isochrone fit.

Since stars OB-type stars are of interest in this work, stars with temperature $T_{\text{eff}} > 10,000$ K need to be chosen. Their temperatures are estimated by choosing their closest neighbour star from the best fitting isochrone. The parameters of this theoretical star is then assigned to the star whose parameters are unknown. The temperature estimates obtained in this manner are generally in agreement with other sources such as the [Skiff \(2014\)](#) catalogue. To confirm these estimates, spectroscopic observations are taken using the FLECHAS instrument [Mugrauer et al. \(2014\)](#) at GSH.

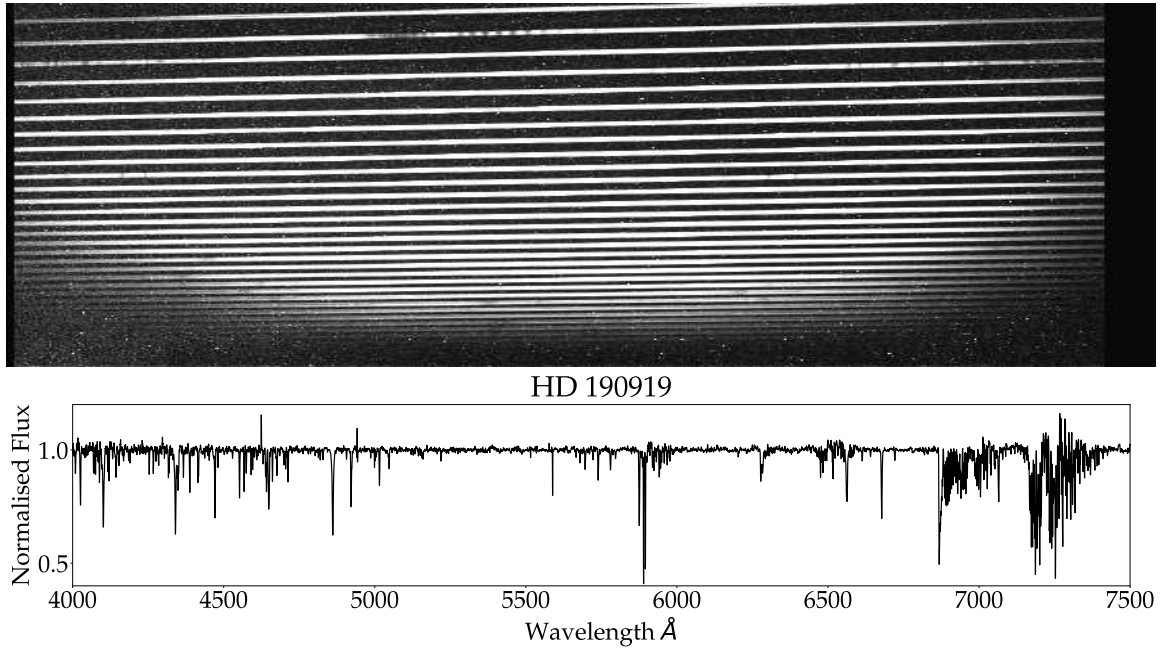


Figure 3.1: *Top:* Raw data from FLECHAS showing the échelle spectrum of a star with the 29 apertures. *Bottom:* Reduced spectrum of the star after the data reduction process described in section A.2.

3.5 Spectroscopic data analysis

Data obtained from FLECHAS was reduced in IRAF (Tody (1986)). The process of obtaining the spectrum of stars involves first removing artefacts from observations like cosmic rays and from CCD readout and biases, calibrating the wavelengths, and subtracting the continuum. For this purpose, 3 different types of images are obtained for each spectrum: Dark, Flat and Science. The science image contains the main data from the star. Additionally, Arc images are obtained that are used to calibrate the spectral lines of the Science images. Darks and Flats are used to gather information about unwanted noise so that they can be removed. For this purpose, a Thorium-Argon lamp is used. Three images of each of Darks, Flats, Sci and Arc are obtained and combined into one master file for each using the `imcombine` command of IRAF.

The goal therefore is to go from raw data as in the upper part of fig 3.1 to the reduced spectrum shown in the lower part. The process of obtaining the spectrum from the raw data has been described in the section A.2.

Comparing with template spectra

The spectra of the stars obtained from FLECHAS were compared to the template spectra from TLUSTY and Kurucz stellar atmosphere models. For this purpose, the `fxcor` command in IRAF was used. These models are described in section 2.6. The models are

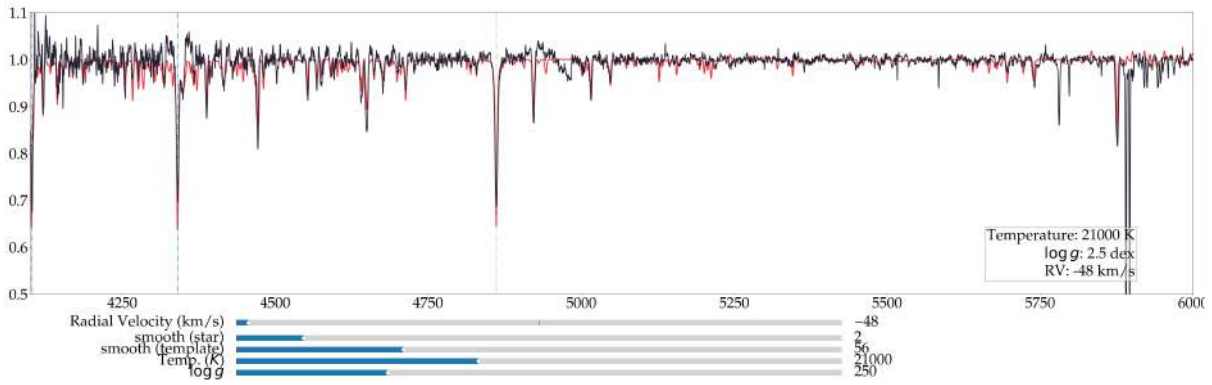


Figure 3.2: Sliders are used to vary the different templates and find the best match. The black spectrum is for a star used in this work and the red spectrum is of a TLUSTY template. The parameters of the template are varied using the sliders.

chosen on the basis of a visual inspection and consideration of the correlation heights with the template spectra. The `specutils`¹ package is used for a visual inspection before performing a fourier cross-correlation. Fig. 3.2 shows the use of the `specutils` package implemented using `matplotlib` to choose the right template for `fxcor` using sliders to vary the different parameters of the templates. Spectral features as described in section 1.9.1 are looked for and used to determine the best-fitting template.

¹<https://specutils.readthedocs.io/en/stable/index.html>

Chapter 4

Results

This chapter contains the results of the automated search for runaway stars near the selected young open clusters (see chapter 2) using the traceback code.

The automated search produced runaway star candidates for 87 clusters. These had to be checked further for the quality of isochrone fits and to see if there is a clustering of background or foreground stars moving with a common proper motion in a direction. If the runaway star moves with such a group, it would be unlikely to be a member of the cluster. The distance of the star and the group with respect to the cluster is often a good indicator to eliminate such candidates. When a suitable candidate is found from this inspection, a more detailed analysis of its cluster members and isochrone fittings were made to adjust the cluster parameters from [Dias et al. \(2021\)](#) which is based on *Gaia* DR2.

In *Gaia* DR3, often more stars have measurements of the radial velocities. This makes it possible to have a better estimate of the radial velocity of the cluster and also to check if the runaway candidate meets the threshold peculiar velocity of 25 km/s in 3D with respect to the cluster. One thing to note is that the availability of radial velocities does not immediately allow for a consideration of a traceback in 3D. This is because to the magnitude of errors in the distance estimates for the stars are large. The runaway stars are chosen as candidates if their distances with errors intersect with that of the clusters.

With a good sample of runaway star candidates, it was checked if they are observable from GSH. Runaway star candidates from two of the clusters have been observed: HD 259991 from Basel 8 and HD 240113 from RSG 8. The runaway star candidate from Berkeley 97 was too faint to be observed from GSH. Two of its cluster members have been observed which help to fix the isochrone of the cluster and hence to estimate the

spectral types of the all stars in the cluster.

Another good candidate from cluster NGC 4103 has been presented in a later section of this chapter. This cluster could not be observed from GSH. Observational data for these candidates and cluster members in the future would be useful to confirm the results for this candidate.

The results of a search for nearby pulsars from the ATNF catalogue has also been discussed for a speculation about a possible association. For this, young pulsars have been chosen. The age estimates of pulsars is usually based on the period and period derivatives: this is called the characteristic age. This age is the upper limit for the age of the pulsar (see topic on pulsars 1.6). Therefore, an almost exact age estimate for the pulsar can be made if an association with the runaway star can be confirmed. Since none of the relevant pulsars in this work has known proper motions, tracing back their trajectories is out of scope. However, an estimate is made for the kinematic age of the pulsars considering a common origin with the runaways.

Each section of the following part of this chapter discusses the clusters and their runaway candidate(s) in more detail. These sections are arranged in an alphabetical order.

4.1 Basel 8

Basel 8 is located at $06^h34^m13.968^s +08^d02^m17.16^s$ at a distance of 1456 ± 54 pc. [Dias et al. \(2021\)](#) provides 260 members in the cluster, out of which 6 have a membership probability > 0.8 and $\frac{\Delta\pi}{\pi} > 10$. Due to this low number, the membership probability filter was changed to > 0.5 , which included 37 members. Two more members HD259597 and HD259787 were added to the member list based on their distances, coordinates and isochrone fits. The cluster's kinematic parameters in table 4.2 are calculated based on these 39 members, brightest 10 of which shown in table 4.1. The selected members are shown in fig. 4.1.

The runaway candidate is HD 259991 which has a G_{mag} of 9.317 ± 0.003 mag. The two members namely HD 259597 and HD 259787 have a G_{mag} of 8.432 ± 0.004 mag and 9.743 ± 0.003 mag respectively. Therefore, all three stars were observable from GSH and their high resolution spectroscopic data was taken using FLECHAS. Details about the observations are given in table A.1. The parameters of the stars derived by fitting stellar atmosphere models are given in table A.2.

The radial velocity of the cluster is calculated to be 22.5 ± 5.0 km/s based on 4 available *Gaia* radial velocity measurements out of the 39 stars. The two additional

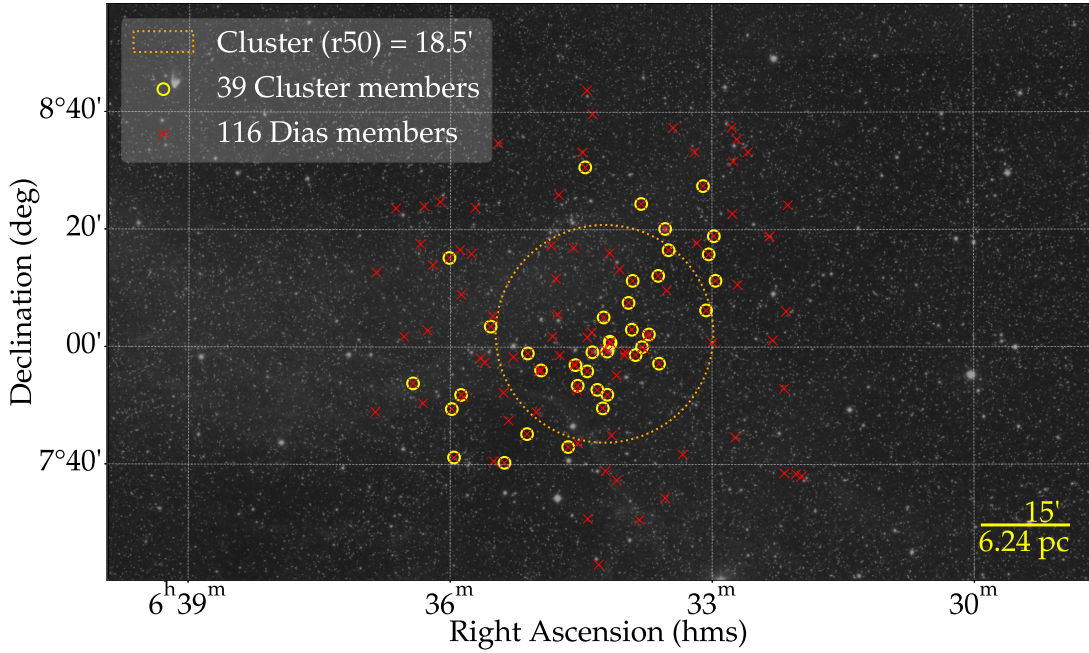


Figure 4.1: 39 selected members (in yellow circles) of Basel 8 out of the 116 members (in red crosses) mentioned in the Dias catalogue.

stars that were added as members have their radial velocities derived from their spectra. Including their measurements makes a total of 6 stars with known radial velocities in this cluster, taking the value to 25.2 ± 3.7 km/s.

Observations

The two members HD259597 and HD259787 were observed on UTC 2024-01-10T02:58 for 900s and on 2024-01-10T23:52 for 1200s respectively, and the data was reduced in the standard procedure described in A.2. The runaway candidate HD259991 was observed on UTC 2024-01-09T22:36 for a 1200s exposure time. The spectrum of these stars are shown in figure 4.2.

Model spectra from TLUSTY Hubeny & Lanz (2011) and Kurucz (1993) were used to fit the T_{eff} and $\log g$ values and to obtain the radial velocities using Fourier cross-correlation `fxcor` command in IRAF. The selected models have the highest correlation heights. The values for the radial velocities thus obtained are shown in table 4.3 along with the spectral types from the literature. HD259597 is a Be star and correlation using templates was poor due to the absence of distinguishable absorption features in the spectra which could be correlated with the templates.

Figure 4.3 compares the CMD generated from the original parameters of Basel 8 given by Dias et al. (2021) to CMD with different parameters. A better fit is obtained

Chapter 4. Results

Table 4.1: Photometric properties of selected members of Basel 8 and the runaway candidate HD 259991 (shown in bold font)

Gaia DR3 Source	r_{geo} pc	G mag	μ_{α}^* mas/yr	μ_{δ} mas/yr	$G_{\text{BP}} - G_{\text{RP}}$ mag
3326048448124872320	1518^{+104}_{-75}	8.432 ± 0.004	-0.76 ± 0.04	-1.06 ± 0.03	0.337 ± 0.020
3326022227346760960	1449^{+39}_{-38}	9.473 ± 0.003	-0.16 ± 0.02	-2.29 ± 0.02	0.266 ± 0.014
3326023842254469120	1409^{+54}_{-55}	9.743 ± 0.003	-0.05 ± 0.03	-2.25 ± 0.03	0.203 ± 0.007
3326024155790212864	1360^{+41}_{-42}	10.758 ± 0.003	-0.02 ± 0.03	-2.29 ± 0.02	0.207 ± 0.007
3326024117132373632	1446^{+48}_{-40}	11.888 ± 0.003	-0.18 ± 0.02	-2.40 ± 0.02	0.295 ± 0.007
3326233883335142272	1458^{+28}_{-32}	13.176 ± 0.003	-0.16 ± 0.01	-2.25 ± 0.01	0.398 ± 0.007
3326028175879616256	1452^{+35}_{-29}	13.195 ± 0.003	-0.12 ± 0.02	-2.32 ± 0.02	0.412 ± 0.007
3326039995629272704	1423^{+37}_{-29}	13.233 ± 0.003	-0.12 ± 0.02	-2.38 ± 0.01	0.394 ± 0.007
3326021303931980544	1477^{+47}_{-51}	13.290 ± 0.003	-0.01 ± 0.02	-2.56 ± 0.01	0.459 ± 0.007
3326081433473431168	1315^{+23}_{-24}	13.326 ± 0.003	-0.34 ± 0.01	-2.66 ± 0.01	0.718 ± 0.007
HD 259991	1458^{+78}_{-80}	9.318 ± 0.003	-1.40 ± 0.04	0.65 ± 0.03	0.329 ± 0.008

Table 4.2: Kinematics of Basel 8 compared to its runaway

Name	α °	δ °	d pc	μ_{α}^* mas yr ⁻¹	μ_{δ} mas yr ⁻¹	v_r km s ⁻¹	v_{trans} km s ⁻¹
Basel 8	98.558	8.038	1429 ± 66	-0.2 ± 0.1	-2.4 ± 0.1	25.2 ± 3.7	
HD 259991	98.691	8.076	1458 ± 158	-1.4 ± 0.0	0.7 ± 0.0	1.9 ± 7.0	21.3 ± 2.9

with $A_V = 1.1$ mag and $\log \tau = 7.8$ instead of the original parameters $A_V = 1.08$ mag and $\log \tau = 7.67$. This isochrone sets the temperature of the runaway to 13240^{+880}_{-970} K, corresponding to a B7V spectral class. This is consistent with the temperature obtained from the spectrum. Further, from the isochrone, the mass estimate for HD 259991 is $5.964 M_{\odot}$ and the surface gravity estimate is 3.562.

The two stars HD 259787 and HD 259597 have their temperature estimates from the isochrone similar to the estimates from their spectra. The difference of the temperature estimate of HD 259787 compared the spectrum comes from its colour appearing bluer. A bluer colour leads to a higher temperature estimate from the isochrone. This could be due to the presence of inhomogeneous dust in the region of the cluster leading to a

Table 4.3: Spectroscopic parameters of HD 259787, HD 259597 and HD 259991

Star	T_{iso} (K)	T_{eff} (K)	$\log g$ (dex)	v_r (km/s)	Sp. Type	Sp. Type (lit.)
HD 259787	17650^{+850}_{-810}	13000 ± 1000	3.50 ± 0.5	21.03 ± 5.47	B7V	A0
HD 259597	13110^{+1250}_{-1360}	13000 ± 1000	3.00 ± 0.5	39.78 ± 8.15	B7V	B0.5V
HD 259991	13240^{+880}_{-970}	15000 ± 1000	3.00 ± 0.25	1.91 ± 7.03	B5V	B3/5

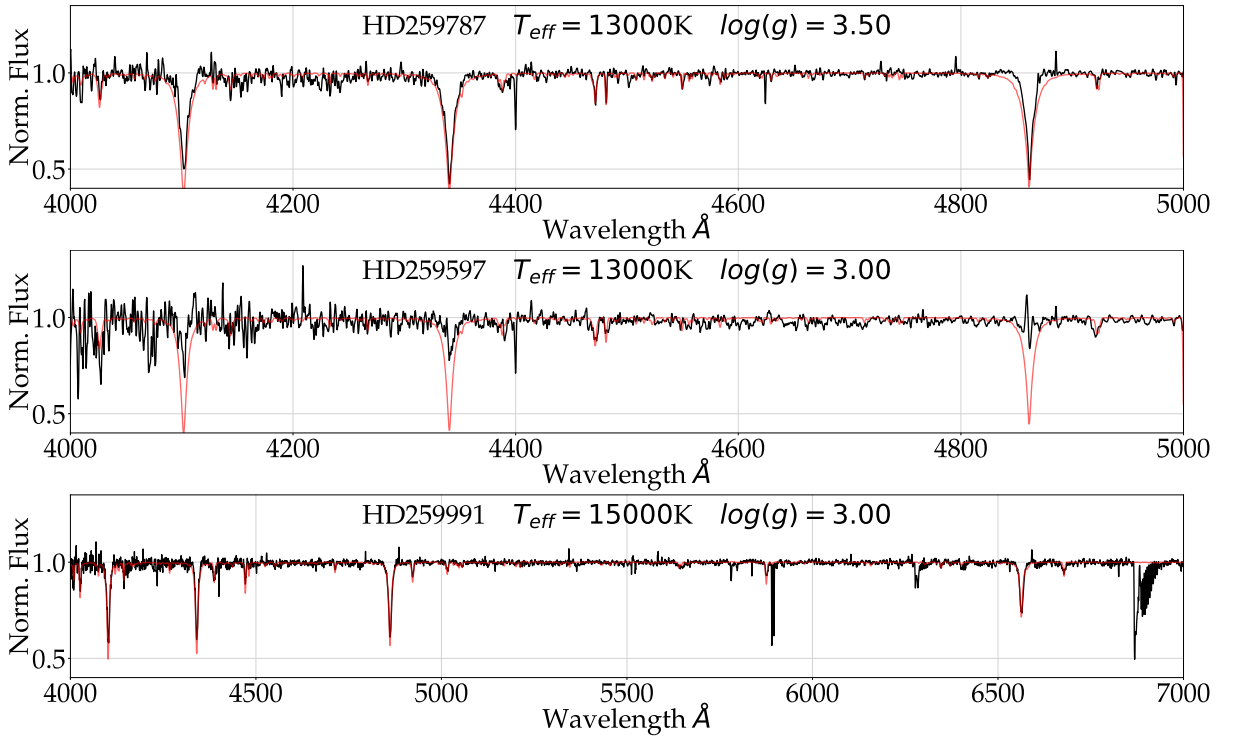


Figure 4.2: Spectrum of HD259787, HD259597 and HD259991 plotted along with their best matching template. TLUSTY models are used for HD 259597 and HD 259991 while KURUCZ models are used for HD 259787. These are used to derive the spectral types as shown in table 4.3.

different measure of extinction towards this star. The surface gravity values HD 259787 and HD 259597 from the isochrone are 3.777 and 3.091. These are in agreement with the values obtained from the template fit. The mass estimates for HD 259787 and HD 259597 are $5.643 M_{\odot}$ and $6.107 M_{\odot}$ respectively.

Possible association of HD 259991 with PSR J0641+07

HD 259991 is a kinematic outlier moving with a relative proper motion of $\Delta\mu_{\alpha}^* = -1.21 \pm 0.26$ mas/yr and $\Delta\mu_{\delta} = 2.83 \pm 0.26$ mas/yr, corresponding to a transverse velocity of 21.3 ± 2.9 km/s. Combining with the relative radial velocity of 23.3 ± 7.91 km/s, the 3D velocity of the HD 259991 is 31.57 ± 6.16 km/s which is above the threshold of 25 km/s for a runaway [Tetzlaff \(2013\)](#).

The traceback plots for the runaway are shown in 4.4. The runaway star candidate HD 259991 is moving north-west with respect to the cluster as shown in the figure. Furthermore, 11 pulsars lie within 80pc of the cluster center, out of which 6 lie in a distance range of 0.71 kpc to 2.14 kpc. Out of these, PSR J0641+07 is interesting since it lies at a distance of 1260 pc (measured using an electron-density model [Yao et al. \(2017b\)](#)), which is close to the cluster distance of 1429 ± 66 pc. Neither the period, nor the pe-

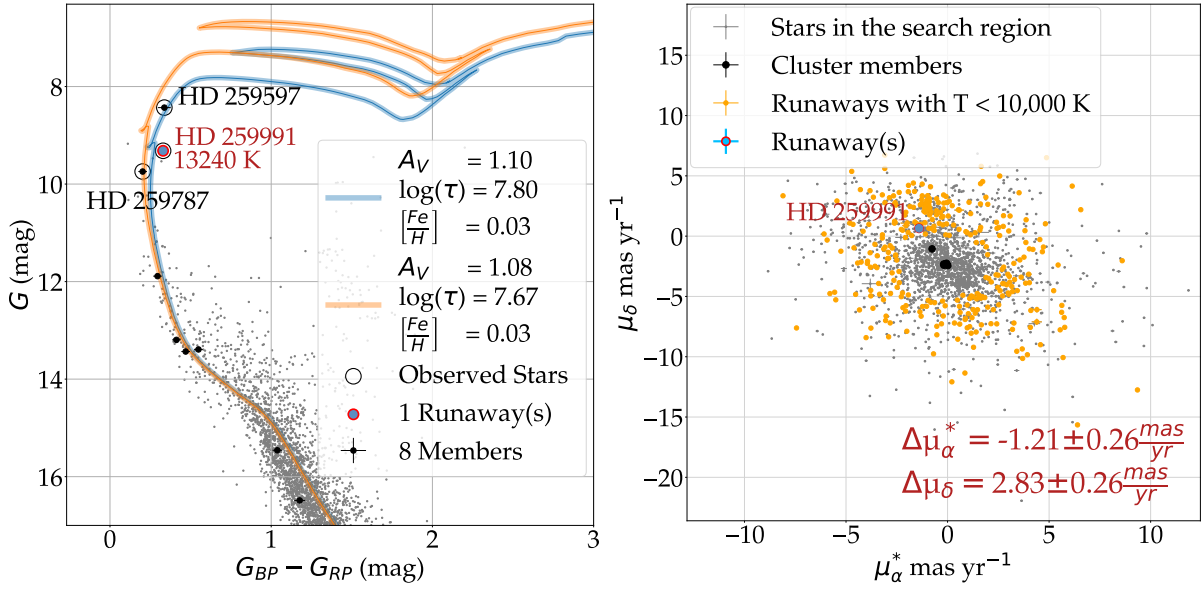


Figure 4.3:

Left: Basal 8 and its best fitting CMD. The temperature estimate of HD 259991 is indicated. The two observed stars are also marked.

Right: Proper motion diagram of Basal 8 showing the relative proper motion of HD 259991 compared to the cluster. The orange scatter points indicate other high transverse velocity stars which could be traced back to the cluster but were not hot enough according to the isochrone fit.

riod derivative is known for the pulsar and therefore an age cannot be determined by the conventional method. The pulsar's proper motion is not known but its trajectory may coincide with the position of HD 259991 in 115 kyr in the past, assuming a typical pulsar velocity of 340 km/s Lyne & Lorimer (1994). This can therefore be the kinematic age of the pulsar.

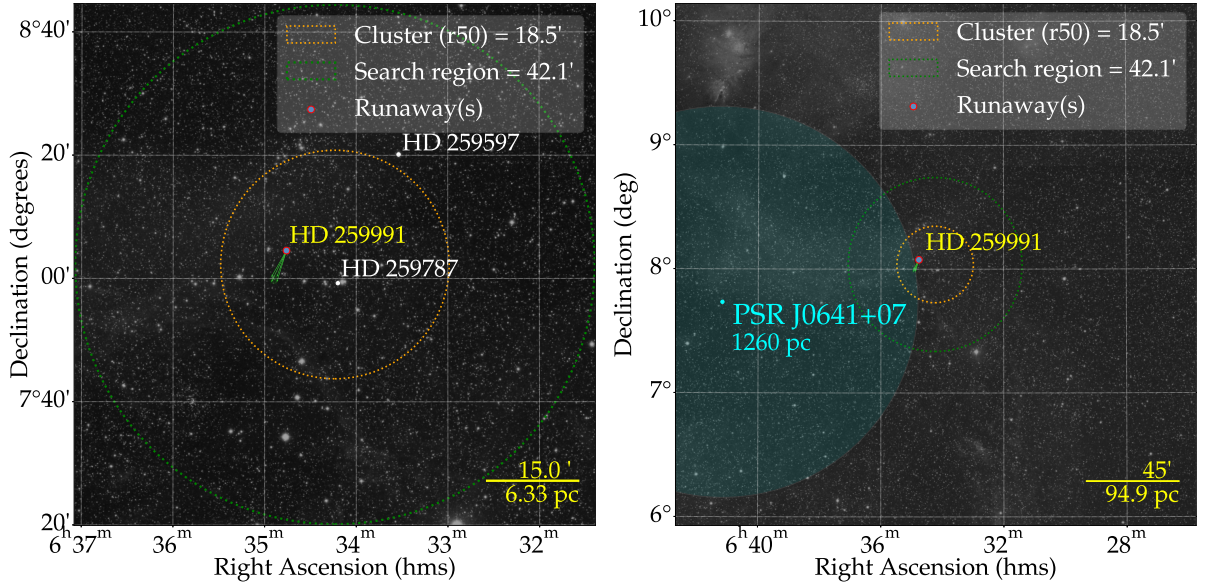


Figure 4.4: *Left:* Motion of HD 259991 with respect to the cluster average. The two observed stars have been also indicated along with the search radius and r50 radius of the cluster. *Right:* Possible association with PSR J0641+07. The pulsar is located at a distance of 1.26 kpc and the circle region indicates a traceback time of 115 kyr in the past with a velocity of 340 km/s.

4.2 Berkeley 97

Berkeley 97 is located at $\alpha = 22^h39^m23.976^s$, $\delta = 59^\circ00'16.56''$ at a distance of 2774 ± 131 pc. Dias et al. (2021) provides 155 members out of which 52 were found inside the search region and 27 met the membership quality criteria. Based on its parameters like distance, position, proper motion values, and isochrone fits, the star HD 240016 was added to the cluster members, making a total of 28 members. The stars belonging to the cluster are shown in fig. 4.5 on the left.

The brightest 10 of the members are shown in table 4.5 along with the runaway star candidate. HD240016 is, in fact, the brightest star in the cluster. With these members, the kinematic parameters of the cluster are computed and compared to the runaway in table 4.4. The radial velocity of the cluster is not present in the literature and none of the selected member stars had radial velocity measurements from *Gaia*. However, the two observed stars were used to estimate the radial velocity. This sample is small to ascertain this value and the binary nature of the two stars cannot be precisely determined with the limited sample of data.

Observations

The two brightest stars of the cluster HD240015 and HD240016 were observed from GSH using FLECHAS and their spectra was obtained over several nights. The among

Table 4.4: Kinematics of Berkeley 97 compared to its runaway

Name	α °	δ °	d pc	μ_{α}^* mas yr ⁻¹	μ_{δ} mas yr ⁻¹	v_r km s ⁻¹	v_{trans} km s ⁻¹
Berkeley 97	339.85	59.005	2769 ± 132	-2.7 ± 0.1	-1.9 ± 0.1	-39.6 ± 2.7	
Runaway	339.896	58.995	2620 ± 298	-0.4 ± 0.0	-1.4 ± 0.0	—	30.0 ± 3.2

Table 4.5: Photometric properties of selected members of Berkeley 97 and the runaway candidate (shown in bold font)

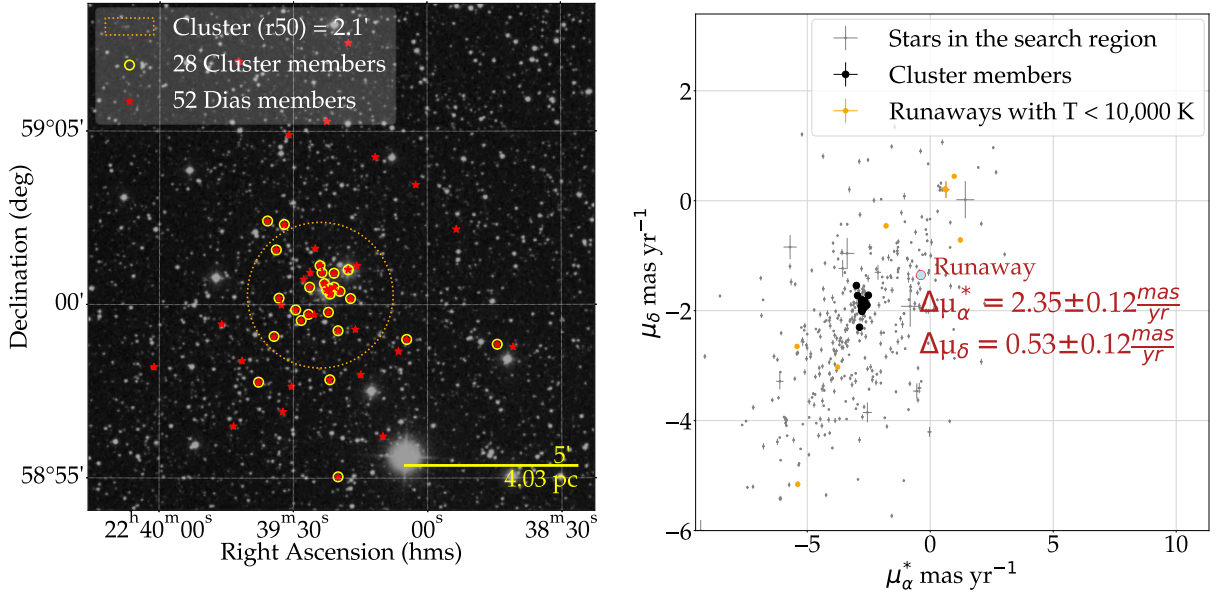
Gaia DR3 Source	r_{geo} pc	G mag	μ_{α}^* mas/yr	μ_{δ} mas/yr	$G_{\text{BP}} - G_{\text{RP}}$ mag
2008383302396655488	2630 ⁺³¹⁰ ₋₂₃₇	9.474 ± 0.003	-2.87 ± 0.04	-2.30 ± 0.04	0.938 ± 0.007
2008430237808841600	2658 ⁺⁷⁷ ₋₇₀	9.840 ± 0.003	-2.75 ± 0.01	-1.91 ± 0.01	0.869 ± 0.007
2008380794144211712	2571 ⁺⁸² ₋₇₃	11.482 ± 0.003	-2.68 ± 0.01	-1.93 ± 0.01	1.201 ± 0.013
2008430203449108224	2525 ⁺⁶⁷ ₋₆₅	13.262 ± 0.003	-2.95 ± 0.01	-1.72 ± 0.01	0.962 ± 0.007
2008383199325604224	2978 ⁺¹⁰⁶ ₋₉₆	13.396 ± 0.003	-2.77 ± 0.01	-2.01 ± 0.01	1.085 ± 0.007
2008430233501492608	2820 ⁺¹³⁴ ₋₁₀₉	13.733 ± 0.003	-2.68 ± 0.01	-1.82 ± 0.01	1.051 ± 0.007
2008383302404812544	2605 ⁺¹¹⁵ ₋₁₃₈	14.199 ± 0.003	-2.63 ± 0.02	-1.91 ± 0.02	1.058 ± 0.008
2008384779873564032	2623 ⁺⁹⁶ ₋₁₂₁	14.491 ± 0.003	-2.78 ± 0.02	-1.83 ± 0.02	1.049 ± 0.007
2008430066010153216	2691 ⁺¹⁴¹ ₋₁₃₉	14.640 ± 0.003	-2.74 ± 0.02	-1.88 ± 0.02	1.057 ± 0.007
2008384814233517312	2824 ⁺¹⁵⁵ ₋₁₂₇	14.664 ± 0.003	-2.52 ± 0.02	-1.72 ± 0.02	1.043 ± 0.007
2008384504995669248	2620 ⁺¹³⁹ ₋₁₅₉	15.384 ± 0.003	-0.38 ± 0.03	-1.35 ± 0.02	1.182 ± 0.008

these, the data with the highest signal-to-noise ration was used for the analysis presented here. These spectra were matched with templates from TLUSTY and are shown in figure 4.7. The parameters of these stars thus obtained are shown in table 4.6. These are used to fit the isochrone for the cluster, as described in the next section. The radial velocities of the two stars were obtained using the Fourier cross-correlation functionality in IRAF with the templates.

To fit the cluster members' positions on the CMD, the values of A_V and $\log \tau$, as given in Dias et al. (2021), needed to be changed from 2.54 to 2.80 and 7.59 to 6.90 respectively. These changes are shown in figure 4.7, in the plot on the left. The values of the temperatures thus obtained for the two observed stars HD 240015 and HD 240016 from the isochrone are in agreement with the temperatures obtained from fitting the

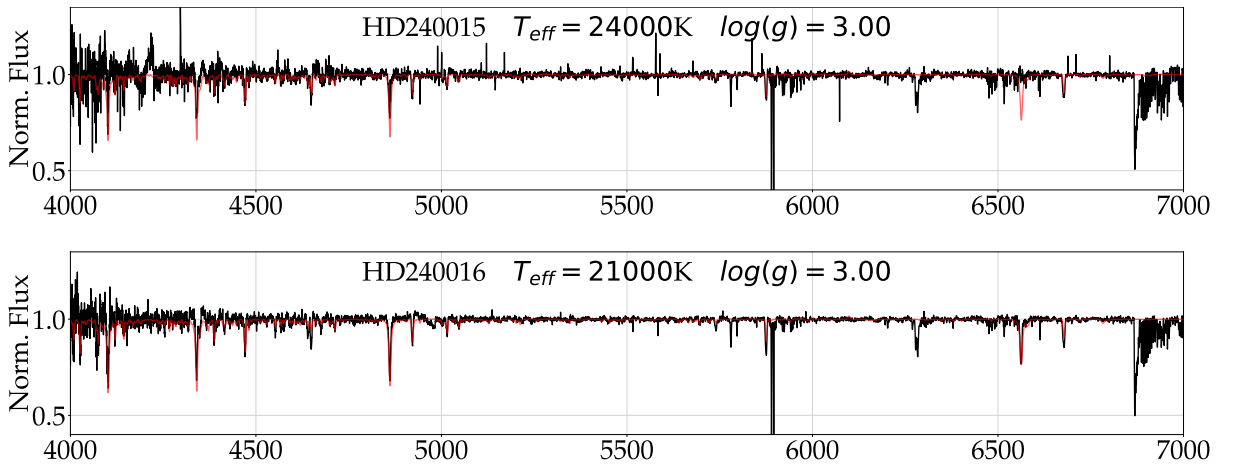
Table 4.6: Spectroscopic parameters of HD 240015 and HD 240016

Star	T_{iso} (K)	T_{obs} (K)	$\log g$ (dex)	v_r (km/s)	SpT	SpT (lit.)
HD 240015	24650 ⁺¹⁵⁸⁰ ₋₁₄₈₀	24000 ± 2000	3.0 ± 0.2	-45.8 ± 4.2	B1.5±0.5II	Be
HD 240016	20410 ⁺²²¹⁰ ₋₁₇₅₀	21000 ± 1000	3.0 ± 0.2	-33.5 ± 3.5	B2±0.5II	B

**Figure 4.5:**

Left: Berkeley 97 and its cluster members. The red stars indicate the stars found in the Dias list of members in the region. The yellow circles are the members selected for this analysis.

Right: Proper motion diagram depicting that the runaway star is an outlier in proper motion in the local rest frame. The relative proper motion of the runaway is marked on the plot. The orange scatter points indicate other high transverse velocity stars which could be traced back to the cluster but were not hot enough according to the isochrone fit.

**Figure 4.6:** Spectra of HD240015 and HD 240016 plotted with their best matching template spectra from TLUSTY Hubeny & Lanz (2011). The parameters obtained are shown in the figure.

spectra using TLUSTY stellar atmosphere models, as shown in table 4.6.

In our analysis presented for this cluster in [Dinçel & Sheth \(2024\)](#), the isochrone was generated using PARSEC version 1.2S using the CMD3.7 interface. In this work, the newer PARSEC version 2.0 was used which generates isochrones which fit the cluster with slightly different parameters. This difference is shown in the plot on the right in 4.7.

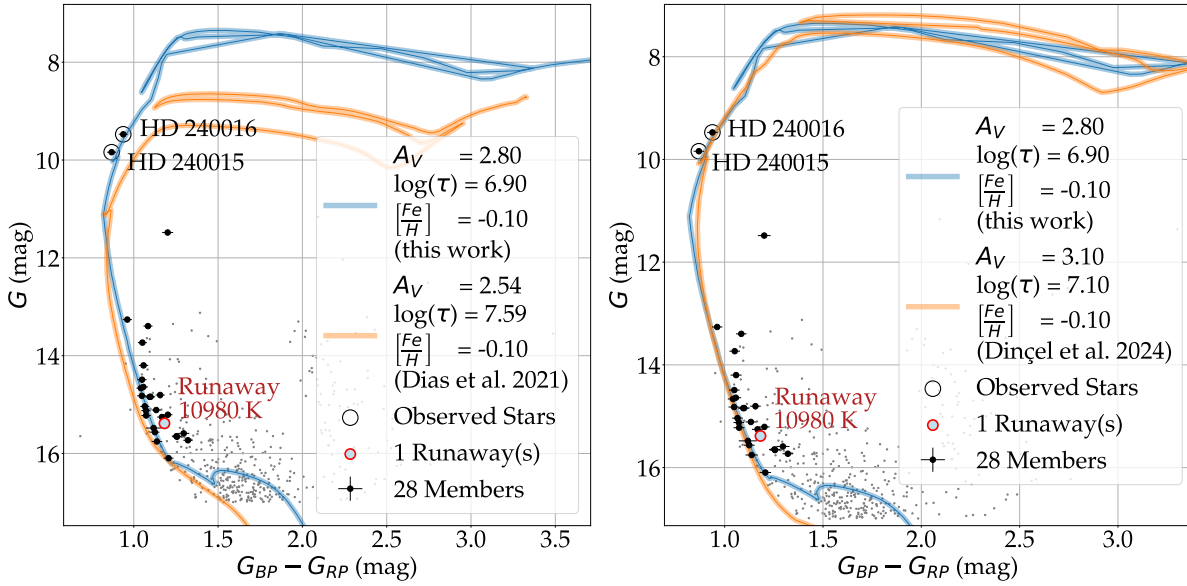


Figure 4.7:

Left: Isochrone generated from the parameters of Berkeley 97 from [Dias et al. \(2021\)](#) (orange) with the parameters estimated in this work (blue).

Right: Isochrone for Berkeley 97 using parameters from [Dinçel & Sheth \(2024\)](#) (orange) generated using PARSEC version 1.2 compared to the isochrone using parameters estimated in this work (blue) using PARSEC version 2.0. Mainly the A_V values differ by 0.3 mag

In the CMDs, the runaway star *Gaia* DR3 2008384504995669248 is marked along with its temperature estimate of 10,980 K using the chosen theoretical isochrone with $A_V = 2.8$ mag, $\log \tau = 6.9$, $[\frac{Fe}{H}] = -0.1$ and at a distance of 2769 ± 132 pc. This isochrone also gives an estimate of the mass of the runaway star to be $2.77 M_{\odot}$ with a surface gravity of 4.392. The masses of the stars HD 240016 and HD 240015 estimated from the isochrone are $13.91 M_{\odot}$ and $13.92 M_{\odot}$, while their $\log g$ values are 2.92 and 3.24 respectively.

Possible association with PSR J2238+5903

The runaway star *Gaia* DR3 2008384504995669248 is a kinematic outlier amongst the members of the cluster with a tangential peculiar velocity with respect to the cluster of $v_{\perp} = 30.0 \pm 3.7$ km s $^{-1}$. This is already higher than the lower limit of $v_{\text{pec}} = 25$ km s $^{-1}$

for a runaway Tetzlaff (2013), without taking the relative radial velocity of the star into account. The radial velocity of the star is not known and hence the 3D peculiar velocity cannot be determined.

The motion of the runaway star relative to the cluster is plotted in figure 4.8. The plot on the right in figure 4.1 shows that the star is an outlier in the local rest frame established by the cluster with a relative proper motion of $\Delta\mu_\alpha^* = 2.35 \pm 0.12 \text{ mas yr}^{-1}$ and $\Delta\mu_\delta = 0.53 \pm 0.12 \text{ mas yr}^{-1}$.

The young radio-quiet, γ -ray pulsar PSR J2238+5903 is located at an angular separation of 7.9 arcmin from the cluster center. The period of pulsation is $P = 162.7 \text{ ms}$ and the period derivative is $\dot{P} = 9.7034 \times 10^{-14} \text{ ss}^{-1}$. From these, the spin-down age of the pulsar is $\tau = 26.6 \text{ kyr}$. The dipole field strength of the pulsar (also deduced from the $P\dot{P}$ relation) is $B = 4.02 \times 10^{12} \text{ G}$. The SNR for such a young pulsar should be visible in radio as a shell of non-thermal synchrotron radiation of accelerated electrons along with thermal radiation in X-ray. Optical counterparts in the form of recombining H atoms and forbidden lines should also be seen. However, no known SNR can be found in the literature in this region so far.

The distance to the pulsar is 2830 pc. This lies within the error bars of the distance of the cluster $2769 \pm 132 \text{ pc}$. Fig. 4.8 shows the pulsar's position next to the cluster and the circle denoting the possible locations of the cluster considering a typical 2D space velocity of 340 km s^{-1} and the age of the pulsar to be 20 kyr.

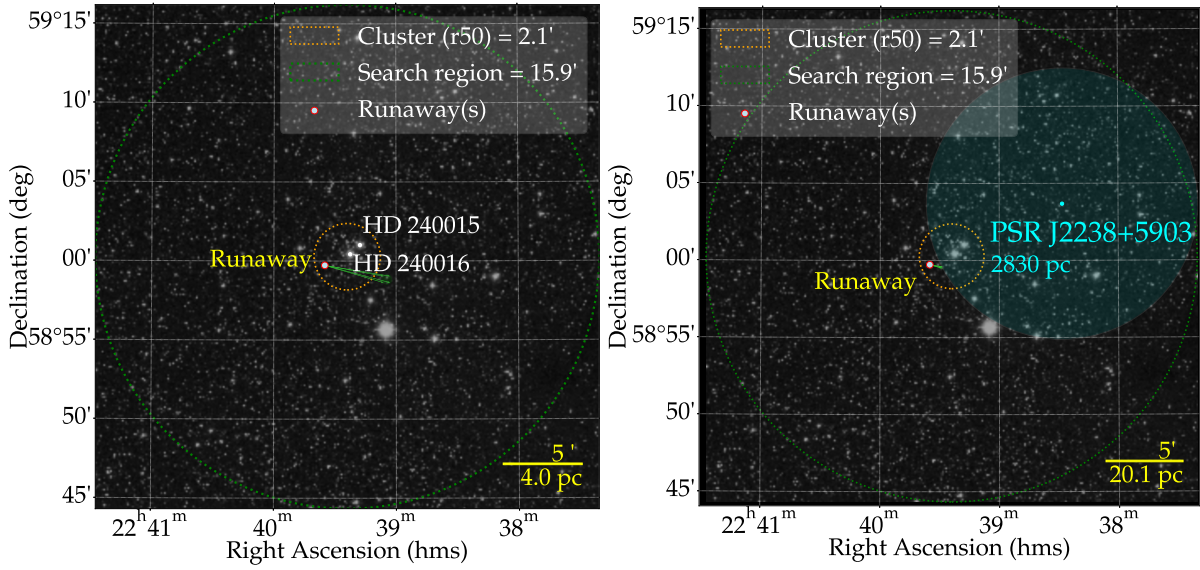


Figure 4.8: *Left:* The runaway star *Gaia* DR3 2008384504995669248's motion with respect to the cluster. The four green lines depict the four extreme cases of the proper motion considering the errors and the green ellipse represents the possible positions of the star 100 kyr ago. The two observed stars of the cluster are also marked in yellow text.

Right: The runaway star and PSR J2238+5903 traced back to a common origin place. The pulsar has a characteristic age of 26 kyr, however, their orbits meet at -20 kyr, making it the kinematic age. This is assuming the typical velocity of 340 km/s of pulsars.

4.3 NGC 4103

NGC 4103 is located at $\alpha = 12^h06^m24.624^s$, $\delta = -61^\circ14'57.12''$ at a distance of 1870 ± 132 pc. 219 out of the 353 mentioned in [Dias et al. \(2021\)](#) have been selected as members. Using these, the kinematic parameters of the cluster have been determined. These are compared to the kinematic parameters of the runaway star in table 4.7. Ten of the brightest members of the cluster are shown in table table 4.8. The cluster and its selected members is shown in fig. 4.9.

Table 4.7: Kinematics of NGC 4103 compared to its runaway

Name	α °	δ °	d pc	μ_α^* mas yr $^{-1}$	μ_δ mas yr $^{-1}$	v_r km s $^{-1}$	v_{trans} km s $^{-1}$
NGC 4103	181.603	-61.249	1869 ± 132	-6.2 ± 0.1	0.1 ± 0.1	-28.2 ± 9.8	
NGC 4103 18	181.588	-61.238	1878 ± 117	-8.0 ± 0.0	1.1 ± 0.0	—	18.2 ± 1.7

CMD and proper motion

The parameters of the cluster's isochrone from [Dias et al. \(2021\)](#) are shown in the CMD fig. 4.10. These parameters fit the cluster members well and did not need modifica-

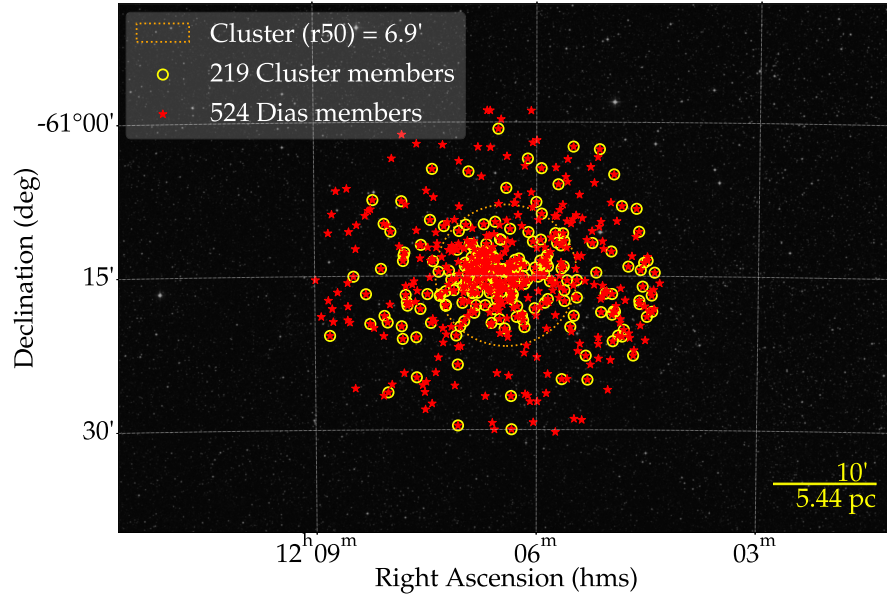


Figure 4.9: Selected cluster members of NGC 4103 (shown in yellow circles) along with the members mentioned in the Dias catalogue (shown in red stars).

tion. Using these, the temperature of the runaway star candidate NGC 4103 18 has been estimated to be 15180^{+990}_{-1350} K, which corresponds to a spectral class of around B5V. The right plot in fig. 4.10 shows the proper motion diagram showing that NGC 4103 18 moves with a relative proper motion $\Delta\mu_{\alpha}^* = -1.78 \pm 0.13$ mas/yr and $\Delta\mu_{\delta} = 1.00 \pm 0.13$ mas/yr with respect to the cluster average. This corresponds to a transverse velocity of 18.2 ± 1.7 km/s, which is higher than the 2D component of 17.6 km/s (corresponding to a 3D velocity of 25 km/s). The radial velocity is not known.

Comparing to the nearest matching star from the isochrone, the runaway star candidate has a mass of $8.846M_{\odot}$ and a $\log g$ of 3.018.

Traceback and possible association with PSR J1151-6108

The runaway star is NGC 4103 18 and its motion relative to the cluster is shown in the left plot of fig. 4.11. The trajectory is relative to the cluster's motion for 100 kyr in past.

PSR J1151-6108 is located at an angular separation of 104 arcmin and located at a distance of 2227 pc. This distance is measured using an electron-density model Yao et al. (2017a). The proper motions of this pulsar are not known. The age of the pulsar from the $P\dot{P}$ relation is 157 kyr. Considering a pulsar velocity of 385 km/s and an age of 150 kyr, a common point origin can be found for the runaway star candidate NGC 4103 18. This is depicted in the plot on the right of fig. 4.11.

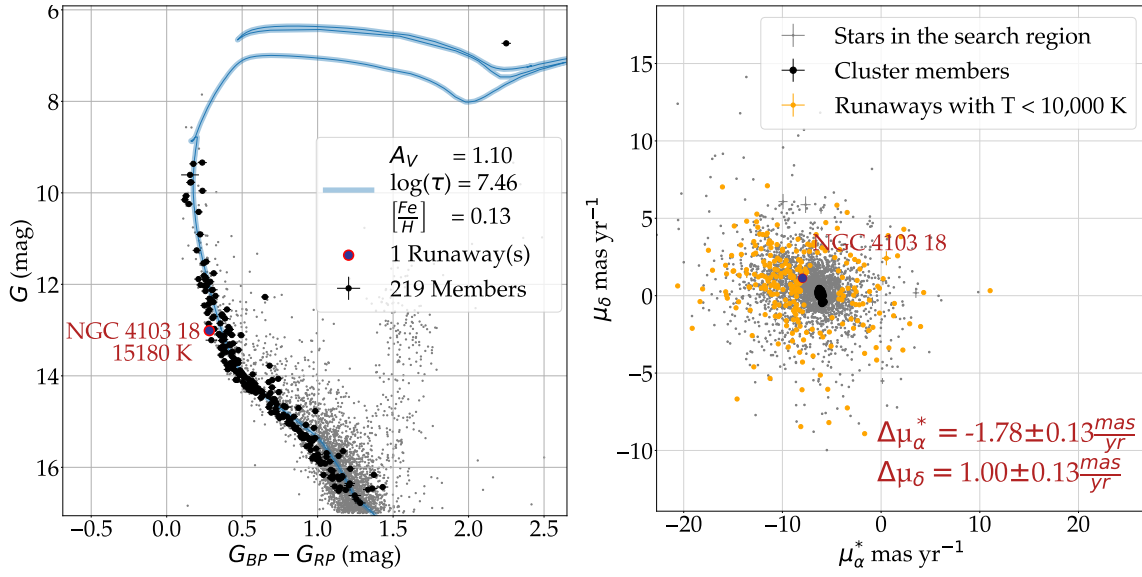


Figure 4.10:

Left: The CMD showing the temperature estimate of the runaway. The isochrone for the temperature estimate is shown in blue.

Right: Proper motion diagram depicting that the star is an outlier in proper motion in the local rest frame. The orange scatter points indicate other high transverse velocity stars which could be traced back to the cluster but were not hot enough according to the isochrone fit.

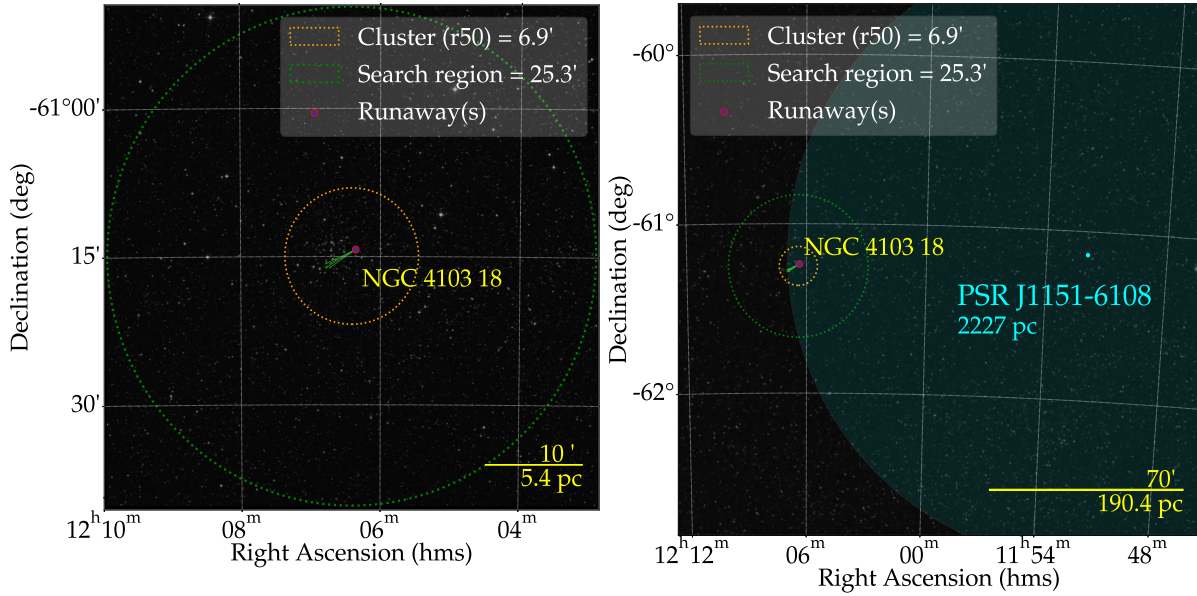


Figure 4.11:

Left: The runaway star's motion with respect to the cluster. The four green lines depict the four extreme cases of the proper motion considering the errors and the green ellipse represents the possible positions of the star 100 kyr ago.

Right: Proper motion diagram depicting that the star is an outlier in proper motion in the rest frame of the cluster.

Table 4.8: Photometric properties of selected members of NGC 4103 and the runaway candidate NGC 4103 18 (shown in bold font)

Gaia DR3 Source	r_{geo} pc	G mag	μ_{α}^* mas/yr	μ_{δ} mas/yr	$G_{\text{BP}} - G_{\text{RP}}$ mag
6058513719987379840	1995^{+105}_{-94}	6.732 ± 0.003	-5.84 ± 0.02	0.00 ± 0.03	2.250 ± 0.011
6058509974775488768	1839^{+52}_{-51}	9.338 ± 0.003	-6.24 ± 0.01	0.32 ± 0.02	0.236 ± 0.010
6058515850291358976	1818^{+60}_{-57}	9.366 ± 0.003	-6.29 ± 0.02	0.18 ± 0.02	0.176 ± 0.007
6058510765049380096	2002^{+87}_{-109}	9.607 ± 0.009	-6.26 ± 0.02	0.26 ± 0.02	0.155 ± 0.039
6058510799409144448	2044^{+78}_{-62}	9.769 ± 0.003	-6.22 ± 0.02	0.35 ± 0.02	0.162 ± 0.007
6058511039927375360	1956^{+62}_{-66}	9.773 ± 0.003	-6.14 ± 0.02	0.31 ± 0.02	0.156 ± 0.007
6058512757914903040	1795^{+50}_{-51}	9.955 ± 0.003	-6.25 ± 0.02	0.17 ± 0.02	0.237 ± 0.008
6058516778004159360	1968^{+87}_{-67}	10.066 ± 0.003	-6.20 ± 0.02	0.27 ± 0.02	0.129 ± 0.007
6058511246085808128	1828^{+71}_{-66}	10.152 ± 0.003	-6.24 ± 0.02	0.26 ± 0.02	0.120 ± 0.007
6058511246085810432	1978^{+70}_{-64}	10.244 ± 0.003	-6.28 ± 0.02	0.22 ± 0.02	0.147 ± 0.007
NGC 4103 18	1878^{+56}_{-61}	13.012 ± 0.003	-7.96 ± 0.01	1.13 ± 0.02	0.283 ± 0.008

4.4 RSG 8

RSG 8 is located at $\alpha = 23^{\text{h}}00^{\text{m}}07.776^{\text{s}}$, $\delta = 59^{\circ}22'10.92''$ at a distance of 450 ± 5 pc. [Dias et al. \(2021\)](#) lists 191 members in the cluster, out of which 96 have been selected. In addition, HD218440 with Gaia source 2013579250758906240 has been added to the cluster members due to its kinematic properties and isochrone fit. The 10 brightest cluster members are shown in table 4.10.

The selected 97 members are used to update the kinematic parameters of the cluster and the updated parameters are shown in table 4.9.

Table 4.9: Kinematics of RSG 8 compared to its runaway

Name	α °	δ °	d pc	μ_{α}^* mas yr ⁻¹	μ_{δ} mas yr ⁻¹	v_r km s ⁻¹	v_{trans} km s ⁻¹
RSG 8	345.032	59.37	442 ± 29	5.3 ± 0.4	-1.6 ± 0.4	-11.7 ± 4.6	
HD 240113	343.691	59.279	428 ± 6	-8.5 ± 0.0	-6.7 ± 0.0	-10.3 ± 0.9	29.8 ± 1.1

Observations

The two brightest cluster members are HD 218440 and HD 217348 with magnitudes 6.423 ± 0.003 and 6.707 ± 0.003 have been observed at GSH using an échelle spectrograph to obtain their spectra. These are shown in 4.13 along with the best fit model spectra from TLUSTY stellar atmosphere models [Hubeny & Lanz \(2011\)](#).

HD 218440 is a known spectroscopic binary ([Pourbaix et al. \(2004\)](#)).

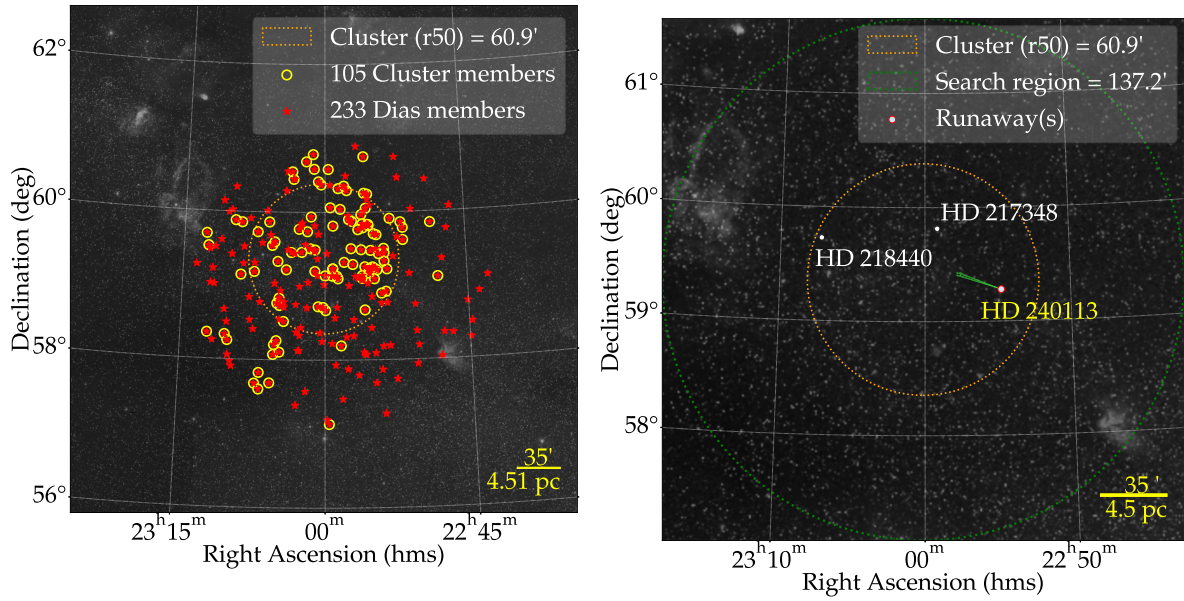


Figure 4.12: *Left:* Selected cluster members of RSG 8 (shown in yellow circles) along with the members mentioned in the Dias catalogue. *Right:* Traceback plot for HD 240113 showing its motion relative to RSG 8.

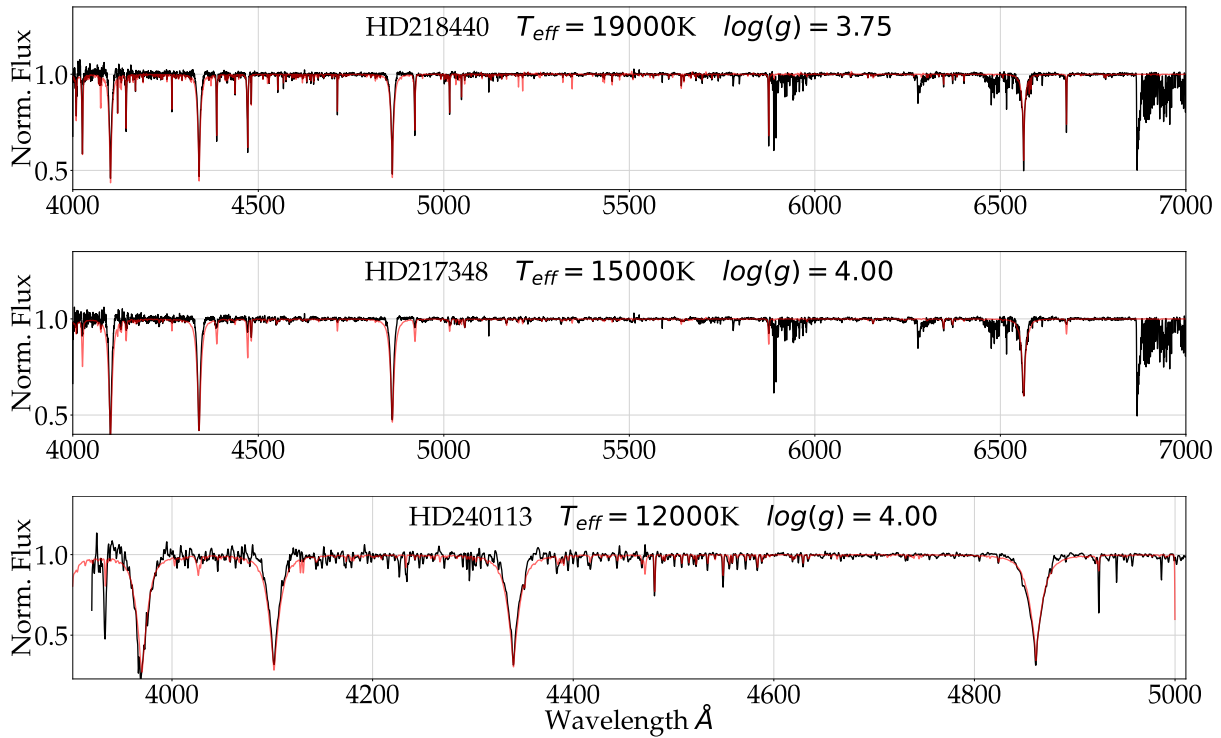


Figure 4.13: Spectra of HD 218440, HD 217348 and the runaway HD240113 plotted with their best matching templates. KURUCZ templates have been used for the HD 240113 while TLUSTY have been used for the others.

The best fit models from TLUSTY for the spectra are determined using the Fourier cross template spectra with `fxcor` command in IRAF. The selected models have the

Table 4.10: Photometric properties of selected members of RSG 8 and the runaway candidate HD 240113 (shown in bold font)

Gaia DR3 Source	r_{geo} pc	G mag	μ_{α}^* mas/yr	μ_{δ} mas/yr	$G_{\text{BP}} - G_{\text{RP}}$ mag
2013579250758906240	461^{+5}_{-7}	6.373 ± 0.003	5.75 ± 0.03	-1.96 ± 0.03	0.004 ± 0.007
2014377526259898240	413^{+4}_{-3}	6.423 ± 0.003	4.75 ± 0.03	-2.24 ± 0.03	0.087 ± 0.007
2014431578426474112	497^{+12}_{-11}	6.707 ± 0.003	5.70 ± 0.05	-1.78 ± 0.04	-0.014 ± 0.007
2013290835115216128	481^{+5}_{-6}	8.234 ± 0.003	5.46 ± 0.03	-1.67 ± 0.03	0.047 ± 0.007
2008292283458395392	424^{+2}_{-3}	8.547 ± 0.003	4.79 ± 0.02	-2.25 ± 0.02	0.196 ± 0.007
2014330285916266624	417^{+3}_{-2}	9.441 ± 0.003	4.73 ± 0.01	-2.14 ± 0.01	0.197 ± 0.007
2014453774818090880	431^{+3}_{-3}	9.728 ± 0.003	4.72 ± 0.02	-2.34 ± 0.01	0.319 ± 0.007
2013261285739804800	481^{+4}_{-6}	9.896 ± 0.003	5.81 ± 0.02	-1.69 ± 0.02	0.116 ± 0.007
2013560215462986368	487^{+3}_{-3}	10.109 ± 0.003	5.49 ± 0.01	-1.64 ± 0.01	0.161 ± 0.007
2014350592522171008	418^{+2}_{-2}	10.623 ± 0.003	4.63 ± 0.01	-2.27 ± 0.01	0.378 ± 0.007
HD 240113	428^{+3}_{-3}	9.061 ± 0.003	-8.51 ± 0.02	-6.73 ± 0.01	0.210 ± 0.007

Table 4.11: Spectroscopic parameters of HD 218440, HD 217348 and HD 240113

Star	T_{iso} (K)	T_{eff} (K)	$\log g$ (dex)	v_r (km/s)	SpT	SpT (lit.)
HD 218440	17960^{+810}_{-570}	23000 ± 1000	3.75 ± 0.25	4.0 ± 1.8	B1V	B2V
HD 217348	14410^{+1280}_{-1090}	15000 ± 1000	3.50 ± 0.25	-8.6 ± 2.8	B5III	B7III
HD 240113	10830^{+630}_{-330}	12000 ± 250	4.00 ± 0.25	-11.2 ± 7.6	B8IV	A0

correlation heights. These are tabulated in table 4.11 along with the spectral types from the literature.

The runaway star candidate is HD 240113 which was also observed. Its spectrum along with a model fit is shown in 4.13. It was found that this star has a temperature of about 12000K which makes it a B8 type star. Kurucz models were used for the correlation of HD 240113. The fit parameters are tabulated in 4.11.

The theoretical isochrone from the parameters mentioned in Dias et al. (2021) fits the cluster well, however the value of the $\log \tau$ had to be changed from 7.57 to 7.7 since this fit the observed temperatures and $\log g$ values of the stars better. This isochrone is plotted for the stars in the region around RSG 8 in 4.14. HD218440 and HD217348 are marked in the CMD along with the runaway star candidate with its temperature estimate. Figure 4.14 also shows the proper motion diagram on the right indicating the runaway being an outlier compared to the proper motion of the cluster. Since HD 218440 is a known spectroscopic binary and HD 217348 has larger deviations from the theoretical isochrone predictions, their radial velocity values are not included in the estimate for the radial velocity of the cluster.

The masses for HD 218440, HD217348 and HD240113 estimated from the isochrone are: $6.771 M_{\odot}$, $3.798 M_{\odot}$, and $2.399 M_{\odot}$ while $\log g$ are 3.615, 4.24, and 4.343 respectively.

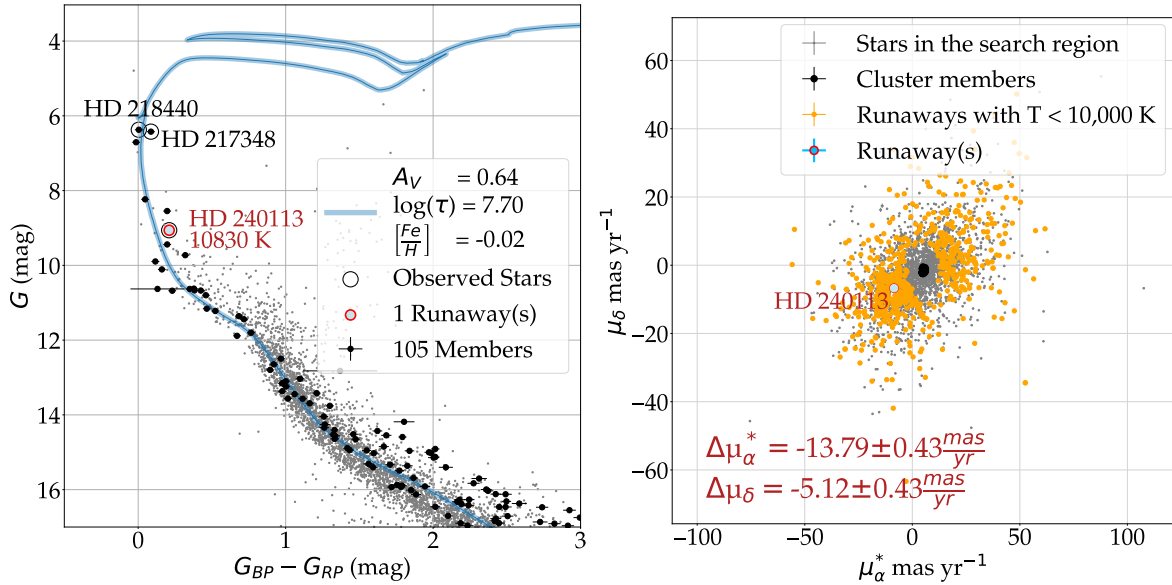


Figure 4.14: Left: CMD fit for RSG 8 showing the runaway and its temperature estimate along with the observed stars.

Right: Plot showing the high proper motion of HD240113 compared to the cluster. The orange scatter points indicate other high transverse velocity stars which could be traced back to the cluster but were not hot enough according to the isochrone fit.

Traceback

The traceback plot for the cluster is shown in 4.12. The runaway star is moving in the south-west direction with a velocity 29.8 km/s. Combined with the radial velocity relative to the cluster of ~ 2 km/s, the 3D space velocity is ~ 30 km/s with respect to the cluster, which makes it a runaway star.

No known pulsars are found within 70 pc from cluster edge and in 40% radial distance tolerance around the cluster distance of 0.441 kpc. While 36 Pulsars within 10.01 degrees (70 pc), none lie in 0.26 kpc to 0.62 kpc which can be associated with the runaway star candidate.

Conclusions

In this work, a comprehensive analysis of young open star clusters has been conducted to identify potential runaway star candidates produced through the Binary Supernova Scenario (BSS). By tracing back the motion of stars around 399 young open clusters using *Gaia* DR3 data, several candidates for runaway stars were identified. New runaway stars have been found and associated with their possible host clusters, and some have been shown to have possible connections to nearby pulsars. These findings provide evidence for BSS events in the past, suggesting that these clusters might have experienced supernova explosions that resulted in the ejection of a companion star as a runaway.

As a byproduct of this study, updated parameters for the candidate runaway stars and their associated clusters have been produced. This includes improved estimates of cluster distances, ages, and other relevant characteristics, enhancing our understanding of the dynamical processes within these star clusters.

Valuable spectroscopic information has been obtained about the 52 stars and their parameters have been derived.

The Python code developed during this work has been made available online. This would allow one to reproduce the results and also to improve upon them when newer data is available.

Future Prospects

The results of this thesis open up several avenues for future research. First, the identified runaway star-pulsar pairs present promising targets for detailed observational follow-up to search for previously undetected supernova remnants (SNRs) in their vicinity. Given the challenges of detecting “invisible” SNRs, these pairs may be key to uncovering remnants that have dissipated faster than usual or in low-density regions of the interstellar medium (ISM).

Furthermore, refining the search for runaway stars in other young open clusters

across the Milky Way, using more detailed astrometric data from future *Gaia* data releases or other missions, could further improve our understanding of the BSS mechanism and its role in shaping the population of runaway stars. Additionally, studying the relationship between the runaway stars and their potential neutron star companions could yield insights into the evolution of binary systems and the final stages of massive star evolution.

Overall, the identification of these runaway-neutron star pairs not only contributes to the study of supernova remnants but also enhances our broader understanding of stellar dynamics and the life cycles of massive stars in our galaxy.

Acknowledgements

The completion of this work has been possible due to the immense support from my supervisors, Prof. Dr. Ralph Neuhäuser and Dr. Baha Dinçel. Their expertise in the field has been invaluable, providing me with insightful guidance and constructive feedback throughout this journey. I am deeply grateful for their encouragement, patience, and unwavering belief in my abilities, which have been instrumental in the successful completion of this project. I have acquired a plethora of knowledge from the work we did together and from the fruitful conversations we had throughout the years.

I would like to thank my excellent friends, Sebastian Alexander Hüttel and Luca Cortese, with whom I have spent the best of times. I cherish every moment in our offices at the institute, working together and spending time with each other. Their comments and suggestions were vital for shaping my work, and their constant support and companionship made this journey more enjoyable and fulfilling. I am truly grateful for their friendship and the many unforgettable memories we have shared.

I would like to thank AIU Jena for providing me with the useful data from GSH [Mugrauer et al. \(2014\)](#) and infrastructure for the completion of this work and for valuable experience in astronomy.

This research has made use of the VizieR catalogue access tool, CDS, Strasbourg, France ([Ochsenbein, 1996](#)). The original description of the VizieR service was published in [Ochsenbein et al. \(2000\)](#). This research has also made use of the SIMBAD database, operated at CDS, Strasbourg, France [Wenger et al. \(2000\)](#).

I also acknowledge the use of Astropy:¹ a community-developed core Python package and an ecosystem of tools and resources for astronomy ([Astropy Collaboration et al., 2013, 2018, 2022](#)) along with its affiliated packages.

I would also like to thank my parents for providing me with the best possible support, even after being a continent apart. None of what I have achieved so far in life would be possible without you.

¹<http://www.astropy.org>

Appendix A

Spectroscopic observations

A.1 Observed stars and their parameters

Table A.1: List of observed stars

Cluster	Star	α (deg)	δ (deg)	Obs. date (UTC)	G (mag)	Exp. (s)
ASCC 127	HD218363	346.577	64.869	2024-01-09T20:05	9.01	1200
ASCC 127	HD218695	347.256	65.193	2024-01-10T04:21	8.70	900
ASCC 127	BD+631957	348.937	64.391	2024-01-10T20:37	9.12	1200
ASCC 16	HD35192A	80.773	1.057	2024-01-19T21:49	7.70	600
ASCC 19	HD36352	82.733	-2.699	2023-10-18T03:05	9.18	1200
Basel 8	HD259597	98.384	8.336	2024-01-10T02:58	8.43	900
Basel 8	HD259787	98.550	7.988	2024-01-10T23:52	9.74	1200
Basel 8*	HD259991	98.691	8.076	2024-01-09T22:36	9.32	1200
Berkeley 97	HD240015	339.824	59.017	2023-08-18T22:12	9.84	2400
Berkeley 97	HD240016	339.844	59.007	2023-09-07T20:11	9.47	2400
Biurakan 2	HD191495	302.223	35.513	2023-09-04T01:49	8.36	900
Biurakan 2	HD191612	302.369	35.734	2023-08-22T01:50	7.71	600
Dolidze 3	HD228602	303.894	36.832	2023-09-14T22:38	9.70	1200
FSR 0498	LSI+62110	7.269	62.416	2023-10-18T00:51	10.23	2400
FSR 0198	HD190429	300.872	36.025	2023-08-23T01:08	7.00	600
Gulliver 6	HD36646	83.281	-1.717	2024-01-08T20:32	6.50	300
IC 1396	HD206183	324.610	56.974	2023-08-22T02:23	7.40	600
IC 1396	LSIII+5712	324.736	57.494	2023-08-22T02:57	7.97	600
IC 1396	HD206773	325.601	57.736	2023-08-22T00:39	6.73	300

Table A.1 continued

Cluster	Star	α (deg)	δ (deg)	Obs. date (UTC)	G (mag)	Exp. (s)
IC 1396	BD+463471	328.142	47.229	2023-09-15T00:53	10.01	2400
IC 1396	BD+463474	328.370	47.267	2023-09-14T23:48	9.40	1200
LP 1821	HD217943	345.849	60.445	2023-08-22T01:14	6.71	300
NGC 1980	HD36959	83.754	-6.009	2024-01-19T22:22	5.68	300
NGC 1980	HD37000	83.796	-5.927	2024-01-19T22:40	7.42	600
NGC 6871	HD190919	301.484	35.672	2023-08-23T01:41	7.21	600
NGC 6871	BD+353955	301.495	35.797	2023-08-23T23:38	7.29	600
NGC 869	61And	34.769	57.135	2023-09-07T04:08	6.45	300
NGC 869	HD14143	34.808	57.169	2023-09-07T04:30	6.51	300
NGC 869	BD+56502	34.637	57.211	2023-09-19T22:17	9.23	1200
Pismis Moreno 1	BD+622060	334.604	63.220	2023-09-04T03:13	8.51	900
Pismis Moreno 1	HD211880	334.616	63.223	2023-08-22T03:29	7.66	600
RSG 8	HD218440	346.794	59.727	2023-08-22T01:31	6.37	300
RSG 8	HD217348	344.788	59.815	2023-08-22T00:56	6.42	300
RSG 8*	HD 240113	343.692	59.279	2024-04-23T01:46	9.06	2400
SAI 4	BD+6164	6.041	62.623	2023-09-12T03:38	9.42	1200

Table A.2: List of parameters T_{eff} and $\log g$ derived using stellar atmosphere models. The model used are indicated using k for Kurucz and t for TLUSTY. Kurucz models have errors of $\pm 250\text{K}$ in temperature below 13000K and $\pm 1000\text{K}$ above that, and ± 0.5 in $\log g$ while TLUSTY models have $\pm 1000\text{K}$ in temperature and ± 0.25 in $\log g$

Cluster	Star	T_{eff} (K)	$\log g$ (dex)	v_r (km/s)	cor. height	model
ASCC 127	HD218363	11000	2.50	-8.19 ± 3.55	0.71	k
ASCC 127	HD218695	12000	3.00	-14.49 ± 5.49	0.97	k
ASCC 127	BD+631957	12000	3.00	-16.35 ± 3.06	0.98	k
ASCC 16	HD35192A	10500	3.00	-19.05 ± 10.85	0.87	k
ASCC 19	HD36352	10000	3.00	14.6 ± 5.27	0.83	k
Basel 8	HD259597	13000	3.00	39.78 ± 8.15	0.81	k
Basel 8	HD259787	13000	3.50	21.03 ± 5.47	0.96	k
Basel 8*	HD259991	15000	3.00	1.90 ± 7.03	0.88	t
Berkeley 97	HD240015	26000	2.75	-45.75 ± 4.20	0.85	t
Berkeley 97	HD240016	21000	3.00	-33.46 ± 3.51	0.91	t

Table A.2 continued

Cluster	Star	T_{eff} (K)	$\log g$ (dex)	v_r (km/s)	cor. height	model
Biurakan 2	HD191495	30000	3.25	-38.32 ± 7.81	0.83	t
Biurakan 2	HD191612	30000	3.50	-33.02 ± 6.54	0.84	t
Dolidze 3	HD228602	30000	3.50	-7.51 ± 7.20	0.74	t
FSR 0498	LSI+62110	30000	3.50	-65.18 ± 12.25	0.65	t
FSR 0198	HD190429	24000	2.50	-35.78 ± 4.45	0.93	t
Gulliver 6	HD36646	24000	3.75	-2.615 ± 12.38	0.78	t
IC 1396	HD206183	30000	3.50	-12.60 ± 5.32	0.91	t
IC 1396	LSIII+5712	30000	3.50	-14.25 ± 4.98	0.79	t
IC 1396	HD206773	30000	3.50	-15.05 ± 2.49	0.82	t
IC 5146	BD+463471	11000	3.00	-94.16 ± 14.02	0.84	k
IC 5146	BD+463474	10500	2.50	-18.78 ± 12.34	0.91	k
LP 1821	HD217943	23000	3.00	-31.09 ± 7.74	0.82	t
NGC 1980	HD36959	23000	3.25	27.17 ± 2.57	0.80	t
NGC 1980	HD37000	25000	3.75	46.09 ± 8.47	0.71	t
NGC 6871	HD190919	26000	3.00	-15.29 ± 4.075	0.74	t
NGC 6871	BD+353955	30000	3.00	-9.74 ± 4.07	0.81	t
NGC 869	61And	18000	3.00	-50.91 ± 3.67	0.82	t
NGC 869	HD14143	16000	2.00	-47.39 ± 5.65	0.71	t
NGC 869	BD+56502	25000	3.00	-35.72 ± 6.61	0.79	t
Pismis Moreno 1	BD+622060	14000	3.00	-25.51 ± 10.33	0.94	k
Pismis Moreno 1	HD211880	29000	3.75	-16.74 ± 4.38	0.87	t
RSG 8	HD218440	23000	3.25	-12.61 ± 2.96	0.85	t
RSG 8	HD217348	15000	3.25	-20.52 ± 4.01	0.91	t
RSG 8*	HD 240113	12000	4.00	-8.16 ± 9.32	0.95	k
SAI 4	BD+6164	30000	4.25	-96.27 ± 7.81	0.79	t

A.2 Data Reduction

Dark subtraction

Darks are the images which are obtained when the telescope (or more specifically, here spectrograph) is exposed to no light at all. These have the same exposure time as the science images. Short-exposure darks are also obtained for Arc images. Darks capture

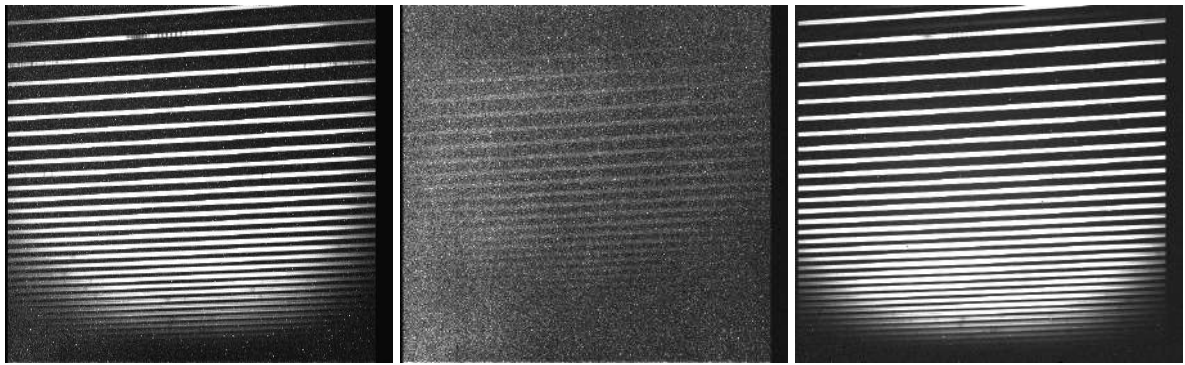


Figure A.1:

Left: Science image obtained for the spectrum of star HD 190919 using FLECHAS with 600s exposure.

Middle: Dark image with 600s exposure. This is subtracted from the left image.

Right: Science image after subtracting the Dark.

the artefacts from the CCD readouts. A typical dark image is shown in fig. A.1, along with the before and after image of the result of subtracting dark.

Tracing the 29 apertures

The spectrum is divided into 29 apertures. These are traced from the skyflat image using the `apall` command in IRAF. This reference is used further to trace the 29 apertures in arcs and flats. The before and after images of the tracing process are shown in figure A.2. A vignetting effect is seen at the corners, these are removed by dividing by the flat.

Flat division

The purpose of flat corrections is to address any unevenness in the CCD's response to light. This non-uniformity can arise from various factors, such as differences in pixel sensitivity across the detector, dust on the filter or the glass window covering the detector, vignetting, or dimming in the corners of the image. These artefacts are removed from the science image by dividing by the flat image. The result is shown in section A.2.

Wavelength calibration

The pixel values are now assigned wavelengths and fluxes to obtain a 1-dimensional spectrum for the star. This process involves comparing to a standard star to identify the lines in the spectrum and combining the 29 apertures. Each of the 29 apertures

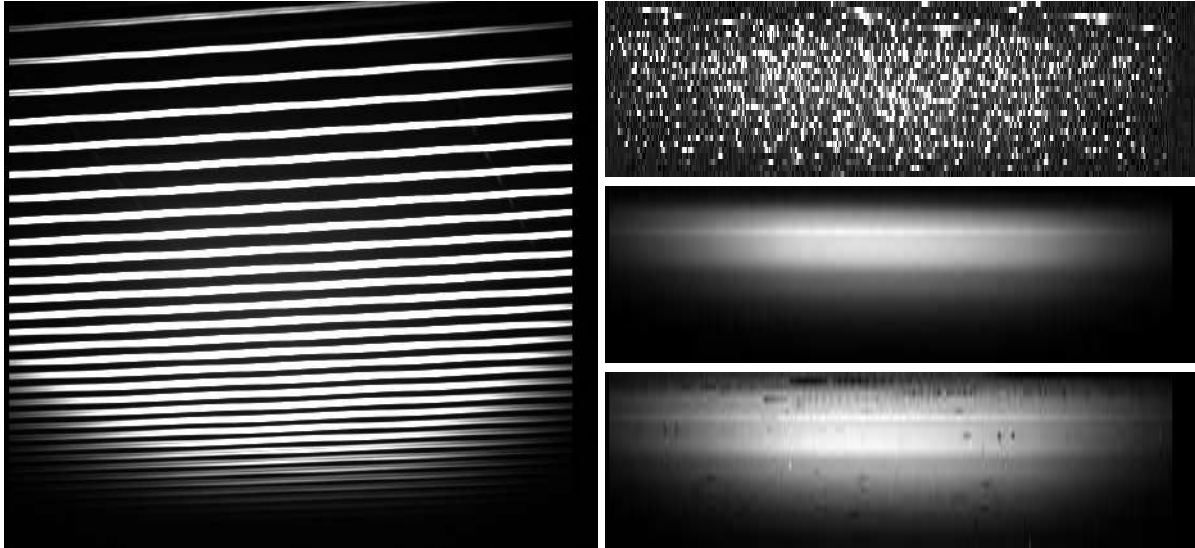


Figure A.2:

Left: Flat image containing the 29 apertures. Note the uneven spacing between the lines. These are traced and stacked in the right image.

Right: From top to bottom is the arc, flat and science images with their 29 apertures stacked in horizontal lines. Note the vignetting effect at the corners.

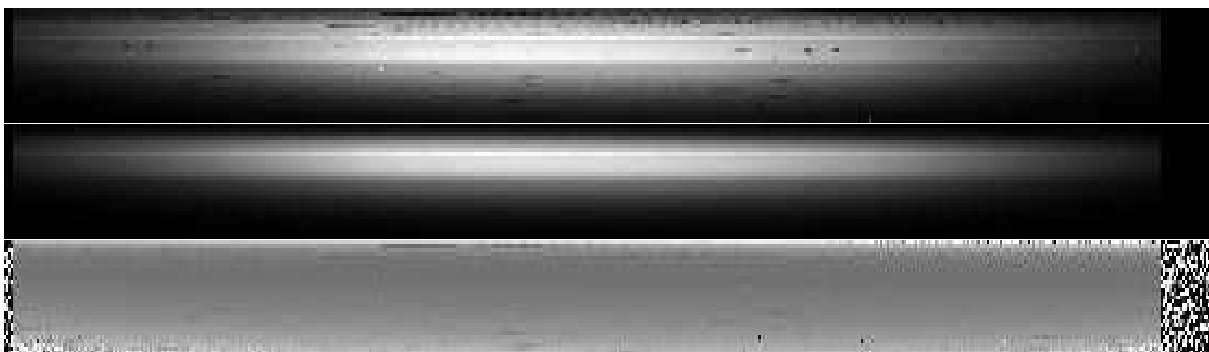


Figure A.3:

Top: Science image without flat division

Middle: Flat image

Bottom: Science image after flat division

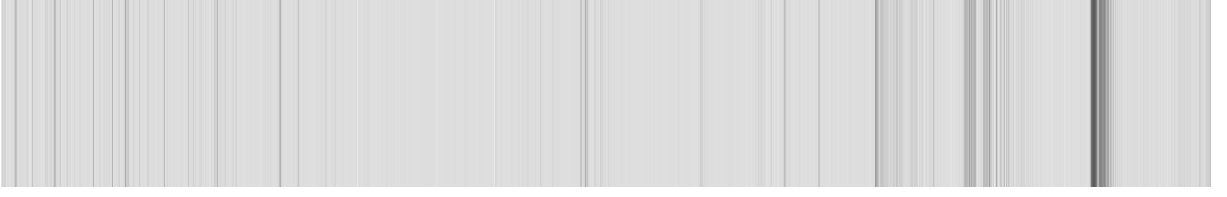


Figure A.4: Final spectrum of the star.

is continuum subtracted by polynomial fitting. After subtracting the continuum and stacking, the final spectrum form looks like figure fig. A.4

A.3 HR Diagrams

One of the most important plots in all of astronomy is the Hertzsprung-Russell (HR) diagram, also known as a color-magnitude diagram (CMD). This is a plot of the brightness or luminosity vs temperature of a sample of stars. A typical HR diagram looks as follows:

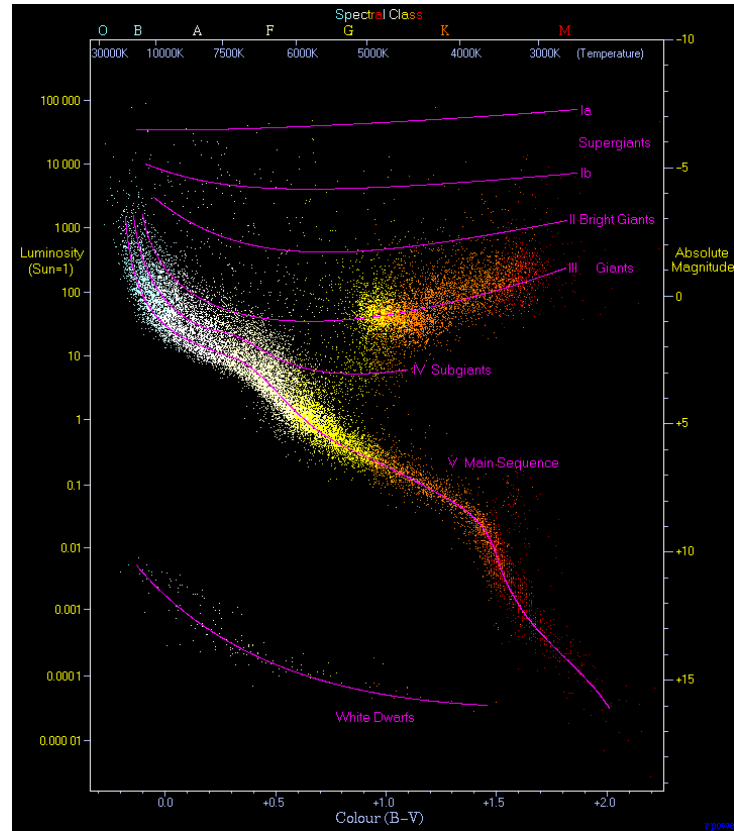


Figure A.5: An observational Hertzsprung–Russell diagram with 22,000 stars plotted from the Hipparcos Catalogue and 1,000 from the Gliese Catalogue of nearby stars. Credit: [Richard Powell](#), “The Hertzsprung Russell Diagram.” Licensed under [CC BY-SA 2.5](#).

Depending on the temperature, the spectral class is determined. The hottest being O, corresponding to temperatures $> 30,000\text{K}$ and the coolest being M, corresponding to a temperature of 3000 K . A handy mnemonic to remember the order of the spectral classes from hottest to coolest is: *“Only Bad Astronomers Forget Generally Known Mnemonics”* These are further subdivided its spectral class. It is seen that stars tend to clump in certain combinations of brightness and temperature. The location of a star in this diagram gives us information about its evolutionary stage, along with its brightness and temperature. Therefore, stars are classified into the following spectral classes depending on their location on the HR diagram.

The Main sequence (V)

After stars are fully formed and start ignition of hydrogen, they enter the main sequence. In the main sequence, they keep burning hydrogen until the star runs out of hydrogen in the core. The Sun is an example of a main sequence star. Towards the right bottom are the stars which are cooler (appearing red) and lower in luminosity. Towards the top left lie the stars which are hot (appearing bluer) and bright. Depending on the mass of the star, the time for which a star stays on the main sequence, known as the main sequence lifetime varies. [A.3](#) shows the main sequence lifetimes of stars from different spectral classes.

Table A.3: Main-sequence spectral class properties

Sp. Cl.	Mass $M_{\odot}$	Luminosity $L_{\odot}$	Temp. (K)	Radius $R_{\odot}$	MS Lifetime (Myr)
O5	40	400,000	40,000	13	1.0
B0	15	13,000	28,000	4.9	11
A0	3.5	80	10,000	3.0	440
F0	1.7	6.4	7,500	1.5	2,700
G0	1.1	1.4	6,000	1.1	8,000
K0	0.8	0.4	5,000	0.9	17,000
M0	0.5	0.08	3,500	0.8	56,000

After the main fuel is exhausted, the stars leave the main sequence and move towards the giant branches. This point is called the main sequence turnoff point. This can be clearly seen in older clusters as shown in [fig. 1.4](#).

Subgiant Branch (IV)

Once a star exhausts the hydrogen in its core, nuclear fusion can no longer sustain the core pressure against gravitational collapse. As a result, the core begins to contract

under its gravity, causing it to heat up. This increased core temperature causes the outer layers of the star to expand and cool, moving the star off the main sequence and into the subgiant branch.

Subgiant stars are characterized by their relatively low luminosity compared to giants and supergiants, but they are still brighter than main-sequence stars of the same spectral type. These stars are typically still fusing hydrogen in a shell around the helium core. The temperature of the star's surface decreases as it expands, leading to a cooler and redder appearance.

Giant Branch (III)

As the core continues to contract and heat up, hydrogen fusion in the surrounding shell intensifies, causing the outer layers of the star to expand even further. This leads the star into the giant branch, where it becomes significantly more luminous and larger in size than it was as a subgiant.

Stars on the giant branch are much cooler on the surface than when they were main sequence stars, giving them a reddish hue. Their radii can expand to tens or even hundreds of times the Sun's radius. The core eventually becomes hot enough to ignite helium fusion, which temporarily halts the core contraction and can cause the star to stabilize as a red giant.

Bright Giant Branch (II)

Bright giants represent an intermediate evolutionary stage between giants and supergiants. These stars have expanded significantly and possess high luminosity, although not as extreme as that of supergiants. They are generally cooler, with surface temperatures lower than those of main sequence stars of the same spectral type.

Bright giants have exhausted the hydrogen in their cores and are usually in a phase of helium burning or shell hydrogen burning. The expansion of their outer layers leads to a further increase in luminosity, but the core remains relatively small and dense.

Supergiant Branch (I)

Supergiants are among the most massive and luminous stars in the universe. These stars have left the main sequence after exhausting the hydrogen in their cores and have expanded to enormous sizes, sometimes exceeding hundreds of solar radii. Supergiants can vary in temperature, with some appearing blue and others red, depending on their mass and stage of evolution.

Supergiants undergo complex processes in their cores, with layers of different fusion occurring simultaneously. They have very short lifespans relative to other stars, typically only a few million years, as they rapidly burn through their nuclear fuel. Due to their immense mass and energy output, supergiants are key contributors to the chemical enrichment of the galaxy, often ending their lives in spectacular supernova explosions.

A.4 Cluster Sample

The 399 clusters selected for this work are tabulated below. These contain the original parameters from Dias et al. (2021). These parameters have been updated for some clusters along with isochrones to fit the updated parameters.

Refer to <https://github.com/surodeep26/Runaway-Stars> for the updates.

Table A.4: Selected clusters from the Dias catalogue

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	$[Fe/H]$	A_V (mag)
ASCC 107	297.16	22.01	9.4	59	864 ± 30	7.4 ± 0.1	0.35 ± 0.10	1.4 ± 0.1
ASCC 114	324.98	54.00	10.8	149	911 ± 12	7.6 ± 0.3	0.04 ± 0.08	1.2 ± 0.1
ASCC 127	347.18	64.92	32.5	113	365 ± 10	7.5 ± 0.1	0.15 ± 0.12	0.7 ± 0.1
ASCC 13	78.31	44.42	33.8	110	1066 ± 26	7.6 ± 0.1	-0.08 ± 0.08	0.9 ± 0.0
ASCC 16	81.20	1.63	22.0	207	348 ± 3	7.1 ± 0.1	-0.06 ± 0.07	0.2 ± 0.0
ASCC 19	82.00	-1.96	36.8	173	356 ± 2	7.1 ± 0.0	0.08 ± 0.08	0.2 ± 0.0
ASCC 21	82.14	3.48	25.1	102	343 ± 5	7.1 ± 0.0	-0.01 ± 0.03	0.2 ± 0.0
ASCC 32	105.71	-26.58	38.8	255	792 ± 11	7.4 ± 0.0	0.00 ± 0.05	0.2 ± 0.0
ASCC 67	175.29	-60.99	9.9	46	1921 ± 89	7.5 ± 0.2	0.22 ± 0.10	0.8 ± 0.0
ASCC 79	229.69	-60.81	49.1	128	831 ± 4	7.4 ± 0.2	0.04 ± 0.08	0.7 ± 0.1
ASCC 9	41.68	57.76	11.0	88	2322 ± 99	6.9 ± 0.1	-0.16 ± 0.08	2.6 ± 0.0
Alessi 20	2.59	58.74	12.6	110	423 ± 2	7.0 ± 0.1	-0.10 ± 0.14	1.0 ± 0.1
Alessi 43	132.64	-41.76	16.0	283	937 ± 4	6.8 ± 0.1	-0.05 ± 0.08	0.9 ± 0.1
Alessi Teutsch 5	332.27	61.10	20.2	147	869 ± 14	7.1 ± 0.1	-0.05 ± 0.10	1.3 ± 0.1
Antalova 2	262.47	-32.52	4.9	28	1079 ± 36	6.9 ± 0.2	0.01 ± 0.09	1.2 ± 0.4
Archinal 1	283.72	5.56	2.7	54	1741 ± 35	7.0 ± 0.1	0.12 ± 0.11	4.6 ± 0.1
Aveni Hunter 1	354.19	48.15	30.8	73	420 ± 3	7.0 ± 0.1	-0.14 ± 0.13	0.5 ± 0.1
BDSB91	96.33	-10.55	2.0	26	912 ± 27	6.7 ± 0.0	0.01 ± 0.07	1.5 ± 0.2
BDSB93	104.30	-8.21	1.7	27	1022 ± 177	7.1 ± 0.4	-0.11 ± 0.12	2.2 ± 0.5
BDSB96	106.35	-12.32	6.0	45	1155 ± 84	6.9 ± 0.1	-0.14 ± 0.16	1.5 ± 0.5
BH 121	174.66	-63.38	7.9	171	2250 ± 89	6.9 ± 0.1	-0.09 ± 0.11	1.3 ± 0.0
BH 164	222.23	-66.44	28.9	196	416 ± 2	7.7 ± 0.1	0.13 ± 0.05	0.4 ± 0.0
BH 200	252.48	-44.18	3.2	111	1997 ± 121	7.5 ± 0.3	0.12 ± 0.09	2.6 ± 0.0

Table A.4 continued

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	$[Fe/H]$	A_V (mag)
BH 205	254.06	-40.64	5.5	82	1612 ± 79	6.9 ± 0.1	0.11 ± 0.14	1.4 ± 0.0
BH 217	259.08	-40.81	2.2	119	2133 ± 306	7.1 ± 0.6	0.24 ± 0.11	3.3 ± 0.1
BH 221	259.65	-32.31	8.6	124	1050 ± 53	7.1 ± 0.1	0.00 ± 0.08	1.0 ± 0.0
BH 23	123.51	-36.24	23.3	78	436 ± 1	7.4 ± 0.1	-0.05 ± 0.05	0.2 ± 0.0
BH 245	266.56	-29.70	1.0	40	2637 ± 523	7.1 ± 1.4	0.12 ± 0.16	4.7 ± 0.7
BH 54	132.39	-44.35	2.7	49	2121 ± 128	6.8 ± 0.1	0.31 ± 0.19	2.6 ± 0.0
BH 56	134.36	-43.16	13.9	116	865 ± 23	7.2 ± 0.1	-0.02 ± 0.08	0.8 ± 0.2
BH 87	151.16	-55.38	3.5	194	1899 ± 80	7.4 ± 0.0	0.03 ± 0.12	1.8 ± 0.1
BH 92	154.78	-56.43	1.0	42	2254 ± 115	7.5 ± 0.4	-0.10 ± 0.13	1.9 ± 0.1
Barkhatova 1	313.41	46.05	4.3	57	1859 ± 74	7.1 ± 0.1	0.05 ± 0.18	1.8 ± 0.1
Basel 18	201.98	-62.31	3.0	65	1797 ± 100	7.3 ± 0.1	0.24 ± 0.07	1.1 ± 0.0
Basel 8	98.56	8.04	17.9	31	1456 ± 54	7.7 ± 0.2	0.03 ± 0.06	1.1 ± 0.1
Berkeley 11	65.13	44.92	2.1	89	2653 ± 220	7.1 ± 0.5	-0.06 ± 0.13	2.7 ± 0.1
Berkeley 15	75.51	44.50	3.4	136	2101 ± 170	7.2 ± 0.4	-0.18 ± 0.09	2.8 ± 0.1
Berkeley 47	292.13	17.36	1.7	59	1974 ± 217	7.1 ± 0.2	0.14 ± 0.13	3.6 ± 0.0
Berkeley 59	8.65	67.46	185.8	223	1026 ± 26	6.8 ± 0.1	0.00 ± 0.10	3.8 ± 0.2
Berkeley 62	15.31	63.92	4.7	112	2496 ± 125	7.1 ± 0.0	-0.10 ± 0.08	2.5 ± 0.1
Berkeley 65	39.76	60.41	1.4	45	2208 ± 168	7.0 ± 0.1	-0.16 ± 0.10	2.9 ± 0.1
Berkeley 7	28.55	62.37	2.3	103	2392 ± 240	7.6 ± 0.1	0.09 ± 0.26	2.1 ± 0.0
Berkeley 79	281.24	-1.15	3.1	74	2190 ± 209	7.0 ± 0.4	0.02 ± 0.12	3.0 ± 0.1
Berkeley 86	305.10	38.71	4.3	31	1719 ± 102	6.8 ± 0.1	0.18 ± 0.31	2.6 ± 0.2
Berkeley 87	305.39	37.44	5.6	130	1708 ± 104	6.8 ± 0.1	-0.23 ± 0.41	3.8 ± 0.0
Berkeley 97	339.85	59.00	1.9	89	2485 ± 110	7.6 ± 0.4	-0.10 ± 0.03	2.5 ± 0.0
Bica 2	308.32	41.31	3.6	140	1550 ± 85	6.7 ± 0.0	0.10 ± 0.15	4.1 ± 0.2
Biurakan 2	302.29	35.50	8.6	64	1638 ± 126	6.9 ± 0.1	0.15 ± 0.12	1.4 ± 0.0
Bochum 10	160.50	-59.13	8.9	264	2365 ± 19	7.2 ± 0.1	0.18 ± 0.08	1.2 ± 0.1
Bochum 11	161.76	-60.10	2.5	41	2379 ± 146	6.7 ± 0.1	0.21 ± 0.25	1.8 ± 0.1
Bochum 13	259.32	-35.53	6.0	59	1611 ± 61	7.0 ± 0.0	0.30 ± 0.06	2.5 ± 0.0
COIN-Gaia 16	80.17	37.42	3.7	43	1550 ± 84	7.3 ± 0.2	-0.05 ± 0.07	1.1 ± 0.1
COIN-Gaia 21	84.76	28.39	6.4	35	1222 ± 72	7.4 ± 0.3	0.00 ± 0.11	1.5 ± 0.1
COIN-Gaia 35	33.47	60.04	2.2	27	2354 ± 290	7.0 ± 0.6	-0.14 ± 0.17	2.7 ± 0.1
COIN-Gaia 37	45.00	58.44	40.9	25	956 ± 24	7.4 ± 0.1	0.06 ± 0.15	1.9 ± 0.1
COIN-Gaia 41	89.83	19.03	8.7	74	1764 ± 181	7.1 ± 0.2	-0.07 ± 0.22	2.2 ± 0.1
Collinder 104	99.16	4.82	8.6	179	1599 ± 35	7.2 ± 0.1	-0.10 ± 0.08	2.1 ± 0.2
Collinder 106	99.20	5.93	15.5	110	1355 ± 94	7.2 ± 0.0	-0.06 ± 0.12	0.9 ± 0.1
Collinder 107	99.23	4.97	23.4	157	1466 ± 85	7.2 ± 0.1	-0.10 ± 0.16	1.5 ± 0.1
Collinder 132	108.60	-30.70	77.3	98	648 ± 11	7.5 ± 0.2	0.04 ± 0.05	0.2 ± 0.0
Collinder 135	109.23	-37.10	65.3	325	307 ± 8	7.7 ± 0.0	0.11 ± 0.06	0.2 ± 0.1
Collinder 140	110.85	-32.00	33.2	142	377 ± 8	7.6 ± 0.0	0.01 ± 0.05	0.1 ± 0.0
Collinder 197	131.22	-41.29	8.5	213	833 ± 64	7.0 ± 0.1	0.07 ± 0.20	1.9 ± 0.2
Collinder 205	135.09	-48.98	2.9	99	1400 ± 224	7.0 ± 0.2	-0.17 ± 0.17	2.5 ± 0.1

Appendix A. Spectroscopic observations

Table A.4 continued

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	$[Fe/H]$	A_V (mag)
Collinder 272	202.58	-61.32	6.1	285	1885 ± 98	7.4 ± 0.1	-0.05 ± 0.14	1.5 ± 0.0
Collinder 419	304.53	40.75	3.5	41	1000 ± 44	6.8 ± 0.1	0.09 ± 0.15	1.1 ± 0.1
Collinder 421	305.84	41.70	7.4	161	1141 ± 28	7.5 ± 0.4	-0.06 ± 0.09	2.5 ± 0.1
Collinder 69	83.77	9.85	60.7	532	400 ± 1	6.9 ± 0.0	-0.01 ± 0.01	0.4 ± 0.0
Collinder 95	97.79	9.91	10.4	103	676 ± 14	6.8 ± 0.1	-0.04 ± 0.02	0.9 ± 0.3
Czernik 1	1.91	61.48	1.3	39	2624 ± 168	7.3 ± 0.4	-0.09 ± 0.06	2.3 ± 0.1
Czernik 31	114.23	-20.51	2.5	88	2731 ± 162	7.4 ± 0.3	-0.11 ± 0.06	1.7 ± 0.1
Czernik 41	297.76	25.29	4.7	227	2620 ± 212	6.8 ± 0.1	0.00 ± 0.11	3.8 ± 0.0
Czernik 6	30.53	62.84	2.8	26	2280 ± 278	7.3 ± 0.2	-0.02 ± 0.13	1.5 ± 0.1
Czernik 8	38.24	58.76	1.4	37	2699 ± 330	7.4 ± 0.5	-0.03 ± 0.17	3.0 ± 0.1
Dias 1	10.64	64.07	2.9	62	2627 ± 127	7.0 ± 0.1	-0.19 ± 0.11	3.0 ± 0.0
Dias 5	274.36	-19.70	4.9	67	1257 ± 65	6.9 ± 0.1	0.22 ± 0.11	1.3 ± 0.1
Dolidze 16	78.66	32.69	7.9	49	1256 ± 63	7.1 ± 0.1	-0.09 ± 0.15	1.3 ± 0.1
Dolidze 3	303.80	36.83	5.8	36	1907 ± 103	6.9 ± 0.1	-0.12 ± 0.18	2.2 ± 0.1
Dolidze 32	280.27	-4.05	5.7	35	1991 ± 158	7.1 ± 0.6	0.04 ± 0.12	3.1 ± 0.1
Dolidze 5	305.02	39.32	4.8	143	2005 ± 123	7.4 ± 0.7	-0.03 ± 0.39	2.5 ± 0.2
Dolidze 53	296.59	24.61	2.3	39	1984 ± 222	6.8 ± 0.1	0.05 ± 0.18	2.7 ± 0.1
Dolidze 8	306.12	42.30	5.5	33	942 ± 27	6.7 ± 0.0	0.37 ± 0.10	1.0 ± 0.1
ESO 134 12	221.21	-59.15	1.8	61	2178 ± 135	7.1 ± 0.2	0.07 ± 0.07	2.8 ± 0.0
ESO 332 08	253.69	-40.73	4.8	201	1693 ± 22	6.9 ± 0.1	0.23 ± 0.11	1.2 ± 0.1
ESO 332 13	254.17	-40.59	3.5	52	1487 ± 84	6.8 ± 0.1	0.17 ± 0.10	1.4 ± 0.0
FSR 0165	299.53	26.78	4.4	84	2414 ± 189	7.0 ± 0.1	0.11 ± 0.10	3.5 ± 0.0
FSR 0198	300.61	35.67	7.2	90	1828 ± 97	6.9 ± 0.1	0.21 ± 0.17	2.6 ± 0.2
FSR 0224	306.35	40.22	1.3	19	1706 ± 127	6.7 ± 0.1	0.24 ± 0.22	3.1 ± 0.1
FSR 0236	308.17	41.44	2.9	89	1610 ± 91	6.9 ± 0.1	0.09 ± 0.13	3.6 ± 0.1
FSR 0284	314.14	49.36	0.8	17	2693 ± 180	7.2 ± 0.5	0.00 ± 0.14	2.7 ± 0.0
FSR 0306	322.12	51.69	3.9	81	2172 ± 245	7.7 ± 0.8	0.18 ± 0.34	3.1 ± 0.1
FSR 0336	327.71	55.24	5.8	192	2312 ± 189	7.1 ± 0.4	0.04 ± 0.32	3.0 ± 0.1
FSR 0398	339.10	64.15	3.6	57	915 ± 23	7.3 ± 0.1	0.29 ± 0.06	2.0 ± 0.1
FSR 0451	357.94	60.93	9.8	241	2602 ± 120	7.1 ± 0.0	-0.10 ± 0.06	2.3 ± 0.0
FSR 0498	7.34	62.39	5.0	51	2463 ± 127	7.5 ± 0.4	-0.07 ± 0.10	1.5 ± 0.1
FSR 0551	24.96	64.87	4.6	52	909 ± 35	7.2 ± 0.1	-0.04 ± 0.15	1.6 ± 0.2
FSR 0686	66.35	46.10	9.0	30	894 ± 22	7.3 ± 0.2	-0.02 ± 0.18	1.7 ± 0.2
FSR 0852	88.37	25.20	2.8	35	2542 ± 156	7.5 ± 0.6	-0.19 ± 0.07	2.1 ± 0.2
FSR 0904	91.77	19.02	5.2	107	1967 ± 97	7.5 ± 0.3	-0.10 ± 0.09	2.1 ± 0.0
FSR 0968	92.85	11.85	2.9	34	2390 ± 243	7.3 ± 0.3	-0.10 ± 0.09	1.6 ± 0.1
Gulliver 10	123.02	-38.66	13.4	38	521 ± 30	7.1 ± 0.2	0.13 ± 0.11	0.3 ± 0.1
Gulliver 15	272.60	-16.71	5.5	74	1880 ± 134	7.1 ± 0.1	-0.04 ± 0.12	2.5 ± 0.1
Gulliver 19	344.21	61.11	7.9	142	1427 ± 118	7.4 ± 0.6	0.04 ± 0.23	3.4 ± 0.2
Gulliver 2	122.87	-37.42	4.9	57	1357 ± 45	7.0 ± 0.0	-0.04 ± 0.12	1.0 ± 0.0
Gulliver 26	80.71	35.28	4.4	62	2554 ± 273	7.5 ± 0.8	-0.02 ± 0.36	2.2 ± 0.2

Table A.4 continued

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	$[Fe/H]$	A_V (mag)
Gulliver 31	301.91	38.24	4.1	54	2137 ± 45	7.6 ± 0.2	0.15 ± 0.08	1.6 ± 0.1
Gulliver 40	163.10	-58.40	3.9	35	1585 ± 69	7.5 ± 0.3	0.13 ± 0.10	0.8 ± 0.1
Gulliver 43	296.28	24.56	4.4	74	2452 ± 238	7.1 ± 0.1	0.19 ± 0.18	3.0 ± 0.0
Gulliver 5	132.61	-45.49	5.5	27	2272 ± 117	6.9 ± 0.2	0.07 ± 0.20	2.0 ± 0.1
Gulliver 6	83.28	-1.66	31.1	273	413 ± 4	7.1 ± 0.0	-0.10 ± 0.01	0.4 ± 0.1
Gulliver 8	80.55	33.80	5.9	35	1114 ± 24	7.5 ± 0.1	0.07 ± 0.09	1.0 ± 0.1
Gulliver 9	126.74	-47.82	52.3	236	477 ± 7	7.3 ± 0.0	-0.04 ± 0.07	0.4 ± 0.1
Haffner 13	115.24	-30.07	18.4	203	561 ± 4	7.6 ± 0.1	0.09 ± 0.05	0.2 ± 0.0
Harvard 16	262.78	-36.81	7.5	354	1331 ± 89	7.4 ± 0.0	0.18 ± 0.09	2.3 ± 0.1
Hogg 10	167.69	-60.39	2.3	16	2460 ± 160	6.8 ± 0.5	0.02 ± 0.12	1.4 ± 0.1
Hogg 16	202.30	-61.21	2.8	46	1943 ± 131	7.5 ± 0.3	0.11 ± 0.14	1.4 ± 0.1
Hogg 18	222.69	-52.29	3.8	25	1619 ± 72	7.5 ± 0.3	-0.02 ± 0.08	1.8 ± 0.1
Hogg 19	247.23	-49.11	2.4	45	2514 ± 252	7.1 ± 0.4	0.10 ± 0.21	2.3 ± 0.1
Hogg 22	251.66	-47.08	2.6	117	2354 ± 171	7.1 ± 0.1	0.12 ± 0.08	2.1 ± 0.0
IC 1396	324.74	57.56	15.0	418	914 ± 14	6.7 ± 0.1	-0.02 ± 0.11	1.6 ± 0.1
IC 1590	13.20	56.63	2.5	73	2346 ± 240	7.1 ± 0.2	-0.16 ± 0.14	1.3 ± 0.1
IC 1805	38.25	61.48	5.9	132	2187 ± 211	6.8 ± 0.1	0.31 ± 0.02	2.3 ± 0.0
IC 1848	42.91	60.42	11.0	365	2004 ± 68	7.1 ± 0.1	-0.10 ± 0.11	1.9 ± 0.0
IC 2157	91.19	24.08	2.5	39	1885 ± 155	7.6 ± 0.2	-0.10 ± 0.06	1.7 ± 0.1
IC 2391	130.38	-53.00	42.4	207	150 ± 1	7.7 ± 0.0	0.00 ± 0.01	0.2 ± 0.0
IC 2395	130.47	-48.07	11.8	269	710 ± 6	7.1 ± 0.0	-0.01 ± 0.06	0.5 ± 0.0
IC 2581	156.86	-57.63	3.2	123	2449 ± 153	7.1 ± 0.2	0.01 ± 0.14	1.4 ± 0.0
IC 2602	160.42	-64.42	67.1	268	152 ± 1	7.7 ± 0.0	-0.02 ± 0.01	0.3 ± 0.0
IC 2948	174.80	-63.45	5.1	51	2371 ± 148	6.9 ± 0.4	0.03 ± 0.22	1.3 ± 0.1
IC 348	56.14	32.18	5.2	80	321 ± 3	6.8 ± 0.1	0.30 ± 0.26	0.3 ± 0.3
IC 4996	304.14	37.65	1.7	79	1876 ± 104	7.1 ± 0.1	0.18 ± 0.24	1.8 ± 0.1
IC 5146	328.35	47.24	4.9	76	657 ± 34	6.9 ± 0.1	-0.01 ± 0.02	1.7 ± 0.2
Ivanov 4	105.15	-8.87	1.6	15	2507 ± 248	7.1 ± 0.4	-0.20 ± 0.20	1.4 ± 0.1
Juchert 20	62.69	46.87	2.3	85	2649 ± 277	7.2 ± 0.6	-0.14 ± 0.08	2.4 ± 0.1
King 14	8.00	63.16	2.7	109	2213 ± 64	7.7 ± 0.2	-0.05 ± 0.10	1.6 ± 0.1
King 21	357.47	62.71	2.0	112	2591 ± 130	7.7 ± 0.5	0.03 ± 0.11	2.4 ± 0.1
King 26	292.23	14.89	2.1	91	2169 ± 198	7.7 ± 0.5	0.32 ± 0.24	3.0 ± 0.1
Kronberger 1	82.09	34.78	1.0	34	1919 ± 155	7.0 ± 0.1	-0.18 ± 0.07	1.7 ± 0.1
Kronberger 69	300.49	31.42	1.9	45	2651 ± 97	7.1 ± 0.1	-0.11 ± 0.22	2.8 ± 0.0
LP 0288	273.48	-19.51	3.2	233	2070 ± 475	7.1 ± 0.3	0.14 ± 0.09	2.4 ± 0.4
LP 0503	306.38	40.86	6.7	201	1770 ± 83	7.0 ± 0.1	-0.07 ± 0.12	2.8 ± 0.1
LP 0506	310.36	46.14	4.3	193	2165 ± 134	7.1 ± 0.1	-0.07 ± 0.10	2.2 ± 0.1
LP 0733	111.07	-20.25	6.8	314	2706 ± 188	7.7 ± 0.2	-0.11 ± 0.08	1.0 ± 0.1
LP 1049	160.57	-59.95	4.4	50	3098 ± 135	6.9 ± 0.2	0.19 ± 0.24	0.9 ± 0.4
LP 1209	271.30	-23.80	4.7	223	2931 ± 247	6.8 ± 0.1	0.10 ± 0.13	3.0 ± 0.1
LP 1211	271.24	-23.71	6.5	161	2389 ± 186	7.1 ± 0.1	0.38 ± 0.14	2.0 ± 0.1

Appendix A. Spectroscopic observations

Table A.4 continued

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	$[Fe/H]$	A_V (mag)
LP 1218	273.97	-18.09	2.8	168	2578 ± 201	6.9 ± 0.4	0.26 ± 0.17	2.7 ± 0.2
LP 1329	25.34	61.49	20.5	1243	2817 ± 69	7.1 ± 0.0	0.17 ± 0.15	2.3 ± 0.1
LP 1342	23.50	61.68	12.6	650	2761 ± 301	7.0 ± 0.1	-0.09 ± 0.21	2.3 ± 0.1
LP 1355	46.02	57.77	17.6	233	2731 ± 158	7.0 ± 0.6	-0.11 ± 0.10	2.7 ± 0.0
LP 1487	160.49	-58.37	5.0	195	2829 ± 189	7.1 ± 0.1	0.00 ± 0.11	1.8 ± 0.1
LP 1490	157.71	-58.26	4.9	127	2449 ± 112	7.2 ± 0.1	0.06 ± 0.07	1.1 ± 0.1
LP 1614	241.56	-52.14	3.2	153	2481 ± 209	7.1 ± 0.6	0.17 ± 0.17	3.3 ± 0.1
LP 1641	277.92	-10.18	3.5	124	3397 ± 325	6.7 ± 0.5	0.15 ± 0.09	4.1 ± 0.1
LP 1768	310.63	45.52	8.5	245	2525 ± 259	7.2 ± 0.5	-0.05 ± 0.09	2.4 ± 0.2
LP 1771	307.37	44.64	4.7	115	2097 ± 116	7.1 ± 0.6	0.14 ± 0.22	3.0 ± 0.1
LP 1775	316.73	47.09	10.0	188	2804 ± 38	7.5 ± 0.3	0.14 ± 0.06	1.2 ± 0.1
LP 1780	321.53	52.71	4.9	233	1858 ± 104	7.1 ± 0.8	0.02 ± 0.16	3.2 ± 0.1
LP 1807	334.11	57.81	5.4	164	2819 ± 339	7.2 ± 0.7	-0.09 ± 0.11	2.8 ± 0.2
LP 1821	347.61	60.97	8.5	250	2987 ± 181	7.0 ± 0.1	-0.06 ± 0.11	2.6 ± 0.1
LP 1864	8.71	62.99	8.5	238	2496 ± 90	7.3 ± 0.2	-0.10 ± 0.09	1.4 ± 0.2
LP 1888	14.63	63.28	19.6	303	2487 ± 90	7.4 ± 0.3	0.15 ± 0.09	1.7 ± 0.2
LP 1937	41.08	61.87	21.5	158	2491 ± 158	7.5 ± 0.4	-0.06 ± 0.10	1.8 ± 0.1
LP 2106	263.50	-35.06	5.4	241	2025 ± 149	6.9 ± 0.7	-0.07 ± 0.14	3.6 ± 0.2
LP 2113	274.70	-18.08	3.5	57	2013 ± 88	7.2 ± 0.1	0.14 ± 0.17	1.6 ± 0.3
LP 2172	282.48	-0.32	4.0	54	1918 ± 127	7.6 ± 0.7	0.14 ± 0.20	4.0 ± 0.1
LP 2219	130.78	-45.78	19.4	206	1402 ± 38	7.4 ± 0.2	-0.01 ± 0.11	1.2 ± 0.0
LP 2221	129.44	-40.03	21.2	319	925 ± 18	7.0 ± 0.1	-0.06 ± 0.12	1.2 ± 0.2
LP 2249	344.82	62.84	35.6	690	819 ± 17	6.7 ± 0.1	-0.03 ± 0.12	2.2 ± 0.3
Lynga 4	233.34	-55.24	2.6	70	2113 ± 303	7.6 ± 0.9	0.19 ± 0.34	3.9 ± 0.1
Mamajek 1	130.57	-78.95	12.2	19	98 ± 0	7.2 ± 0.1	-0.10 ± 0.12	0.1 ± 0.1
Markarian 38	273.82	-19.01	2.5	26	1770 ± 102	7.0 ± 0.1	0.06 ± 0.07	1.2 ± 0.0
Markarian 50	348.81	60.45	1.8	43	2677 ± 113	7.1 ± 0.0	-0.05 ± 0.10	2.2 ± 0.2
Mayer 1	5.52	61.76	5.0	55	2380 ± 118	7.1 ± 0.1	0.01 ± 0.20	1.6 ± 0.1
Moffat 1	240.41	-54.15	1.1	19	2506 ± 118	6.8 ± 0.1	0.50 ± 0.29	2.4 ± 0.1
NGC 1220	47.92	53.35	1.1	57	2405 ± 145	7.4 ± 0.1	-0.14 ± 0.09	2.5 ± 0.0
NGC 1348	53.53	51.41	3.5	92	2090 ± 185	7.1 ± 0.1	-0.20 ± 0.11	2.8 ± 0.1
NGC 1444	57.34	52.65	3.1	20	1098 ± 48	7.2 ± 0.1	0.10 ± 0.12	2.0 ± 0.1
NGC 146	8.26	63.33	3.8	66	2685 ± 146	7.3 ± 0.1	-0.05 ± 0.13	1.8 ± 0.0
NGC 1502	61.95	62.34	3.9	148	1056 ± 35	6.9 ± 0.1	-0.01 ± 0.15	2.2 ± 0.0
NGC 1579	67.49	35.37	8.0	35	561 ± 102	6.9 ± 1.2	0.19 ± 0.55	3.7 ± 1.2
NGC 1960	84.08	34.13	4.9	285	1086 ± 37	7.5 ± 0.1	-0.03 ± 0.09	0.9 ± 0.0
NGC 1977	83.77	-4.79	9.5	84	381 ± 9	6.7 ± 0.1	-0.18 ± 0.08	0.3 ± 0.1
NGC 1980	83.79	-5.93	9.2	106	316 ± 19	7.0 ± 0.0	-0.24 ± 0.09	0.1 ± 0.1
NGC 2129	90.28	23.34	3.0	79	1810 ± 146	7.2 ± 0.3	-0.07 ± 0.07	2.2 ± 0.1
NGC 2169	92.13	13.94	5.3	58	1004 ± 20	6.9 ± 0.1	-0.09 ± 0.10	0.8 ± 0.1
NGC 2183	92.71	-6.23	9.1	53	777 ± 61	6.9 ± 0.2	-0.06 ± 0.09	1.4 ± 0.7

Table A.4 continued

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	$[Fe/H]$	A_V (mag)
NGC 2232	96.86	-4.71	30.7	172	322 ± 1	7.5 ± 0.0	0.02 ± 0.07	0.2 ± 0.1
NGC 2244	98.09	4.92	16.3	442	1254 ± 89	7.1 ± 0.1	-0.21 ± 0.08	1.6 ± 0.1
NGC 2264	100.21	9.88	4.4	131	702 ± 7	6.8 ± 0.1	-0.18 ± 0.11	0.4 ± 0.1
NGC 2362	109.67	-24.95	3.1	142	1233 ± 57	7.2 ± 0.1	-0.10 ± 0.07	0.4 ± 0.0
NGC 2367	110.03	-21.86	2.8	41	1722 ± 244	7.5 ± 0.4	-0.07 ± 0.13	1.0 ± 0.2
NGC 2384	111.29	-21.02	3.8	80	2494 ± 179	7.3 ± 0.2	-0.15 ± 0.21	1.0 ± 0.1
NGC 2451B	116.01	-37.86	38.3	258	360 ± 3	7.7 ± 0.1	0.05 ± 0.05	0.3 ± 0.1
NGC 2547	122.48	-49.18	16.1	222	386 ± 0	7.6 ± 0.0	-0.01 ± 0.05	0.3 ± 0.1
NGC 2571	124.72	-29.77	7.3	163	1293 ± 46	7.5 ± 0.1	0.06 ± 0.11	0.4 ± 0.0
NGC 2645	129.78	-46.23	2.5	54	1737 ± 50	7.3 ± 0.1	0.13 ± 0.09	1.4 ± 0.1
NGC 2659	130.63	-45.00	2.3	91	1815 ± 75	7.6 ± 0.2	-0.08 ± 0.14	1.5 ± 0.1
NGC 3228	155.46	-51.76	24.4	113	483 ± 6	7.7 ± 0.1	0.14 ± 0.08	0.2 ± 0.0
NGC 3293	158.97	-58.23	3.4	343	2334 ± 52	7.1 ± 0.0	0.03 ± 0.04	1.0 ± 0.0
NGC 3324	159.37	-58.63	3.4	38	2489 ± 135	6.8 ± 0.1	0.29 ± 0.13	1.6 ± 0.1
NGC 3572	167.63	-60.30	4.3	112	2276 ± 130	6.8 ± 0.0	0.07 ± 0.07	1.4 ± 0.1
NGC 3590	168.26	-60.79	1.7	167	2420 ± 97	7.1 ± 0.1	0.06 ± 0.07	1.7 ± 0.0
NGC 366	16.62	62.22	1.9	132	2611 ± 65	7.1 ± 0.3	-0.03 ± 0.17	3.3 ± 0.1
NGC 3766	174.06	-61.61	5.7	632	1862 ± 60	7.5 ± 0.1	0.05 ± 0.08	0.8 ± 0.0
NGC 4103	181.60	-61.25	5.5	353	1873 ± 50	7.5 ± 0.0	0.13 ± 0.07	1.1 ± 0.0
NGC 433	18.79	60.13	2.9	110	1802 ± 89	7.6 ± 0.3	0.05 ± 0.12	2.0 ± 0.1
NGC 4463	187.46	-64.80	2.9	85	1638 ± 41	7.6 ± 0.2	0.16 ± 0.10	1.6 ± 0.0
NGC 457	19.85	58.30	6.7	584	2540 ± 133	7.4 ± 0.1	-0.03 ± 0.15	1.6 ± 0.0
NGC 4755	193.42	-60.37	4.3	617	1887 ± 68	7.3 ± 0.1	0.06 ± 0.08	1.3 ± 0.0
NGC 5606	216.95	-59.63	1.7	44	2344 ± 134	7.0 ± 0.1	0.14 ± 0.10	1.7 ± 0.0
NGC 581	23.33	60.66	3.2	149	2292 ± 89	7.5 ± 0.1	-0.02 ± 0.10	1.5 ± 0.0
NGC 6178	248.93	-45.65	4.0	41	844 ± 15	7.1 ± 0.2	0.07 ± 0.16	0.8 ± 0.0
NGC 6193	250.30	-48.79	9.7	304	1080 ± 58	6.7 ± 0.0	0.14 ± 0.13	1.6 ± 0.1
NGC 6200	251.03	-47.46	6.5	433	2352 ± 205	7.1 ± 0.1	0.17 ± 0.12	1.9 ± 0.0
NGC 6216	252.33	-44.72	3.2	180	2065 ± 218	7.2 ± 0.4	0.14 ± 0.06	2.6 ± 0.1
NGC 6231	253.54	-41.80	8.6	606	1429 ± 79	6.9 ± 0.1	0.11 ± 0.07	1.5 ± 0.0
NGC 6249	254.44	-44.81	2.6	59	1198 ± 51	7.3 ± 0.0	0.12 ± 0.13	1.5 ± 0.1
NGC 6250	254.51	-45.95	5.3	84	954 ± 39	7.3 ± 0.0	-0.01 ± 0.10	1.3 ± 0.0
NGC 6318	259.04	-39.42	2.6	169	2219 ± 320	6.9 ± 0.1	0.02 ± 0.14	3.5 ± 0.0
NGC 6322	259.63	-42.94	3.5	68	1315 ± 63	7.0 ± 0.2	0.01 ± 0.16	2.1 ± 0.1
NGC 637	25.77	64.04	1.4	75	2519 ± 143	7.1 ± 0.1	0.10 ± 0.07	1.9 ± 0.0
NGC 6383	263.66	-32.57	5.6	197	1086 ± 104	7.1 ± 0.1	0.00 ± 0.12	1.2 ± 0.1
NGC 6396	264.40	-35.01	1.9	50	2144 ± 357	7.1 ± 0.5	0.29 ± 0.08	2.9 ± 0.1
NGC 6404	264.92	-33.22	3.6	256	2767 ± 199	6.7 ± 0.0	0.29 ± 0.14	3.7 ± 0.1
NGC 6451	267.67	-30.20	4.5	195	2794 ± 268	6.9 ± 0.1	0.46 ± 0.20	2.4 ± 0.1
NGC 6520	270.86	-27.88	1.7	42	1699 ± 87	7.4 ± 0.3	0.03 ± 0.14	1.5 ± 0.1
NGC 6530	271.11	-24.36	5.2	80	1206 ± 39	6.7 ± 0.0	0.37 ± 0.14	1.2 ± 0.0

Appendix A. Spectroscopic observations

Table A.4 continued

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	$[Fe/H]$	A_V (mag)
NGC 6531	271.03	-22.52	8.1	186	1161 ± 49	7.1 ± 0.1	0.09 ± 0.02	1.0 ± 0.0
NGC 654	26.01	61.88	2.8	353	2516 ± 211	7.2 ± 0.1	-0.10 ± 0.08	2.6 ± 0.1
NGC 6561	272.68	-16.74	5.5	68	1461 ± 86	7.2 ± 0.1	0.33 ± 0.16	1.4 ± 0.0
NGC 6604	274.51	-12.24	2.9	88	1885 ± 75	6.8 ± 0.1	0.10 ± 0.16	2.8 ± 0.1
NGC 6611	274.69	-13.80	5.5	153	1582 ± 134	7.0 ± 0.1	0.07 ± 0.14	2.2 ± 0.0
NGC 6613	275.00	-17.09	3.8	48	1352 ± 31	7.5 ± 0.2	0.13 ± 0.07	1.5 ± 0.1
NGC 663	26.57	61.22	7.3	612	2353 ± 148	7.4 ± 0.3	-0.13 ± 0.08	2.3 ± 0.0
NGC 6649	278.36	-10.40	3.4	518	1917 ± 80	7.6 ± 0.3	0.44 ± 0.16	3.7 ± 0.0
NGC 6664	279.12	-8.19	6.2	202	1944 ± 211	7.1 ± 0.3	-0.14 ± 0.27	2.4 ± 0.1
NGC 6823	295.80	23.33	4.4	155	2081 ± 142	6.8 ± 0.1	0.13 ± 0.24	2.5 ± 0.1
NGC 6871	301.59	35.77	21.1	580	1741 ± 69	7.1 ± 0.1	0.10 ± 0.13	1.4 ± 0.0
NGC 6910	305.79	40.77	5.0	154	1722 ± 111	6.9 ± 0.1	0.04 ± 0.10	3.2 ± 0.1
NGC 6913	305.91	38.48	6.2	89	1785 ± 107	6.9 ± 0.0	0.21 ± 0.26	2.8 ± 0.1
NGC 7039	317.66	45.59	8.1	110	755 ± 10	7.2 ± 0.1	0.09 ± 0.10	0.6 ± 0.1
NGC 7129	325.78	66.13	3.5	20	894 ± 14	6.8 ± 0.3	0.13 ± 0.20	2.2 ± 0.1
NGC 7160	328.44	62.59	3.7	45	919 ± 17	7.1 ± 0.2	0.12 ± 0.16	1.3 ± 0.1
NGC 7281	336.28	57.87	6.9	56	2013 ± 104	7.4 ± 0.2	0.06 ± 0.12	1.7 ± 0.0
NGC 7380	341.80	58.15	5.6	112	2394 ± 199	7.0 ± 0.1	-0.06 ± 0.14	1.9 ± 0.1
NGC 869	34.74	57.14	6.9	705	2246 ± 64	7.1 ± 0.0	-0.10 ± 0.08	1.7 ± 0.0
NGC 884	35.63	57.15	6.7	475	2150 ± 103	7.2 ± 0.1	-0.08 ± 0.09	1.7 ± 0.0
NGC 957	38.36	57.55	7.0	158	2070 ± 99	7.4 ± 0.2	-0.05 ± 0.04	2.2 ± 0.0
Pismis 11	138.97	-50.01	0.9	44	2340 ± 114	6.8 ± 0.1	0.05 ± 0.19	3.3 ± 0.1
Pismis 27	92.72	20.62	3.7	37	1744 ± 139	7.0 ± 0.2	0.10 ± 0.14	1.8 ± 0.3
Pismis 5	129.37	-39.64	7.3	88	921 ± 24	6.9 ± 0.1	-0.03 ± 0.13	1.4 ± 0.1
Pismis 8	130.41	-46.27	1.7	53	1756 ± 73	7.1 ± 0.1	-0.03 ± 0.10	2.1 ± 0.1
Pismis Moreno 1	334.68	63.24	4.2	60	917 ± 12	6.9 ± 0.2	-0.01 ± 0.13	1.9 ± 0.0
Platais 8	136.86	-58.36	140.5	194	134 ± 0	7.6 ± 0.0	-0.05 ± 0.06	0.2 ± 0.1
Pozzo 1	122.33	-47.30	37.1	327	345 ± 3	7.1 ± 0.1	-0.07 ± 0.06	0.2 ± 0.0
RSG 8	345.03	59.37	48.2	191	450 ± 5	7.6 ± 0.1	-0.02 ± 0.13	0.6 ± 0.1
Riddle 4	31.86	60.26	1.9	52	2638 ± 100	7.0 ± 0.1	-0.11 ± 0.15	3.0 ± 0.0
Roslund 2	296.27	23.96	9.4	131	1835 ± 136	7.1 ± 0.0	0.19 ± 0.13	2.7 ± 0.0
Ruprecht 120	248.79	-48.28	2.7	75	2101 ± 38	7.3 ± 0.2	0.27 ± 0.17	2.5 ± 0.1
Ruprecht 127	264.47	-36.31	3.7	74	2630 ± 273	6.8 ± 0.2	0.23 ± 0.15	2.9 ± 0.1
Ruprecht 138	269.98	-24.67	3.4	113	2258 ± 186	7.1 ± 0.0	0.20 ± 0.15	2.4 ± 0.0
Ruprecht 144	278.47	-11.46	8.6	55	1507 ± 156	7.2 ± 0.1	0.23 ± 0.11	2.2 ± 0.0
Ruprecht 170	276.32	-9.99	4.5	88	2050 ± 183	7.1 ± 0.4	0.09 ± 0.07	2.5 ± 0.1
Ruprecht 26	114.26	-15.57	11.2	80	1030 ± 21	7.7 ± 0.2	-0.10 ± 0.03	0.5 ± 0.1
Ruprecht 65	129.86	-44.02	4.8	39	2021 ± 199	7.7 ± 0.4	0.09 ± 0.12	1.7 ± 0.1
Ruprecht 71	132.32	-46.86	3.5	49	1815 ± 84	7.2 ± 0.2	0.17 ± 0.17	1.1 ± 0.1
Ruprecht 94	172.87	-63.46	11.8	71	2332 ± 159	7.2 ± 0.2	0.12 ± 0.16	1.2 ± 0.1
SAI 118	205.80	-63.14	4.1	121	1708 ± 68	7.4 ± 0.2	0.05 ± 0.11	1.1 ± 0.0

Table A.4 continued

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	[Fe/H]	A_V (mag)
SAI 24	44.78	60.56	5.3	107	1928 ± 137	7.1 ± 0.0	0.16 ± 0.13	2.0 ± 0.0
SAI 4	5.89	62.71	2.5	50	2498 ± 184	7.6 ± 0.3	-0.07 ± 0.09	1.8 ± 0.1
Sigma Ori	84.76	-2.59	9.1	82	303 ± 26	7.0 ± 0.1	-0.09 ± 0.05	0.2 ± 0.0
Stephenson 1	283.52	36.93	15.4	92	355 ± 2	7.6 ± 0.0	0.06 ± 0.08	0.2 ± 0.1
Stock 14	175.93	-62.53	4.2	45	2286 ± 162	7.2 ± 0.0	0.12 ± 0.11	0.9 ± 0.0
Stock 20	6.31	62.61	2.3	46	2566 ± 230	7.1 ± 0.1	-0.10 ± 0.12	1.5 ± 0.0
Stock 8	81.96	34.50	10.0	319	2033 ± 133	7.0 ± 0.0	-0.07 ± 0.14	1.7 ± 0.0
Teutsch 30	306.93	36.08	1.3	31	1872 ± 63	7.0 ± 0.1	-0.11 ± 0.13	3.5 ± 0.0
Teutsch 38	131.52	-38.09	33.8	108	652 ± 4	7.6 ± 0.2	0.12 ± 0.04	0.2 ± 0.1
Teutsch 8	300.60	35.31	0.5	19	1808 ± 95	6.7 ± 0.0	0.17 ± 0.10	1.8 ± 0.0
Trapezium-FG	83.84	-5.41	21.1	269	381 ± 12	6.8 ± 0.1	-0.15 ± 0.06	0.2 ± 0.1
Trumpler 1	23.92	61.29	1.8	67	2425 ± 230	7.3 ± 0.1	-0.01 ± 0.12	1.8 ± 0.0
Trumpler 14	160.99	-59.55	2.2	88	2147 ± 264	6.7 ± 0.0	0.19 ± 0.07	1.7 ± 0.0
Trumpler 15	161.18	-59.37	1.9	99	2290 ± 137	6.8 ± 0.1	-0.12 ± 0.14	1.6 ± 0.0
Trumpler 16	161.25	-59.69	2.9	90	2077 ± 103	6.7 ± 0.1	0.35 ± 0.16	1.5 ± 0.1
Trumpler 17	164.10	-59.23	3.5	188	2255 ± 201	7.4 ± 0.3	0.04 ± 0.12	2.0 ± 0.0
Trumpler 28	264.24	-32.47	5.6	124	1355 ± 81	7.3 ± 0.1	0.17 ± 0.14	2.2 ± 0.1
Trumpler 3	48.02	63.22	13.8	219	646 ± 11	7.6 ± 0.1	-0.21 ± 0.02	1.2 ± 0.0
Trumpler 33	276.16	-19.71	3.1	40	1407 ± 68	7.7 ± 0.2	0.15 ± 0.10	1.5 ± 0.1
UBC 121	288.91	12.86	8.0	124	1640 ± 95	7.1 ± 0.4	0.02 ± 0.18	2.8 ± 0.1
UBC 133	303.72	28.36	7.9	150	1555 ± 41	7.6 ± 0.4	0.41 ± 0.20	2.8 ± 0.1
UBC 134	299.73	31.09	4.4	93	2301 ± 83	7.3 ± 0.3	0.12 ± 0.15	2.5 ± 0.1
UBC 148	309.73	46.48	12.1	177	2130 ± 96	7.6 ± 0.3	0.17 ± 0.17	2.0 ± 0.1
UBC 155	318.43	50.14	9.8	137	1636 ± 52	7.1 ± 0.1	0.05 ± 0.21	3.3 ± 0.1
UBC 156	321.74	49.11	5.3	107	2122 ± 145	7.5 ± 0.4	0.04 ± 0.16	2.0 ± 0.2
UBC 166	334.26	54.24	8.2	261	3071 ± 164	7.2 ± 0.5	-0.01 ± 0.09	1.5 ± 0.1
UBC 177	339.65	67.06	10.6	73	1147 ± 41	7.3 ± 0.3	0.25 ± 0.17	2.0 ± 0.2
UBC 178	346.21	63.33	12.1	156	867 ± 15	7.0 ± 0.1	0.25 ± 0.11	2.0 ± 0.1
UBC 17a	83.33	-1.61	85.3	539	355 ± 3	7.0 ± 0.3	-0.23 ± 0.19	0.5 ± 0.5
UBC 182	4.11	52.28	8.7	70	1603 ± 67	7.1 ± 0.1	-0.17 ± 0.10	0.7 ± 0.1
UBC 188	27.46	53.92	15.8	220	2577 ± 161	7.2 ± 0.1	-0.09 ± 0.08	0.7 ± 0.1
UBC 191	37.48	57.96	13.5	476	2242 ± 79	7.2 ± 0.3	-0.04 ± 0.08	2.1 ± 0.0
UBC 192	37.02	57.17	16.4	534	2239 ± 75	7.3 ± 0.2	0.02 ± 0.07	1.8 ± 0.0
UBC 198	82.12	35.70	10.7	161	1741 ± 184	7.3 ± 0.2	-0.15 ± 0.10	1.5 ± 0.1
UBC 245	129.32	-45.23	12.5	113	2186 ± 81	7.2 ± 0.1	0.12 ± 0.20	1.6 ± 0.1
UBC 249	136.96	-52.93	12.5	113	1850 ± 45	7.6 ± 0.4	0.05 ± 0.17	1.3 ± 0.2
UBC 258	158.88	-57.60	10.7	202	2406 ± 56	7.4 ± 0.4	0.20 ± 0.09	0.8 ± 0.1
UBC 267	168.30	-58.91	10.4	215	2877 ± 86	6.9 ± 0.1	0.20 ± 0.12	0.9 ± 0.1
UBC 270	169.79	-59.55	6.8	83	2410 ± 100	7.1 ± 0.5	0.05 ± 0.13	1.2 ± 0.1
UBC 272	168.14	-62.27	9.7	163	2310 ± 57	7.6 ± 0.3	0.19 ± 0.12	1.4 ± 0.1
UBC 280	177.93	-62.65	9.0	189	2255 ± 78	7.5 ± 0.2	0.17 ± 0.08	0.9 ± 0.1

Appendix A. Spectroscopic observations

Table A.4 continued

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	$[Fe/H]$	A_V (mag)
UBC 281	178.12	-64.57	7.1	74	2552 ± 65	7.5 ± 0.4	0.52 ± 0.20	0.9 ± 0.1
UBC 296	216.23	-61.05	4.6	102	1971 ± 100	7.7 ± 0.1	0.38 ± 0.13	1.6 ± 0.1
UBC 31	61.11	32.84	88.6	152	379 ± 2	7.4 ± 0.5	0.43 ± 0.01	1.6 ± 0.2
UBC 322	255.20	-44.20	9.5	200	2881 ± 342	6.8 ± 0.7	0.35 ± 0.20	3.2 ± 0.1
UBC 337	271.74	-24.65	6.1	128	1721 ± 52	6.9 ± 0.1	0.09 ± 0.20	3.0 ± 0.2
UBC 338	271.54	-24.24	5.5	93	1559 ± 59	7.6 ± 0.2	0.31 ± 0.10	2.8 ± 0.0
UBC 341	276.45	-17.08	6.9	150	2041 ± 76	7.1 ± 0.5	0.15 ± 0.20	2.7 ± 0.1
UBC 342	273.87	-14.91	15.0	122	1620 ± 81	7.1 ± 0.0	0.25 ± 0.08	2.0 ± 0.1
UBC 345	275.97	-12.24	9.1	76	2041 ± 114	6.9 ± 0.1	0.23 ± 0.21	3.4 ± 0.0
UBC 354	282.35	-4.41	6.5	165	1824 ± 101	7.2 ± 0.8	0.33 ± 0.18	2.9 ± 0.1
UBC 361	290.01	15.16	4.0	47	1491 ± 51	7.5 ± 0.4	0.35 ± 0.10	2.0 ± 0.1
UBC 373	306.86	37.67	4.6	47	1232 ± 26	7.2 ± 0.2	0.02 ± 0.06	3.0 ± 0.0
UBC 377	314.85	45.85	7.9	57	2984 ± 195	7.1 ± 0.7	0.17 ± 0.14	1.8 ± 0.3
UBC 379	313.27	48.53	4.9	62	2456 ± 113	7.3 ± 0.3	0.45 ± 0.17	2.7 ± 0.1
UBC 386	319.76	61.94	7.9	57	1160 ± 29	6.9 ± 0.0	0.04 ± 0.08	1.5 ± 0.0
UBC 389	338.34	58.37	2.9	43	2257 ± 104	7.1 ± 0.4	0.10 ± 0.16	1.7 ± 0.1
UBC 391	338.89	59.01	11.0	110	2386 ± 98	7.1 ± 0.5	0.16 ± 0.24	2.8 ± 0.1
UBC 396	337.30	66.52	7.1	29	1054 ± 25	7.1 ± 0.4	0.02 ± 0.35	2.5 ± 0.2
UBC 410	6.98	62.91	10.1	171	2663 ± 137	7.3 ± 0.4	0.27 ± 0.10	1.5 ± 0.1
UBC 413	17.65	60.47	3.8	41	2746 ± 89	7.1 ± 0.3	0.10 ± 0.23	2.3 ± 0.2
UBC 415	42.56	72.90	7.1	29	893 ± 26	7.2 ± 0.1	-0.08 ± 0.16	1.6 ± 0.1
UBC 417	29.18	54.42	11.2	67	1926 ± 113	7.6 ± 0.2	-0.03 ± 0.23	0.6 ± 0.1
UBC 422	40.57	58.16	10.1	108	2494 ± 221	7.1 ± 0.6	-0.05 ± 0.17	2.2 ± 0.1
UBC 423	48.30	58.94	3.1	47	2623 ± 157	6.9 ± 0.1	0.14 ± 0.20	2.6 ± 0.0
UBC 432	79.03	39.53	7.3	108	2635 ± 144	7.1 ± 0.5	-0.09 ± 0.10	1.9 ± 0.1
UBC 46	33.74	57.31	11.0	339	2003 ± 77	7.6 ± 0.1	-0.14 ± 0.07	1.8 ± 0.0
UBC 479	124.40	-42.59	9.7	59	1480 ± 49	7.2 ± 0.1	0.31 ± 0.26	1.4 ± 0.4
UBC 482	131.16	-45.97	16.5	232	2001 ± 105	7.3 ± 0.2	-0.01 ± 0.17	1.1 ± 0.1
UBC 483	133.74	-47.44	5.2	26	1877 ± 30	7.5 ± 0.5	0.13 ± 0.10	2.0 ± 0.1
UBC 487	139.75	-51.60	6.4	65	1852 ± 66	7.3 ± 0.3	0.19 ± 0.21	1.9 ± 0.1
UBC 499	158.64	-59.75	7.4	67	2390 ± 29	7.5 ± 0.2	0.35 ± 0.06	1.1 ± 0.1
UBC 51	59.69	52.55	7.7	59	1111 ± 42	7.0 ± 0.1	-0.11 ± 0.08	1.4 ± 0.2
UBC 521	187.52	-64.16	4.7	83	2954 ± 135	6.8 ± 0.5	0.23 ± 0.23	2.7 ± 0.1
UBC 531	211.81	-63.06	4.6	67	2601 ± 115	7.1 ± 0.3	0.25 ± 0.24	2.0 ± 0.1
UBC 532	226.46	-62.24	2.5	57	2036 ± 25	7.1 ± 0.6	0.18 ± 0.15	2.6 ± 0.1
UBC 534	238.66	-57.87	7.9	58	1710 ± 61	7.7 ± 0.6	0.26 ± 0.14	1.4 ± 0.1
UBC 535	237.52	-56.78	5.8	84	3022 ± 99	7.0 ± 0.2	0.36 ± 0.20	2.2 ± 0.0
UBC 536	237.67	-55.62	5.7	48	1975 ± 64	7.4 ± 0.5	0.27 ± 0.21	3.0 ± 0.1
UBC 541	243.05	-50.17	9.2	163	2406 ± 170	7.0 ± 0.1	0.45 ± 0.15	2.8 ± 0.0
UBC 542	244.54	-49.50	5.7	124	1789 ± 106	7.2 ± 0.4	0.24 ± 0.26	2.5 ± 0.2
UBC 545	248.52	-48.92	8.0	118	3144 ± 191	6.8 ± 0.2	0.27 ± 0.13	2.5 ± 0.0

Table A.4 continued

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	$[Fe/H]$	A_V (mag)
UBC 548	250.40	-46.10	6.7	131	2501 ± 130	7.1 ± 0.5	0.13 ± 0.15	3.1 ± 0.1
UBC 549	252.16	-46.92	6.2	71	3131 ± 160	6.9 ± 0.1	0.45 ± 0.13	2.4 ± 0.1
UBC 550	252.63	-45.06	5.9	130	2491 ± 100	7.1 ± 0.6	0.36 ± 0.15	2.6 ± 0.1
UBC 552	254.69	-42.04	5.8	106	2846 ± 159	6.8 ± 0.1	0.41 ± 0.26	3.3 ± 0.1
UBC 559	259.10	-39.66	16.1	329	2302 ± 189	7.0 ± 0.1	0.29 ± 0.17	3.2 ± 0.0
UBC 562	262.77	-37.88	6.6	125	2224 ± 338	7.0 ± 0.7	0.03 ± 0.18	3.4 ± 0.1
UBC 576	284.70	0.42	2.7	28	1332 ± 36	7.4 ± 0.8	0.10 ± 0.22	3.0 ± 0.2
UBC 582	299.84	38.55	3.6	17	1103 ± 13	7.2 ± 0.1	-0.01 ± 0.09	0.6 ± 0.1
UBC 588	311.59	45.40	5.6	50	1732 ± 45	6.9 ± 0.6	0.13 ± 0.23	3.0 ± 0.1
UBC 606	37.46	57.37	5.0	45	2463 ± 54	7.0 ± 0.2	-0.13 ± 0.13	2.2 ± 0.0
UBC 620	97.96	2.53	7.9	36	1464 ± 28	7.4 ± 0.2	0.14 ± 0.11	1.9 ± 0.1
UBC 63	79.65	37.81	8.4	70	1470 ± 57	7.3 ± 0.1	-0.18 ± 0.07	0.8 ± 0.0
UBC 632	121.59	-9.61	9.5	107	4110 ± 194	7.7 ± 0.8	-0.25 ± 0.07	1.5 ± 0.3
UBC 652	162.35	-54.70	6.1	39	2760 ± 75	6.8 ± 0.1	0.09 ± 0.12	0.9 ± 0.0
UBC 653	160.11	-59.01	7.9	132	2553 ± 107	7.0 ± 0.2	0.13 ± 0.13	1.5 ± 0.1
UBC 663	182.75	-63.63	7.4	36	2718 ± 43	6.9 ± 0.1	0.11 ± 0.11	1.3 ± 0.1
UBC 665	242.17	-51.43	2.9	29	2098 ± 84	7.0 ± 0.4	0.18 ± 0.22	2.9 ± 0.1
UBC 668	250.66	-48.00	4.1	68	3290 ± 99	6.8 ± 0.5	0.52 ± 0.13	3.1 ± 0.1
UFMG 22	263.20	-37.64	7.3	86	2327 ± 295	7.1 ± 0.7	0.26 ± 0.09	3.2 ± 0.1
UFMG 3	238.11	-55.42	5.5	242	1897 ± 70	7.4 ± 0.2	0.19 ± 0.12	2.7 ± 0.1
UFMG 45	239.70	-55.80	4.6	27	2716 ± 214	7.6 ± 0.5	0.42 ± 0.14	1.9 ± 0.2
UFMG 53	195.43	-64.03	3.3	43	2888 ± 261	7.1 ± 0.8	0.17 ± 0.16	3.1 ± 0.1
UPK 150	319.54	54.49	39.4	89	976 ± 21	7.1 ± 0.2	-0.10 ± 0.13	2.5 ± 0.9
UPK 166	339.20	46.90	47.8	214	625 ± 16	7.7 ± 0.1	0.00 ± 0.05	0.4 ± 0.0
UPK 169	326.75	59.80	27.6	63	822 ± 17	7.1 ± 0.5	-0.02 ± 0.29	1.8 ± 0.3
UPK 194	336.52	65.01	17.8	114	907 ± 14	7.5 ± 0.2	0.10 ± 0.18	2.1 ± 0.2
UPK 220	350.99	66.51	12.5	194	950 ± 12	7.3 ± 0.3	0.03 ± 0.13	2.7 ± 0.1
UPK 23	282.22	-12.87	7.3	20	951 ± 40	7.5 ± 0.3	0.25 ± 0.14	1.5 ± 0.1
UPK 265	32.03	65.43	20.7	53	846 ± 17	7.4 ± 0.3	-0.12 ± 0.15	1.9 ± 0.2
UPK 28	277.36	-4.98	9.8	50	840 ± 30	7.6 ± 0.2	0.35 ± 0.11	3.1 ± 0.1
UPK 38	280.83	-1.65	4.9	21	589 ± 25	7.1 ± 0.1	-0.03 ± 0.17	2.2 ± 0.1
UPK 385	75.73	13.06	23.0	29	316 ± 4	7.1 ± 0.1	-0.12 ± 0.10	0.7 ± 0.3
UPK 398	85.99	7.06	18.8	36	448 ± 2	7.2 ± 0.1	0.01 ± 0.09	0.5 ± 0.3
UPK 422	87.90	-7.59	47.8	104	297 ± 1	7.4 ± 0.0	0.06 ± 0.06	0.4 ± 0.1
UPK 445	92.38	-15.07	18.1	51	656 ± 12	7.2 ± 0.1	-0.11 ± 0.05	1.2 ± 0.0
UPK 457	104.46	-18.70	14.3	22	960 ± 11	7.4 ± 0.0	0.06 ± 0.17	0.4 ± 0.1
UPK 526	115.27	-52.91	46.9	37	564 ± 7	7.6 ± 0.1	-0.03 ± 0.05	0.6 ± 0.1
UPK 535	127.11	-50.94	66.0	100	319 ± 5	7.6 ± 0.0	-0.03 ± 0.04	0.3 ± 0.1
UPK 540	114.54	-58.43	46.0	48	365 ± 4	7.5 ± 0.0	-0.01 ± 0.07	0.4 ± 0.1
UPK 604	224.32	-59.81	15.6	43	745 ± 9	7.1 ± 0.5	-0.21 ± 0.31	1.8 ± 0.4
UPK 606	216.13	-46.36	43.0	46	167 ± 2	7.2 ± 0.1	-0.05 ± 0.18	0.1 ± 0.3

Table A.4 continued

Cluster	α (deg)	δ (deg)	r_{50} (arcmin)	N	d (pc)	$\log \tau$	$[Fe/H]$	A_V (mag)
UPK 62	289.73	20.83	6.6	33	885 ± 21	7.0 ± 0.2	0.02 ± 0.25	3.4 ± 0.3
UPK 621	237.20	-54.39	25.5	57	878 ± 32	7.6 ± 0.2	0.15 ± 0.13	0.9 ± 0.2
UPK 640	250.41	-39.57	73.9	540	173 ± 1	7.4 ± 0.1	0.15 ± 0.10	0.2 ± 0.1
vdBergh 130	304.46	39.34	2.9	62	1456 ± 240	7.0 ± 0.1	-0.03 ± 0.16	2.4 ± 0.0
vdBergh 80	97.75	-9.62	9.1	60	947 ± 2	6.8 ± 0.0	-0.15 ± 0.09	1.7 ± 0.2
vdBergh 85	101.73	1.33	2.7	29	1720 ± 167	7.1 ± 0.1	-0.06 ± 0.12	1.2 ± 0.3
vdBergh 92	106.04	-11.49	6.8	154	1114 ± 42	6.7 ± 0.1	0.03 ± 0.09	1.0 ± 0.1

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Statement of Authorship

I hereby solemnly declare that I have completed the present work independently and without any unauthorised assistance from third parties. No aids or literature other than those specified have been used. Passages, information, and concepts directly or indirectly taken from other sources are identified with a citation of the source. This work has not been submitted to any other examination authority in the same or similar form. I solemnly affirm that I have told the truth to the best of my knowledge and have not withheld anything.

I have no objections to making this master's thesis available to the public.

Place, Date

Author's signature

Jena, 09.09.2024

Surodeep Sheth